Bangladesh University of Engineering and Technology

Department of Computer Science & Engineering

Electronic Circuits Question Bank

EEE 263 · Electronic Circuits

Year Coverage: 2011-12 to 2024–25

Topics: Diode Characteristics, Wave Shaping & Clipping/Clamping, BJT & MOSFET Amplifiers, Op-Amps (Filters, Schmitt Trigger, Analog Computing), Oscillators & Timers (555) & more

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BUET · EEE 263 · Electronic Circuits

Electronic Circuits

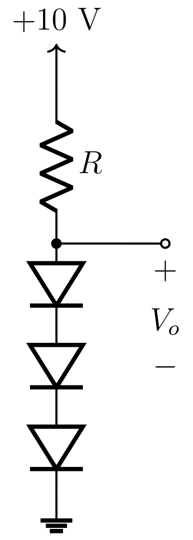
Part I — Semiconductor Device Characteristics

Ideal device models, I-V characteristics, biasing, and circuit analysis for Diode, BJT, and MOSFET

S1 Ideal Diode Characteristics

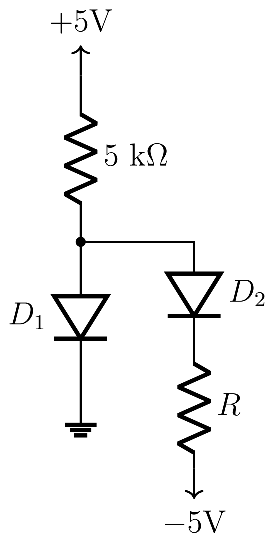
Diode Circuit Analysis

Q1. For the circuit shown, I is the current flowing through each of the diodes and R is unknown. The diodes are identical with $I_S = 10^{-16}$ A and $n = 1$. (i) Find the value of (I, R) such that $V_o = 2.4$ V is maintained. (ii) If a current of 1 mA is drawn away from the output terminated by a load, what is the change in output voltage?



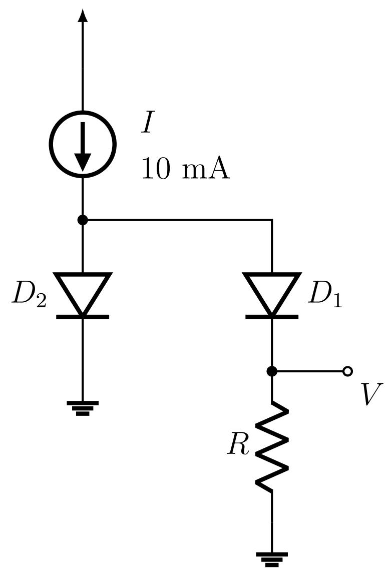
(15 marks) [EEE 263 | L-2/T-1 | 2024–25]

Q2. For the circuit given, determine the range of values of resistor R such that D_1 does not conduct and D_2 conducts. Assume ideal diodes.



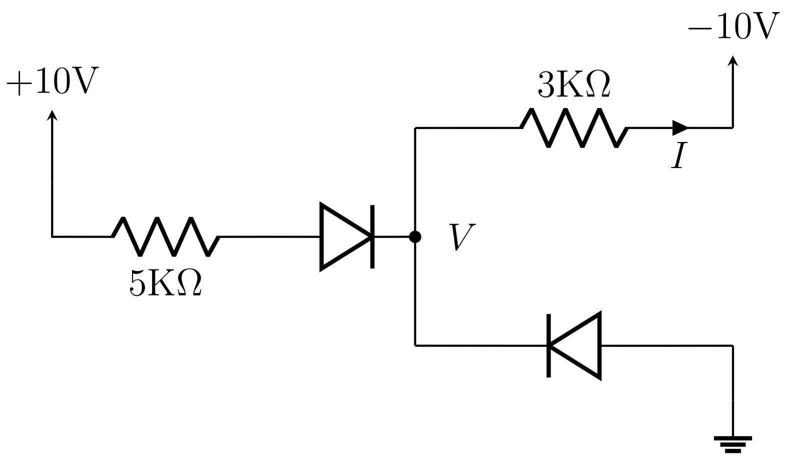
(15 marks) [EEE 263 | L-2/T-1 | 2022–23]

Q3. In the circuit shown, both diodes are identical. Find the value of R for which $V = 50$ mV.



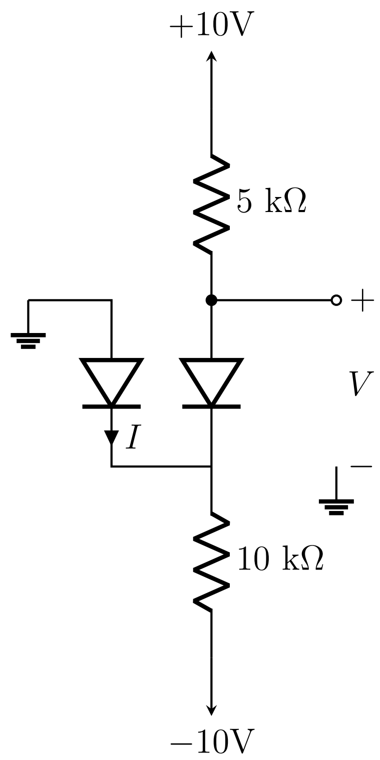
(16 marks) [EEE 263 | L-2/T-1 | 2021–22]

Q4. Find the values of I and V in the circuit shown. Use the 0.7 V drop diode model.

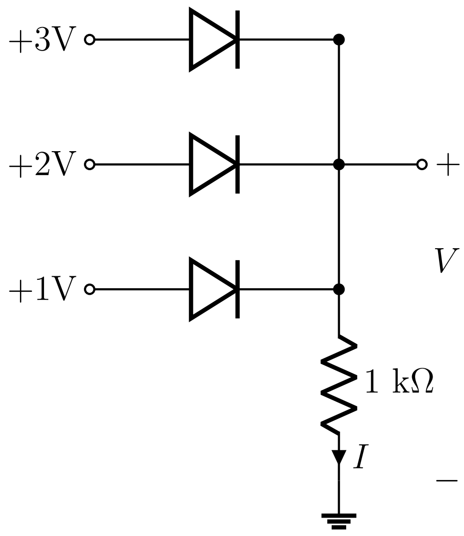


(10.67 marks) [EEE 263 | L-2/T-1 | 2014–15]

Q5. Assuming the diodes to be ideal, find the values of I and V in the circuits shown.



(i)



(ii)

(16 marks) [EEE 263 | L-2/T-1 | 2011–12]

Diode Models (Ideal, CVD, Piecewise Linear)

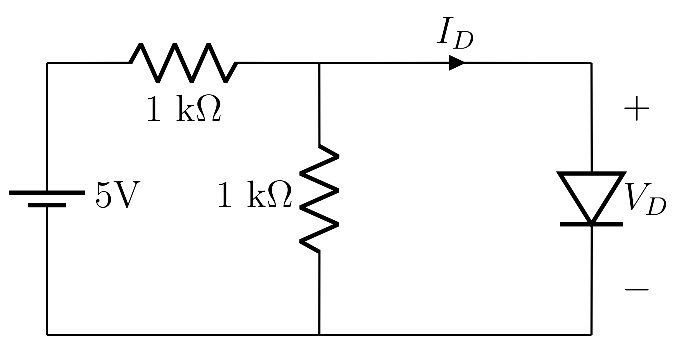
Q1. Describe the different models for diode forward characteristics. **(7 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Q2. Determine the current I_D and diode voltage V_D for the circuit shown using:

(a) Piecewise linear model ($V_{DO} = 0.7\,\text{V}$, $r_D = 20\,\Omega$)

(b) Constant voltage drop model ($V_{DO} = 0.7\,\text{V}$)

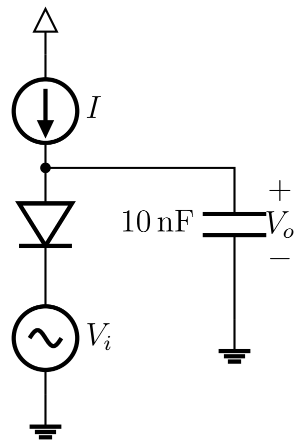
(c) Ideal diode model



(18 marks) [EEE 263 | L-2/T-1 | 2011–12]

Small Signal Model

Q1. In the circuit shown, "I" is a DC current source and "V_i" is a sinusoidal signal with small amplitude ($< 10\,\text{mV}$) and a frequency of $100\,\text{kHz}$. Representing the diode by its small-signal resistance r_d (a function of I):
(i) sketch the small-signal equivalent circuit for determining the sinusoidal output V_o across the capacitor;
(ii) find the phase shift between V_i and V_o;
(iii) find the value of I that will provide a phase shift of -45;
(iv) find the range of phase shift achieved as I is varied from 0.1 to 10 times the value found in (iii). Assume $n = 1$.



(20 marks) [EEE 263 | L-2/T-1 | 2023–24]

Q2. The small-signal model of a diode is said to be valid for voltage variations of about $10\,\text{mV}$. To what percentage current change does this correspond for $n = 1$ (consider both positive and negative signals)? What is the maximum allowable voltage signal (positive or negative) if the current change is limited to 10%? **(10 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Peak Rectifier

Q1. Consider a half-wave rectifier circuit with $R = 1\,\text{k}\Omega$ load operating from a $120\,\text{V}$ (rms), $60\,\text{Hz}$ household supply through a 10-to-1 step-down transformer, augmented with a filter capacitor C across the load. Assume $V_D = 0.7\,\text{V}$ for any current.
(i) Find C for a peak-to-peak ripple of 10% of peak output.
(ii) Find C for a peak-to-peak ripple of 1% of peak output.
(iii) Find the average output voltage for each of the case (i) & (ii) .
(iv) Find the fraction of the cycle for which the diode conducts for each of the case (i) & (ii).
(v) Find the average and peak diode currents for each of the case (i) & (ii).
(20 marks) [EEE 263 | L-2/T-1 | 2024–25]

Q2. Consider a peak rectifier fed by a $60\,\text{Hz}$ sinusoid with a peak value $V_p = 100\,\text{V}$. Let the load resistance $R = 10\,\text{k}\Omega$. Find the value of capacitance C that will result in a peak-to-peak ripple of $2\,\text{V}$. Also calculate the fraction of the cycle during which the diode is conducting and the average and peak values of the diode current. **(10 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Q3. Consider a half-wave peak rectifier fed with a voltage having a triangular waveform with $20\,\text{V}$ peak-to-peak amplitude, zero average and $1\,\text{kHz}$ frequency. Load resistance $R = 100\,\Omega$ and filter capacitor $C = 100\,\mu\text{F}$. Assume the diode has a forward voltage drop of $0.7\,\text{V}$ when conducting. Find: (i) average DC output voltage, (ii) time interval during which the diode conducts, (iii) average diode current during conduction, (iv) maximum diode current. **(20 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Q4. A peak rectifier circuit is operated with a $1\,\text{k}\Omega$ load and a diode that can be modeled to have a $0.7\,\text{V}$ drop at any current. It is fed by a $50\,\text{Hz}$ sinusoid and the capacitor is chosen to provide a peak-to-peak ripple voltage of 10% of the peak output voltage. Consider the load current $I_L = 15\,\text{mA}$.

(a) What is the value of the capacitor?

(b) What fraction of the cycle does the diode conduct?

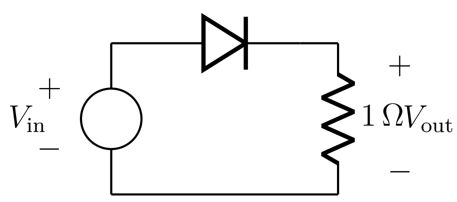
(c) What is the average diode current?

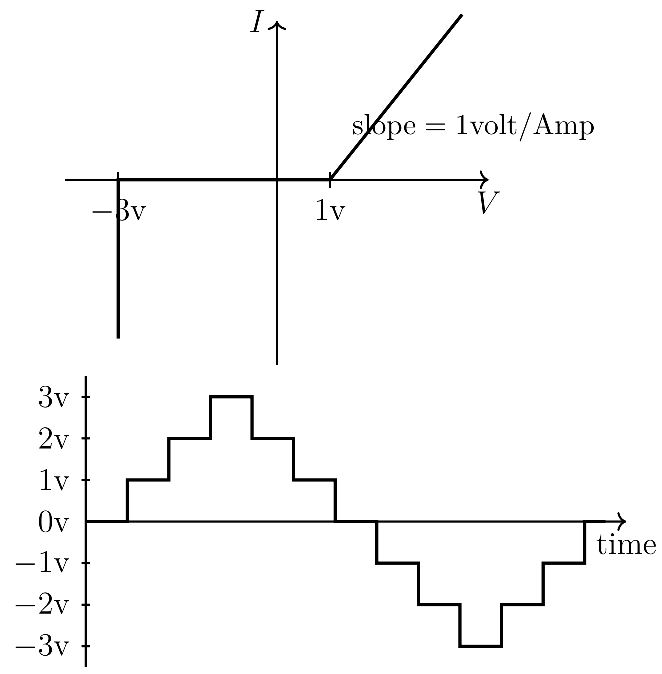
(d) What is the peak diode current?

(16 marks) [EEE 263 | L-2/T-1 | 2014–15]

I-V Characteristics & Waveform Analysis

Q1. (a) What would be the output voltage if the diode and the resistor were connected in parallel? **(10.67 marks)** [EEE 263 | L-2/T-1 | 2013–14]

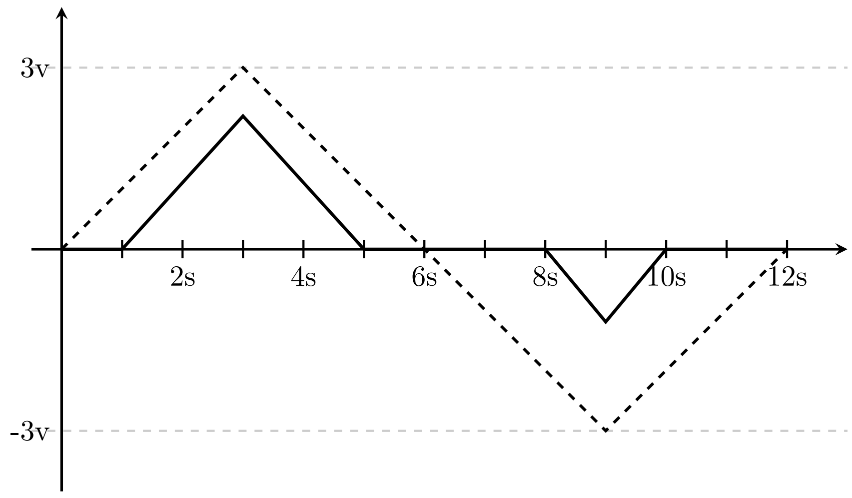




(b) The I-V characteristics of a diode and a sinusoidal input voltage signal V_{in} are given in the figure. Draw the output voltage V_{out} across the resistor. **(16 marks)**

(c) What would be the output voltage V_{out} across the resistor if the diode had a constant 0.7 V voltage drop? **(10 marks)**

Q2. (a) An AC voltage source is applied to a series combination of a diode and a resistance. The dotted line in the figure is the input voltage waveform (V_{ac}) and the solid line is the output voltage waveform (V_R) across the resistance. Copy V_R from the figure to your answer script exactly to scale and draw the voltage waveform across the diode over the same graph. Given that $\frac{dV_{ac}}{dt} = 1\,\text{V/sec}$ when V_{ac} is rising and $\frac{dV_{ac}}{dt} = -1\,\text{V/sec}$ when V_{ac} is falling.



(20 marks) [EEE 263 | L-2/T-1 | 2012–13]

(b) Draw the I-V characteristics curve of the above diode exactly to scale (mention the voltage(s) at which the diode begins conduction). **(16 marks)** [EEE 263 | L-2/T-1 | 2012–13]

(c) Draw the voltage waveform across the diode if it had been connected in parallel to the resistance. **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Diode Thermal Characteristics

Q1. When a 15 A current is applied to a particular diode, the junction voltage immediately becomes 0.7 V. As power dissipation raises its temperature, the voltage eventually reaches 0.6 V. (i) What is the apparent rise in junction temperature? (ii) What is the temperature rise per watt of power dissipation? (iii) What is the power dissipated in the diode in its final state? **(15 marks)** [EEE 263 | L-2/T-1 | 2024–25]

Q2. When a 15 A current is applied to a particular diode, the junction voltage immediately becomes 0.7 V. As power dissipation raises its temperature, the voltage eventually reaches 0.58 V.
(i) What is the apparent rise in junction temperature?
(ii) What is the temperature rise per watt of power dissipation?
(iii) What is the power dissipated in the diode in its final state?
(10 marks) [EEE 263 | L-2/T-1 | 2023–24]

Bridge Rectifier

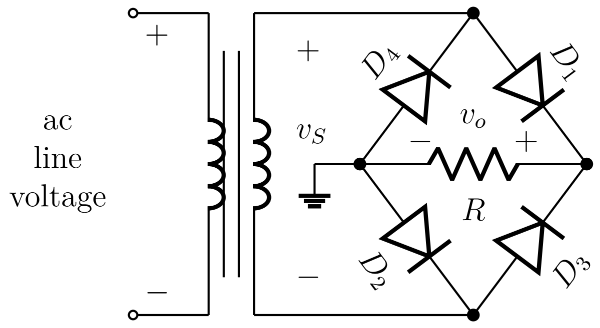
Q1. Consider a bridge rectifier circuit with a filter capacitor C placed across the load resistor $R = 100\,\Omega$. The transformer secondary delivers a sinusoid of 12 V (rms) at 60 Hz. Assume $V_D = 0.8\,\text{V}$.
(i) Find the value of C such that the ripple voltage is no larger than 1 V peak-to-peak.
(ii) What is the DC voltage at the output?

(iii) Find the load current.
(iv) Find the average and peak diode currents.
(v) What is the peak reverse voltage across each diode?
(15 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Q2. A full-wave bridge rectifier circuit with $1\,\text{k}\Omega$ load operates from a $120\,\text{V}$ (rms), $60\,\text{Hz}$ household supply through a 10:1 step-down transformer. Each diode has a voltage drop of $0.7\,\text{V}$ at any current. (i) What is the peak value of the rectified voltage across the load? (ii) For what fraction of a cycle does each diode conduct? (iii) What are the average voltage across and average current through the load? **(10 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

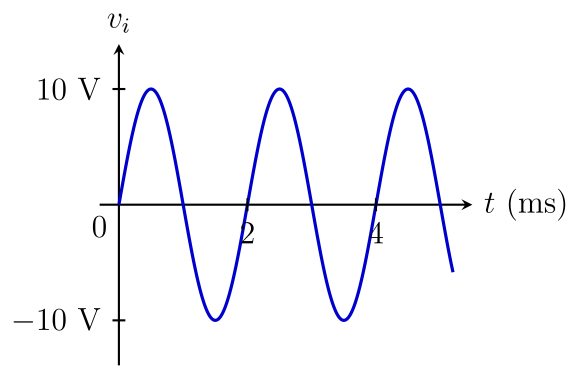
Q3. A full-wave bridge rectifier circuit with a 1-kΩ load operates from a 120-V (rms), 60-Hz household supply through a 10-to-1 step-down transformer having a single secondary winding. Each diode can be modeled to have a 0.7-V drop in forward bias. Find: the peak value of the rectified voltage across the load; the fraction of a cycle for which each diode conducts; the average voltage and average current through the load. **(16 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

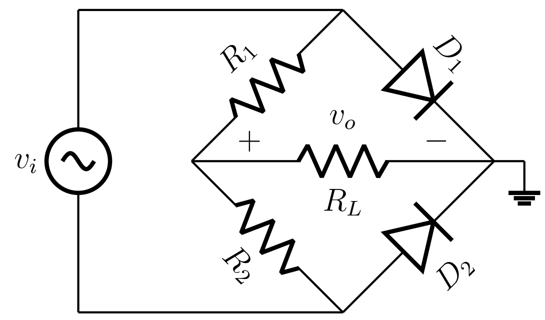
Q4. For the bridge rectifier circuit shown, v_s is a $12\,\text{V}$ (rms) sinusoid, $V_D = 0.7\,\text{V}$, and $R = 100\,\Omega$. Find the average of the output voltage v_o, the peak diode current, and the PIV.



(15 marks) **[EEE 263 | L-2/T-1 | 2021–22]**

Q5. Draw the transfer curve of the circuit shown, assuming all diodes are ideal, $R_1 = R_2 = 20\,\text{k}\Omega$, and $R_L = 10\,\text{k}\Omega$. (i) What is the maximum peak voltage of the output signal? (ii) What is the peak-inverse voltage? (iii) What is the average current through the load?





(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Q6. Draw the circuit of a full-wave bridge rectifier and show the output wave shape for a sinusoidal input. **(16 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Q7. For a full-wave bridge rectifier, find the value of peak current through the diode and PIV for the diode. Given: $V_s = 12\,\text{V}$ (rms), $V_{DO} = 0.7\,\text{V}$, $R = 100\,\Omega$. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

Q8. Comment on the value of RC time constant for a rectifier circuit. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

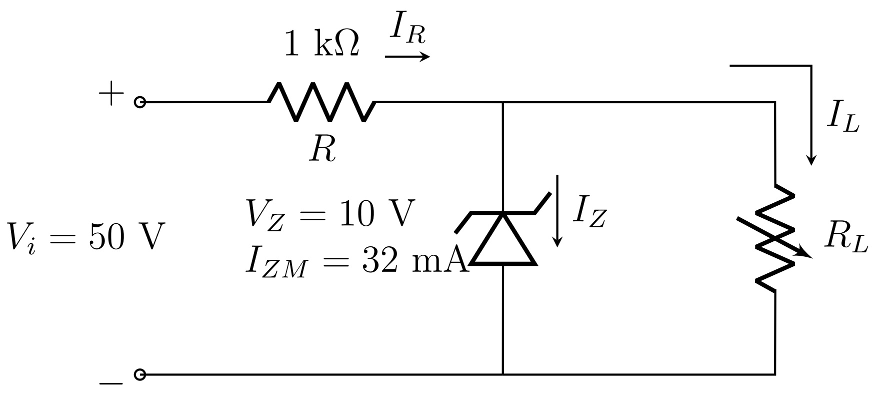
Q9. Show that the average output voltage of a bridge rectifier circuit is $V_o = \frac{2}{\pi}V_s - 2V_{DO}$. If the input voltage is a $12\,\text{V}$ (rms) sinusoid and the load resistance is $100\,\Omega$, calculate: (i) the average output voltage, (ii) the peak diode current, and (iii) the PIV of the diodes. Assume $V_{DO} = 0.7\,\text{V}$. **(16 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Zener Diode Regulator

Q1. A 6.8 V Zener diode exhibits its nominal voltage at a test current of 5 mA. At this current the incremental resistance is $r_z = 20\,\Omega$ and knee current $I_{ZK} = 0.2$ mA. The negative terminal is connected to $V^+ = 10$ V (± 1 V) through a 0.5 kΩ resistor; positive terminal is grounded. (i) Find V_o with no load at nominal supply. (ii) Find the change in V_o as supply varies by ± 1 V. (iii) Find the change in V_o when $R_L = 2$ kΩ is connected; what happens if $R_L = 0.5$ kΩ? (iv) Find the minimum R_L for which the diode still operates in breakdown. **(20 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

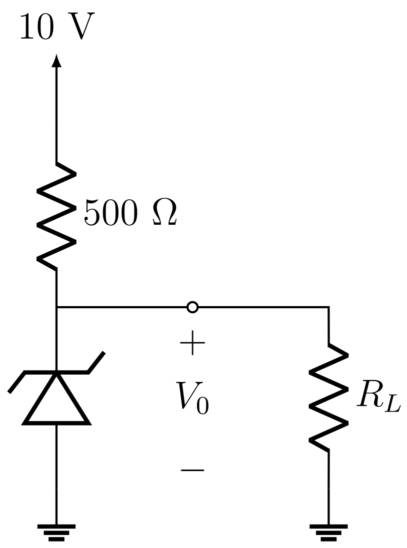
Q2. A 9.1 V Zener diode exhibits its nominal voltage at a test current of 28 mA. At this current the incremental resistance is specified to be 5 Ω. (i) Find V_{z0} in the Zener model. (ii) Find the Zener voltage at currents of 10 mA and 100 mA, respectively. (iii) Comment on the advantages/disadvantages of running the Zener diode at current values other than the nominal value. **(10 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

Q3. For the circuit given, determine the range of R_L and I_L that will result in V_{R_L} being maintained at 10 V. Also determine the maximum wattage rating of the diode. The Zener diode is a 10 V Zener with maximum current rating of 32 mA.



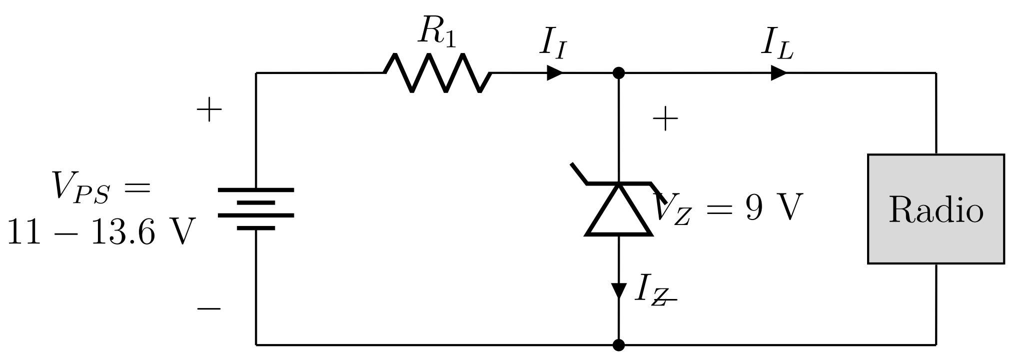
(18 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Q4. The zener diode in the circuit shown is specified to have $V_Z = 6.8$ V at $I_Z = 5$ mA, $r_z = 20\,\Omega$ and $I_{ZK} = 0.2$ mA. (i) Find V_0 at no load conditions. (ii) What is the minimum value of R_L for which the diode will still operate in the breakdown region?



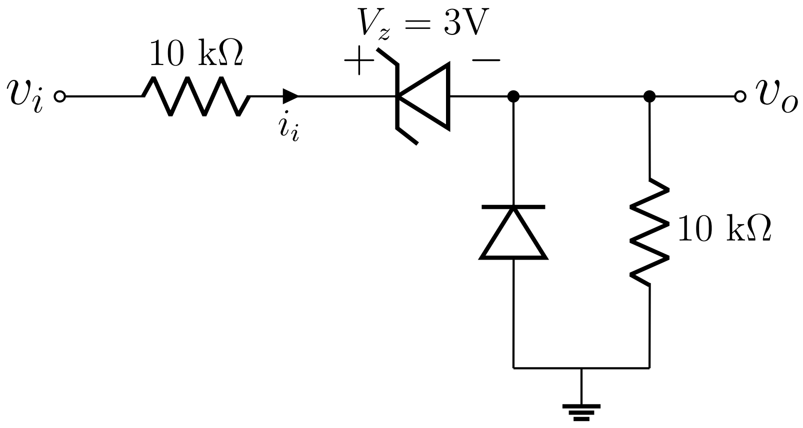
(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Q5. A voltage regulator circuit is to power a car radio at $V_L = 9$ V from an automobile battery whose voltage may vary between 11 and 13.6 V. The current in the radio will vary between 0 (off) and 100 mA (full volume). Determine the value of the current-limiting resistor R_i and the maximum power dissipated in the Zener diode.



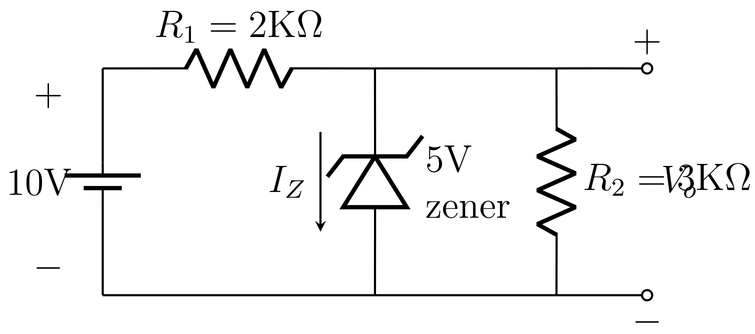
(18 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

Q6. For the circuit shown in Figure , plot v_0 versus v_i and i_1 versus v_i over the range $-10 \le v_\mathrm{i} \le +10$ V. Assume the diodes to be ideal.



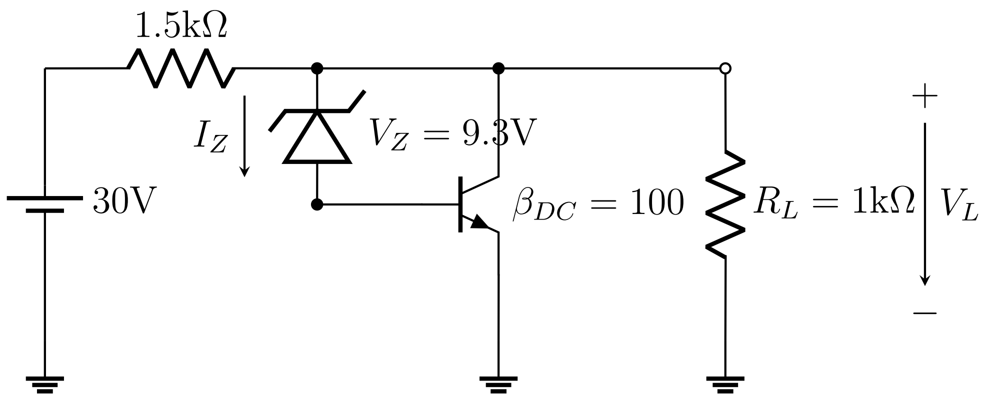
(13.67 marks) [EEE 263 | L-2/T-1 | 2016–17]

Q7. For the circuit shown, find V_a and I_z. Now if R_1 and R_2 are swapped, what will be the value of V_a and I_z?



(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

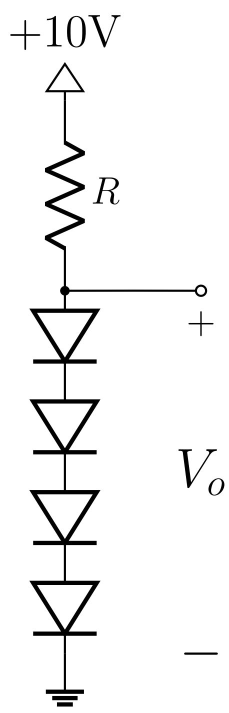
Q8. Calculate the regulated load voltage V_L and the current I_Z flowing through the zener diode in the voltage regulator circuit shown.



(16 marks) [EEE 263 | L-2/T-1 | 2014–15]

Diode Circuit Design

Q1. Design the circuit shown to provide an output voltage $v_a = 3$ V. Assume that the diodes available have 0.7 V drop at 1 mA, thermal voltage = 25 mV, and parameter $n = 1$.



(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

Full Wave Rectifier

Q1. Draw the circuit of a full-wave rectifier with center-tapped secondary winding and draw the input and output voltage waveshapes. Comment on the values of RC time constant for a rectifier circuit. **(20 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Q2. Draw the circuit diagram of a full wave rectifier with a filter capacitor. Assume the input is a sinusoidal signal with 10 kHz frequency and 7.07 V rms voltage, the load is 1 kΩ and capacitor used is 1 μF. Diode has a forward voltage drop of 0.7 V. Find:

(a) Ripple voltage.

(b) PIV and duty cycle.

(c) Peak output voltage.

(16 marks) [EEE 263 | L-2/T-1 | 2015–16]

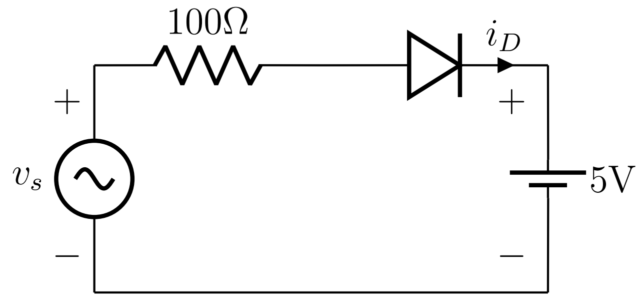
Battery Charger Circuit

Q1. The circuit shown is for charging a 5 V battery with $V_s = 10\sin(100\pi t)$. Find:

(a) Duty cycle (fraction of each cycle during which the diode conducts).

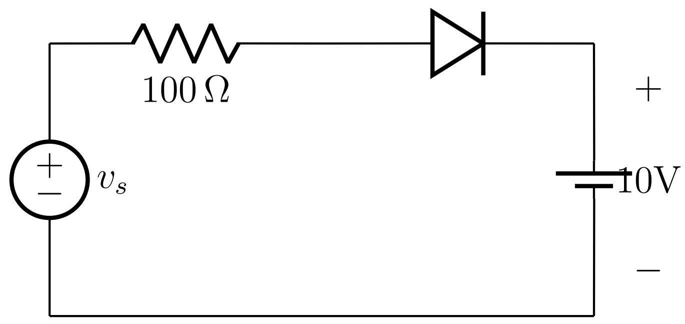
(b) Draw the diode current waveform indicating peak value.

(c) Maximum reverse bias voltage that appears across the diode.



(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

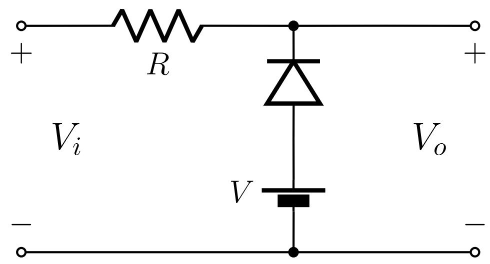
Q2. A circuit for charging a 10 V battery is shown in the figure. If V_s is a sinusoid with 20 V peak amplitude, find the fraction of each cycle during which the diode conducts. Assume the diode is ideal. Also find the peak value of the diode current and the maximum reverse bias voltage that appears across the diode.



(15 marks) [EEE 263 | L-2/T-1 | 2011–12]

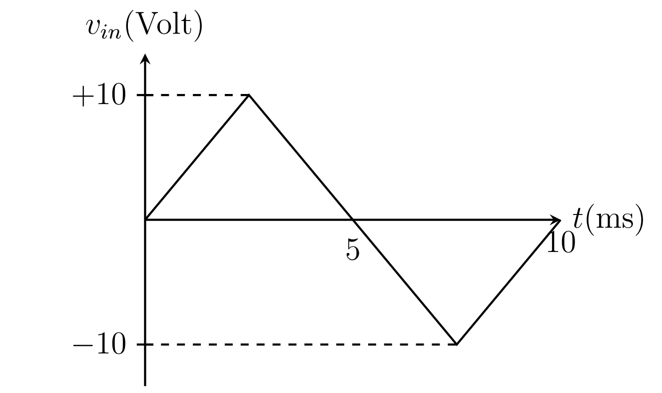
Diode Wave Shaping

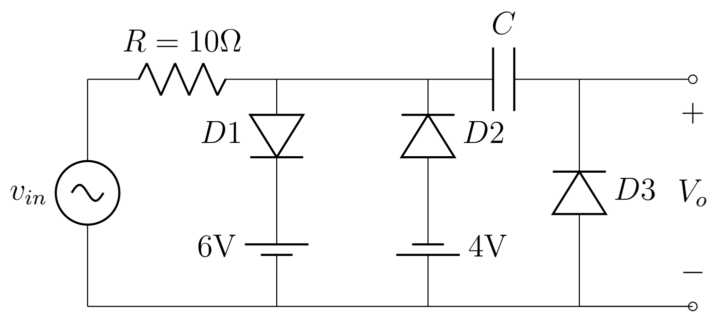
Q1. For a sinusoidal input, draw the output waveshape for the circuit shown.



(10 marks) [EEE 263 | L-2/T-1 | 2018–19]

Q2. Assuming that the diodes are ideal, find the output voltage V_o for the circuit given. Draw the wave shape clearly.





(20 marks) [EEE 263 | L-2/T-1 | 2014–15]

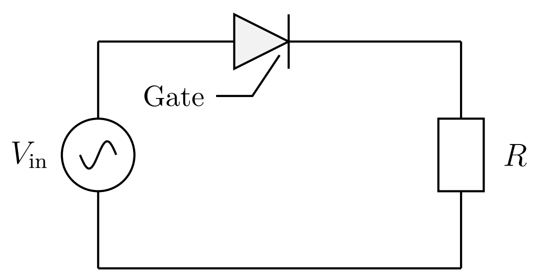
Special Purpose Diodes/Thyristors

Q1. Draw the circuit diagram of controlled full wave rectifier using SCR. Draw the waveforms of sinusoidal input, gate pulses and rectified output. **(11.67 marks)** [EEE 263 | L-2/T-1 | 2014–15]

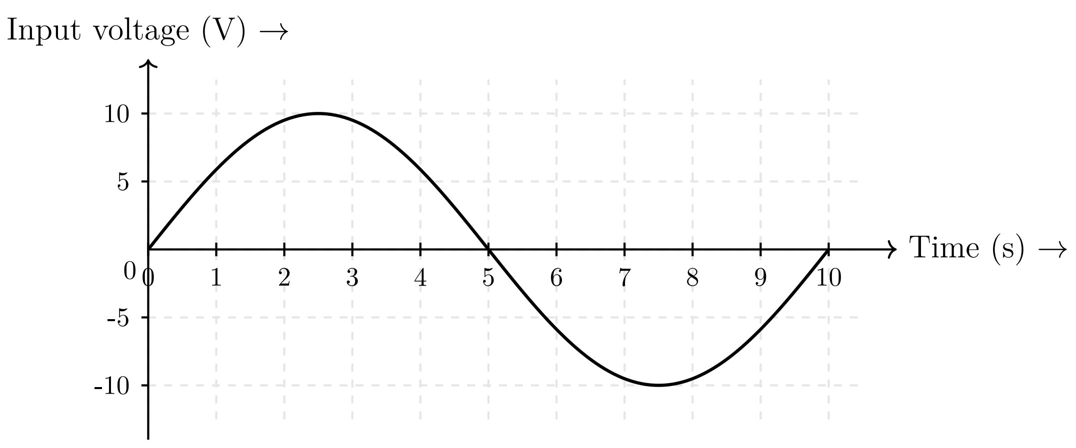
Q2. Explain SCR and TRIAC operation with suitable examples. **(15 marks)** [EEE 263 | L-2/T-1 | 2013–14]

Q3. Sketch the output wave form of the following circuit given in Fig. for Q. 1(c)(i) assuming output wave is obtained across the resistance. The input waveform and gate pulse sequence is given in Fig. for Q. 1(c)(ii).

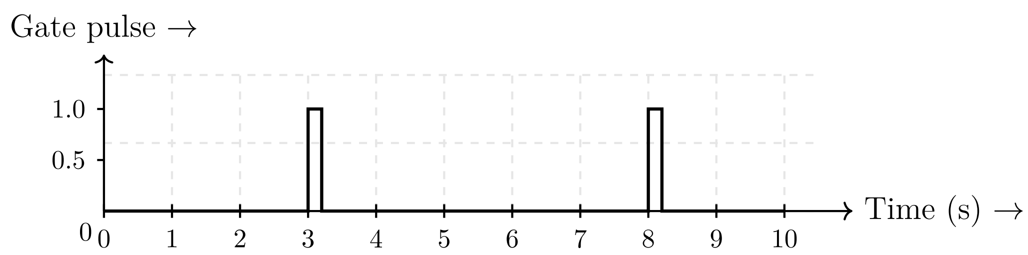
1. **Circuit Diagram [Fig. 1(c)(i)]**



2. **Input Voltage Waveform [Fig. 1(c)(ii)]**



3. **Gate Pulse Sequence [Fig. 1(c)(ii)]**



(20 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Q4. Draw the I-V characteristics of an SCR and a TRIAC. **(10 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

Q5. Explain the operating principle of a Silicon controlled Rectifier with the help of 'Two transistor model'. **(20 marks)** **[EEE 263 | L-2/T-1 | 2011–12]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

S2 — BJT Characteristics

Energy band profiles, DC analysis, biasing stability, circuit design

S2 BJT Characteristics

Energy Band Profile

Q1. Draw the energy band profile of a pnp transistor under zero bias and forward-active mode bias conditions. **(16/23 marks)** [EEE 263 | L-2/T-1 | 2020–21, 2018–19]

Device Structure and Operating Principle

Q1. Why is the third terminal (Base or Gate) used in a transistor? **(10 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Early Effect

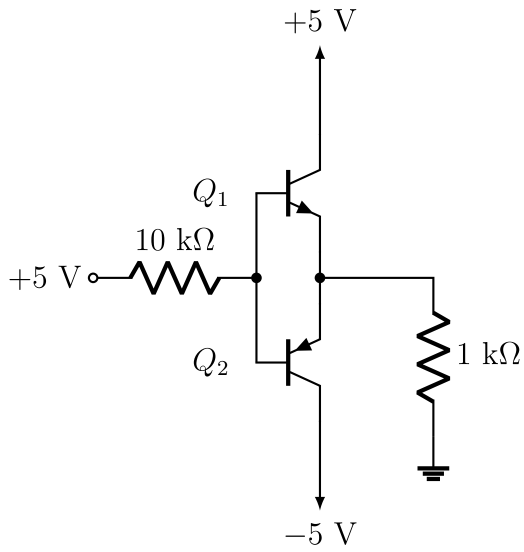
Q1. What is the Early-effect in a BJT? Draw the I–V curve of a practical BJT showing the Early effect. **(10 marks)** [EEE 263 | L-2/T-1 | 2023–24]

Q2. What is the early effect for BJT? What are the consequences of early effect on BJT behavior? **(16 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Q3. Why biasing is needed for an amplifier using BJT? Draw the output characteristics of a BJT connected in CE configuration. Define Early effect. **(10.667 marks)** [EEE 263 | L-2/T-1 | 2011–12]

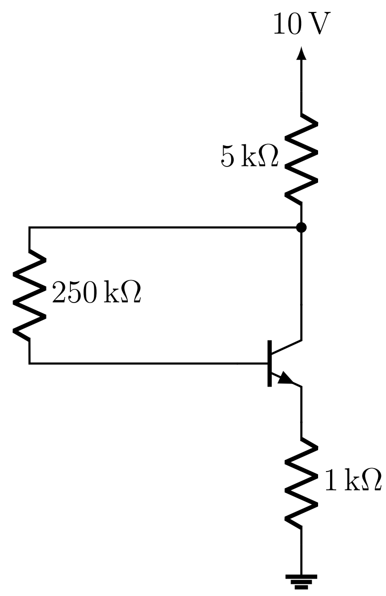
DC Analysis

Q1. For the circuit shown, find the voltages at all nodes and the currents through all branches. Assume $\beta = 100$.



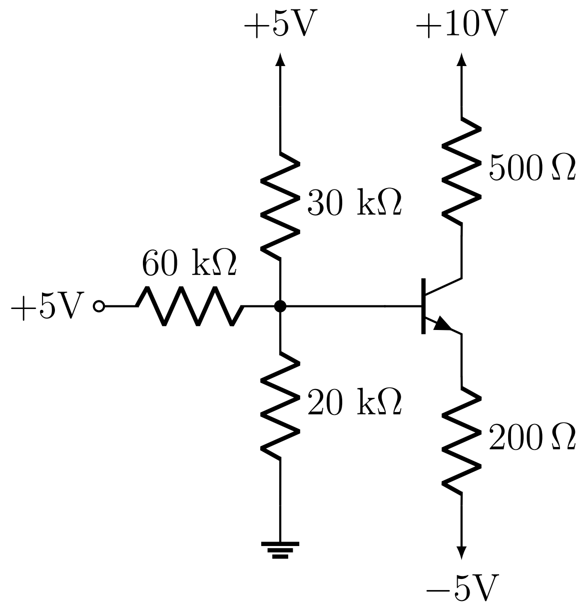
(20 marks) [EEE 263 | L-2/T-1 | 2020–21]

Q2. Calculate the collector current, emitter current and base current for the circuit shown in Figure . Also verify the operating mode of the transistor. Assume $\beta = 100$. (16)



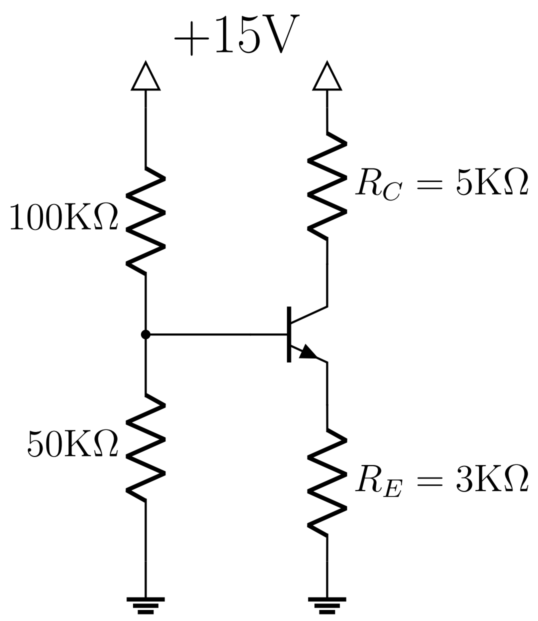
(16 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q3. Find I_C and V_{CE} for the circuit shown. Assume $\beta = 100$.



(16 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Q4. For the circuit shown, determine the voltages of all nodes and current through all branches. Assume $\beta = 100$.



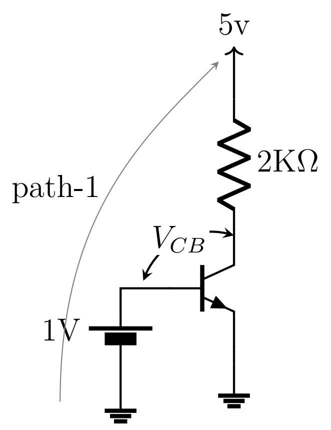
(20 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Q5. (a) Express the collector current I_c as a function of V_{CB} that involves the 2 kΩ resistance (it will be a straight line equation). **(16 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(b) Find out the mathematical equation describing the I_c–V_{CB} characteristic graph of the BJT (it will be a piecewise linear graph). **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(c) Draw the two I_c–V_{CB} graphs over the same plot exactly to scale and determine their intersection point. This intersection point gives the value of I_c and V_{CB} at the bias point. **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**

(d) From the value of V_{CB}, can you determine the value of V_{CE} at the bias point? **(10 marks)** **[EEE 263 | L-2/T-1 | 2012–13]**



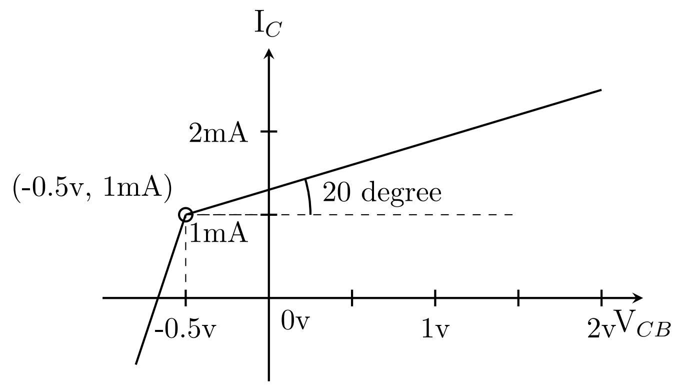
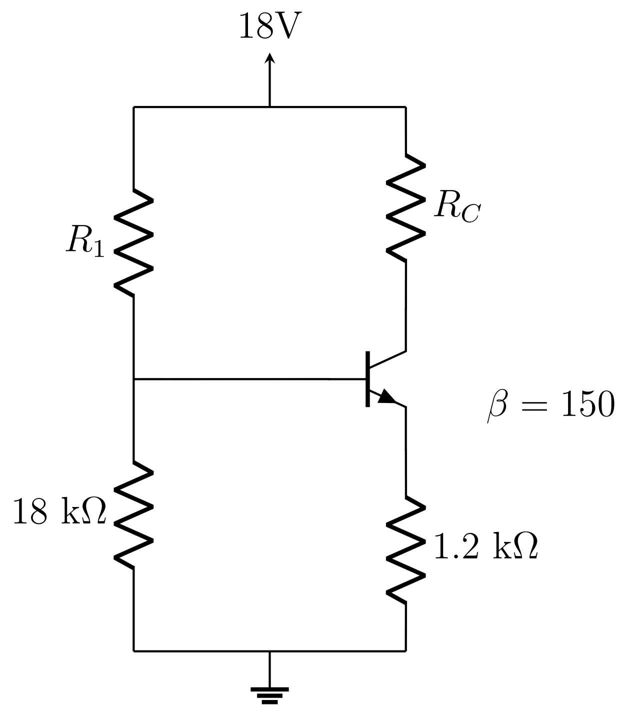


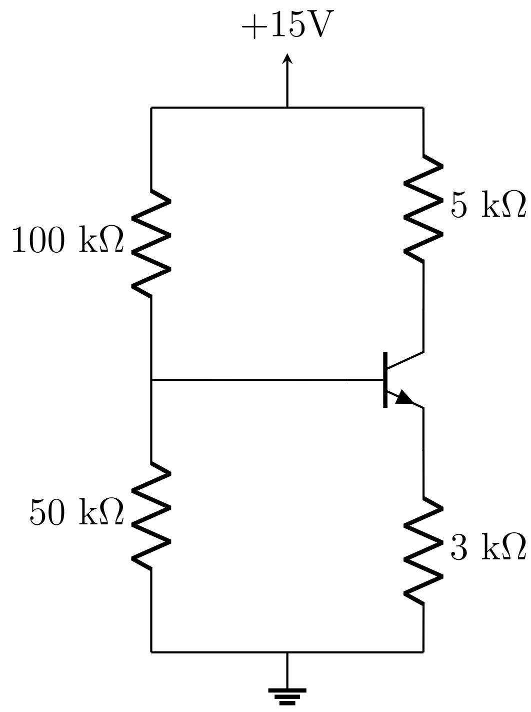
Figure-2

Q6. For the circuit shown, determine R_1 and R_C if $I_B = 0.02\,\text{mA}$ and $V_{CE} = 10V$.



(16 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Q7. Determine all node voltages and branch currents for the circuit shown. Assume $\beta = 100$ and $V_{BE} = 0.7\,\text{V}$.



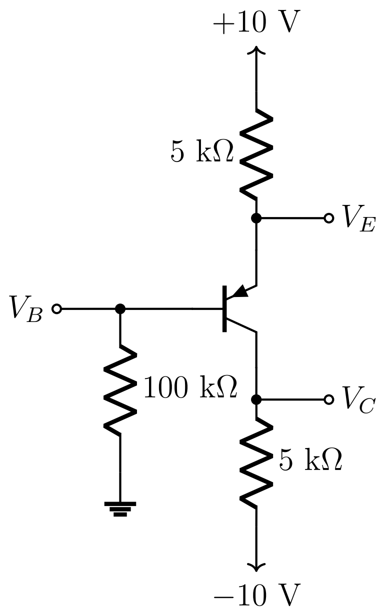
(18 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

BJT Saturation Parameters

Q1. A BJT for which $I_B = 0.5\,\text{mA}$ has $v_{CEsat} = 140\,\text{mV}$ at $I_C = 10\,\text{mA}$ and $v_{CEsat} = 170\,\text{mV}$ at $I_C = 20\,\text{mA}$. Estimate the values of: (i) saturation resistance R_{CEsat}; (ii) offset voltage V_{CEoff}; (iii) β_f; (iv) β_r. **(10 marks)** **[EEE 263 | L-2/T-1 | 2023–24]**

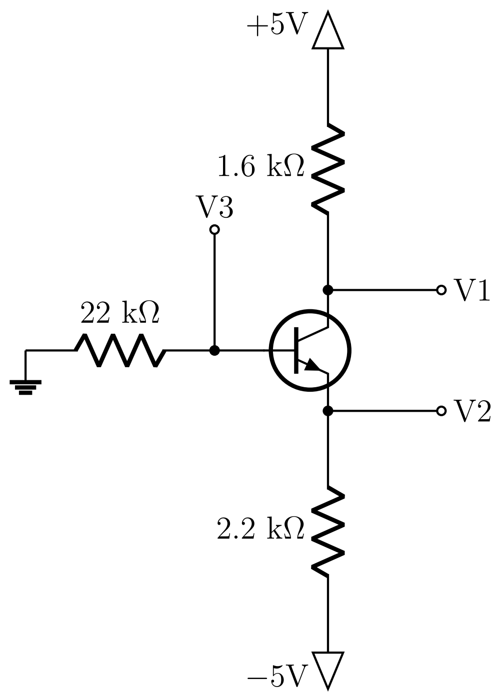
Multi-BJT Circuit Analysis

Q1. For the circuit shown, the emitter voltage was measured to be $V_E = +1.7\,\text{V}$. Assume $|V_{BE}| = 0.7\,\text{V}$. Find the values of the labeled node voltages and branch currents. What are the values of α and β for the pnp transistor?



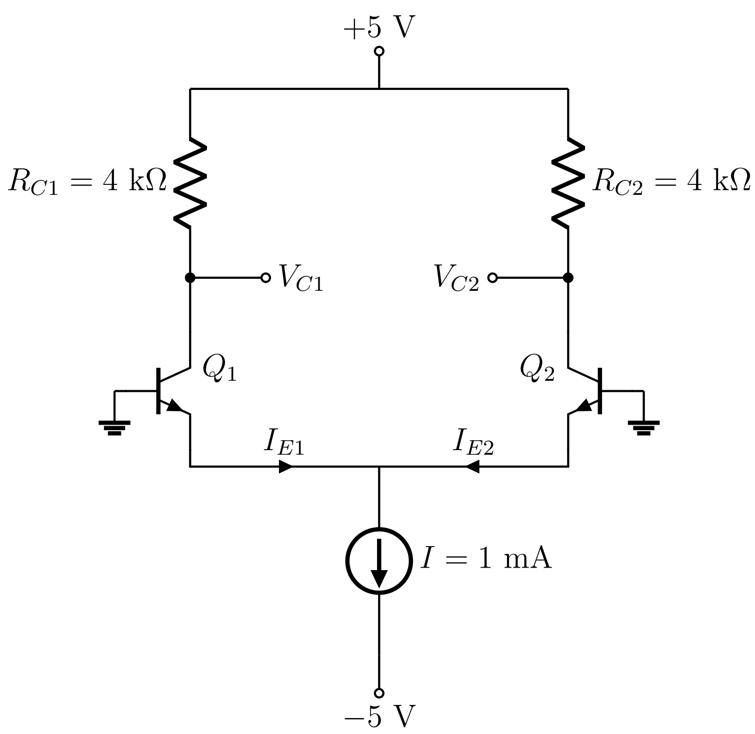
(15 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Q2. For the circuit shown, find the values of the labeled node voltages and branch currents. Assume β to be very high and $|V_{BE}| = 0.7\,\text{V}$. What will happen if β is reduced to a value of 100?



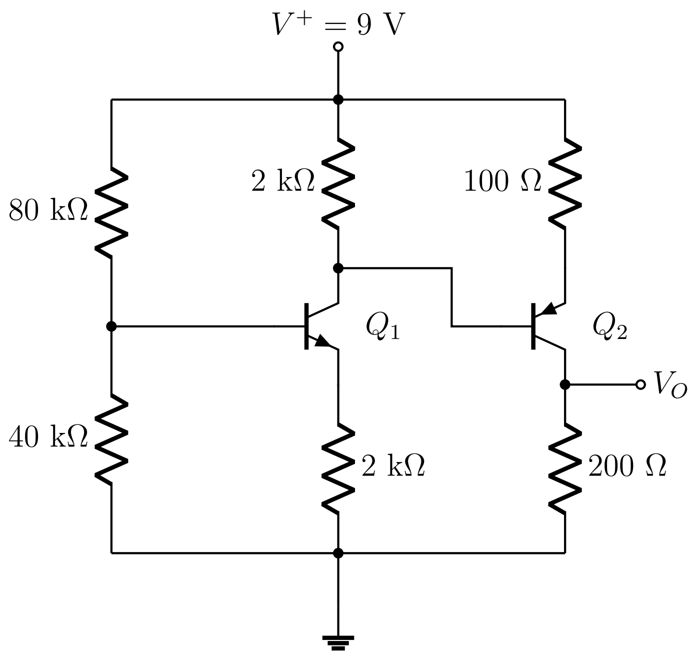
(15 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Q3. For the circuit shown with $\beta = 200$ for each transistor, determine: (i) I_{E1}, (ii) I_{E2}, (iii) V_{C1}, and (iv) V_{C2}.



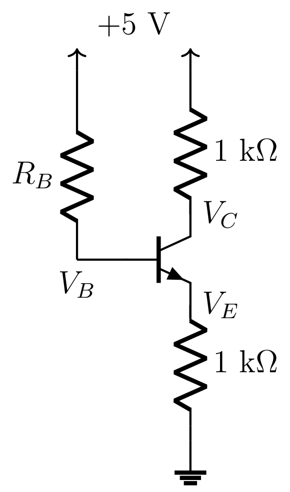
(15 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Q4. For the circuit shown, $\beta_n = 120$ (npn) and $\beta_p = 80$ (pnp). Determine I_{B1}, I_{C1}, I_{B2}, I_{C2}, V_{CE1}, and V_{EC2}.



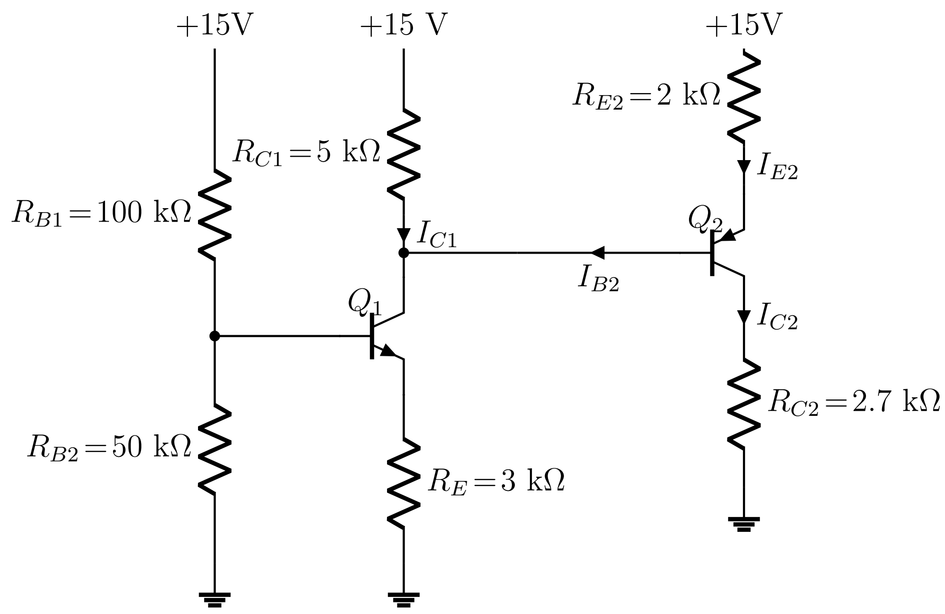
(20 marks) [EEE 263 | L-2/T-1 | 2022–23]

Q5. For the circuit shown, find V_B, V_E, and V_C for $R_B = 10\,\text{k}\Omega$. Let $\beta = 100$.



(16 marks) [EEE 263 | L-2/T-1 | 2021–22]

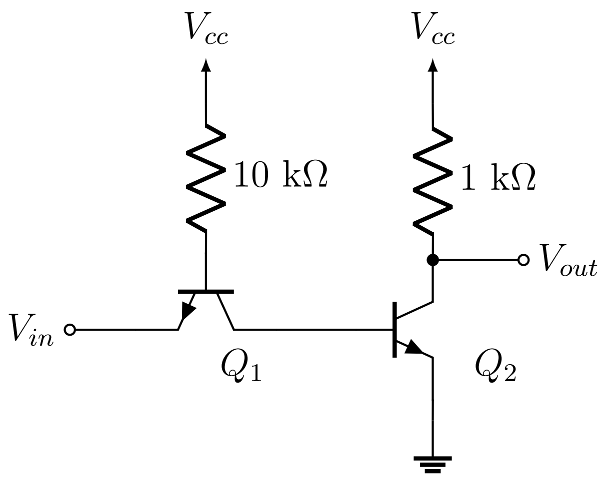
Q6. Determine the currents I_{C1} and I_{C2} for the multi-BJT circuit shown.



(30 marks) [EEE 263 | L-2/T-1 | 2017–18]

Voltage Transfer Characteristics

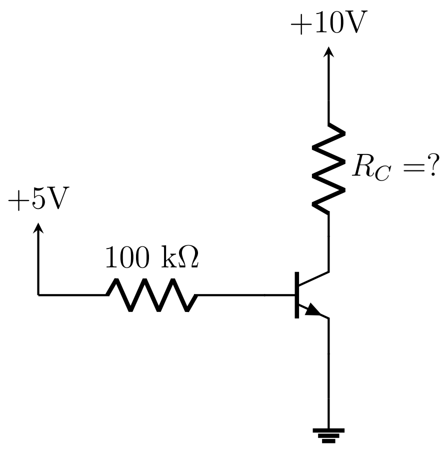
Q1. Draw the voltage transfer characteristics by qualitatively analyzing the circuit shown.



(15 marks) [EEE 263 | L-2/T-1 | 2016–17]

Operating Mode Design

Q1. The circuit shown is to be fabricated using a transistor type whose β can vary from 50 to 150. Design the circuit (find R_c) so that all fabricated circuits are guaranteed to be in the active mode. What is the range of collector voltages that the fabricated circuits may exhibit?



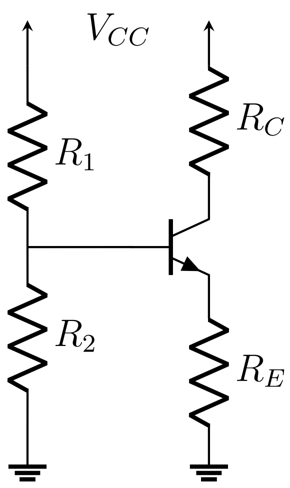
(20 marks) [EEE 263 | L-2/T-1 | 2018–19]

Biasing and Stability

Q1. What is transistor biasing? Why do we need transistor biasing? How can you achieve biasing stability? **(10 marks)** [EEE 263 | L-2/T-1 | 2020–21]

Q2. Comment on the desired values of input impedance, output impedance, voltage gain, and current gain of a BJT amplifier. Comment on the values of these four parameters for CB, CE, and CC configurations. **(16 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

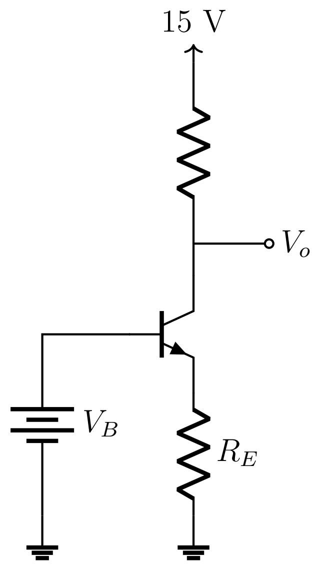
Q3. (a) Find the expression for the emitter current in the circuit shown. What are the conditions that should be satisfied? How does the feedback operation of R_E work? **(16 marks)**



(b) In the circuit of the previous question, if $R_1 = 40\,\text{k}\Omega$, $R_2 = 10\,\text{k}\Omega$, $R_c = 1.5\,\text{k}\Omega$, $R_E = 1\,\text{k}\Omega$, $V_{cc} = 12\,\text{V}$, and $\beta = 1000$, calculate the biasing point. How can the values of R_1 and R_2 be changed for a good biasing design? **(10 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Q4. Why do we need to bias a BJT at a proper Q-point before using it as an amplifier? **(5 marks)** [EEE 263 | L-2/T-1 | 2012–13]

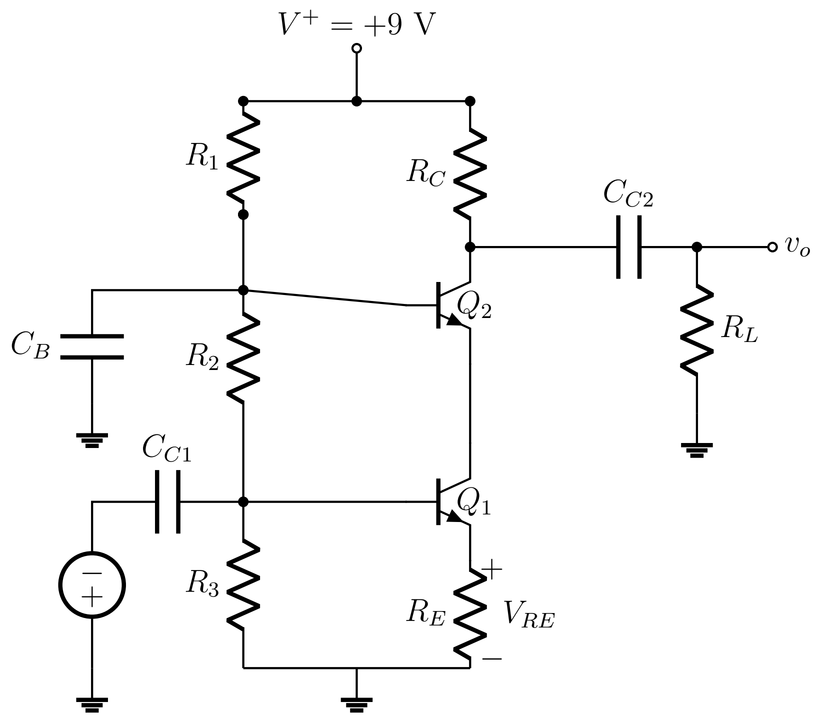
Q5. Graphically explain the benefit of having $R_E > 0$ in the biasing scheme shown.



(16 marks) **[EEE 263 | L-2/T-1 | 2012–13]**

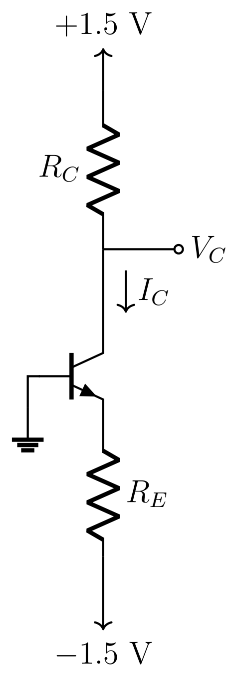
BJT Circuit Design

Q1. Design the circuit shown to meet the following specifications: $V_{CE1} = V_{CE2} = 2.5\,\text{V}$, $V_{RE} = 0.7\,\text{V}$, $I_{C1} \cong I_{C2} \cong 1\,\text{mA}$, and $I_{R1} \cong I_{R2} \cong I_{R3} = 0.10\,\text{mA}$.



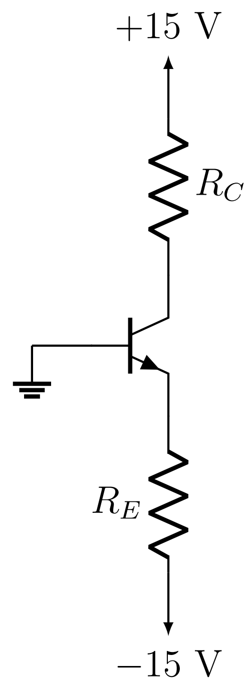
(15 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Q2. Design the circuit shown to establish $I_C = 0.2\,\text{mA}$ and $V_C = 0.5\,\text{V}$. The transistor exhibits $V_{BE} = 0.8\,\text{V}$ at $i_c = 1\,\text{mA}$, and $\beta = 100$.



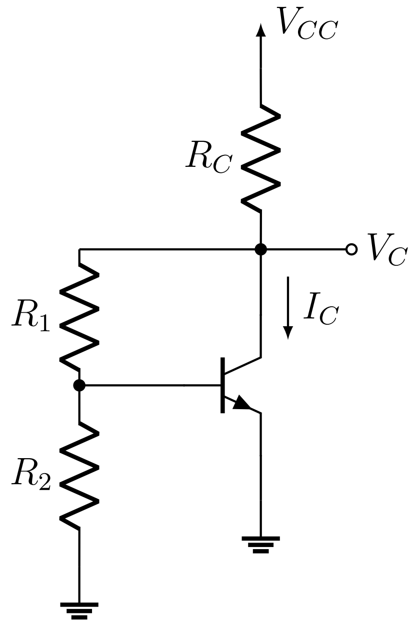
(20 marks) **[EEE 263 | L-2/T-1 | 2021–22]**

Q3. The transistor in the circuit has $\beta = 100$ and exhibits $V_{BE} = 0.7\,\text{V}$ at $i_c = 1\,\text{mA}$. Design the circuit so that a current of 2 mA flows through the collector and a voltage of +5 V appears at the collector.



(20 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

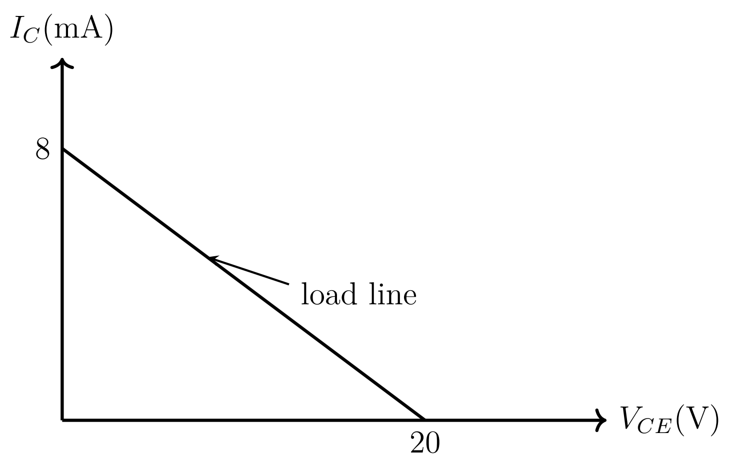
Q4. Design the circuit shown to provide $I_C = 3\,\text{mA}$ and $V_{CE} = 1.5\,\text{V}$ for $\beta = 90$. Use $V_{CC} = 3\,\text{V}$ and a current through R_2 equal to the base current.



(15 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Linear Amplifier Constraints

Q1. For the figure shown, find out the maximum collector current that the device can support. If the operating point is set for that maximum current, what kind of problem will we face if we want to use the device as a linear amplifier?



(12 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

S3 — MOSFET Characteristics

Structure, threshold voltage, DC analysis, biasing stability, CMOS logic

S3 MOSFET Characteristics

Device Characteristics

Q1. Explain the different operation regions of an enhancement PMOS transistor with properly labeled I_D–V_{SD} plots. Derive the threshold conditions for operating in those regions. **(8 marks)** [EEE 263 | L-2/T-1 | 2024–25]

Q2. For a particular MOSFET operating in the saturation region at a constant v_{gs}, $i_D = 2\,\text{mA}$ at $v_{ds} = 4\,\text{V}$. The i_D increases by 10% when v_{ds} is doubled. Find the corresponding values of r_0, V_A, and λ. **(10 marks)** [EEE 263 | L-2/T-1 | 2023–24]

Q3. Briefly explain the I-V characteristics of an n-channel enhancement MOSFET with necessary plots. **(7 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Q4. What is the difference between enhancement-type and depletion-type MOSFETs? Draw I_D vs. V_{GS} and I_D vs. V_{DS} characteristics for different values of V_{GS} for both enhancement-type and depletion-type MOSFETs. **(26 marks)** [EEE 263 | L-2/T-1 | 2018–19]

Q5. Showing different operating regions, describe the output characteristics of a MOSFET. Why does current saturation occur in a MOSFET as V_{DS} is increased? **(18 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Comparison with BJT

Q1. Compare MOSFET and BJT in terms of their operating principles. Mention each of their advantages and disadvantages. **(10 marks)** [EEE 263 | L-2/T-1 | 2013–14]

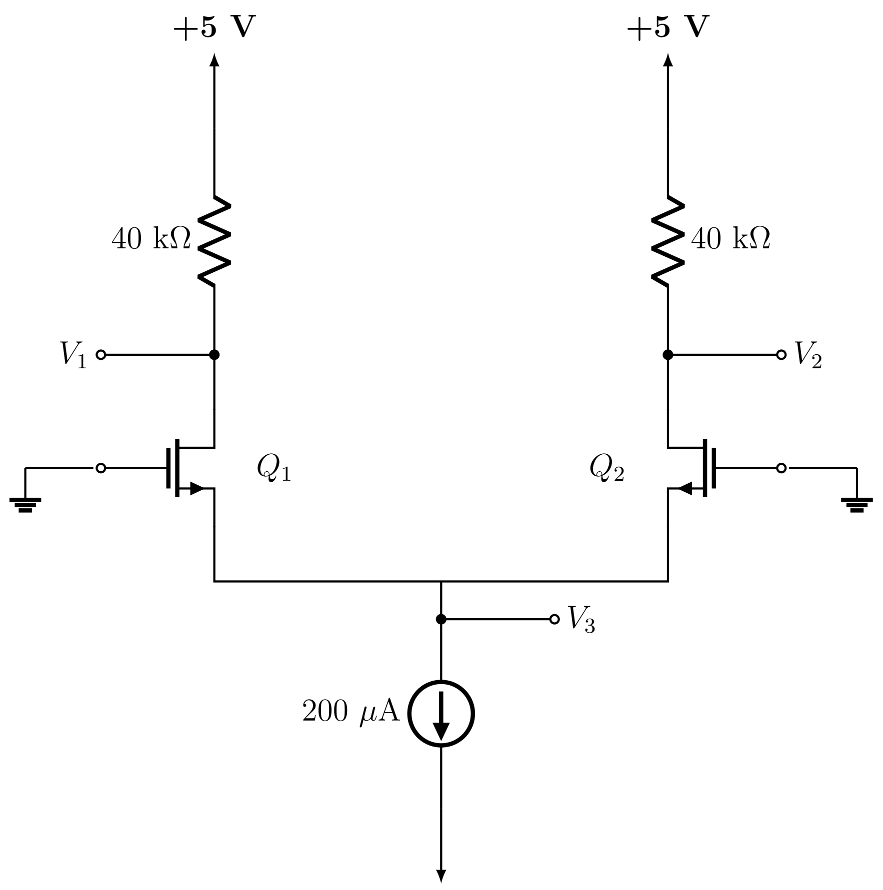
Q2. What are the mutual advantages and disadvantages of BJT and MOSFET? **(12 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Threshold Voltage

Q1. Explain the concept of threshold voltage (V_t) of a MOSFET. What happens to V_t and oxide capacitance (C_{ox}) if: (i) the dielectric thickness is increased/decreased? (ii) the gate insulator is replaced with a more high-k dielectric of the same thickness? **(20 marks)** [EEE 263 | L-2/T-1 | 2012–13]

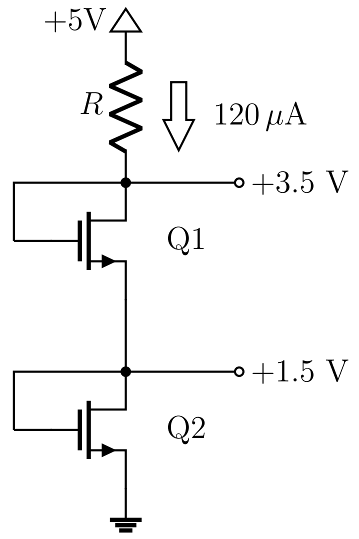
DC Analysis & Transistor Sizing

Q1. In the circuit shown, the aspect ratios of Q_1 and Q_2 are $(W/L)_1 = 2(W/L)_2 = 20$. Both transistors have $V_t = 1\,\text{V}$ and $\mu_n C_{ox} = 100\,\mu\text{A/V}^2$. Find the node voltages V_1, V_2, and V_3.



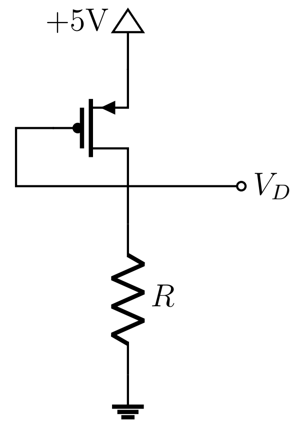
(17 marks) [EEE 263 | L-2/T-1 | 2024–25]

Q2. The NMOS transistors in the circuit have $V_t = 1\,\text{V}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, $\lambda = 0$, and $L_1 = L_2 = 1\,\mu\text{m}$. Find the required values of gate width W_1 and W_2 for Q_1 and Q_2, and the value of R, for obtaining the voltage values indicated. The current in R will be $120\,\mu\text{A}$.



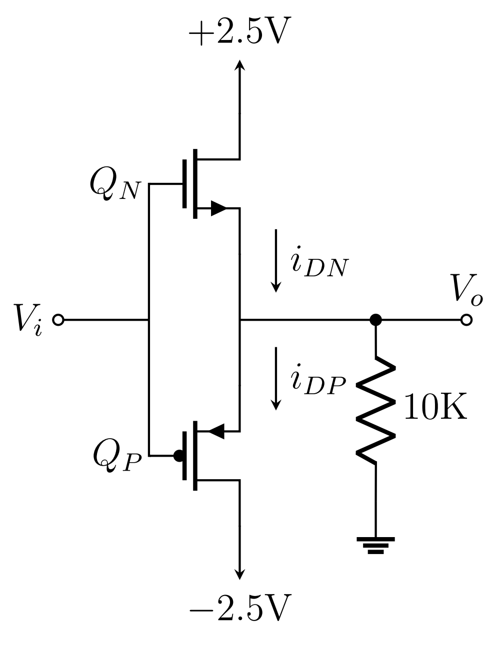
(15 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Q3. The PMOS transistor in the circuit shown has $V_t = -0.7\,\text{V}$, $\mu_p C_{ox} = 60\,\mu\text{A/V}^2$, $L = 0.8\,\mu\text{m}$, and $\lambda = 0$. Find the value of W and R in order to establish a drain current of $115\,\mu\text{A}$ and a voltage $V_D = 3.5\,\text{V}$.



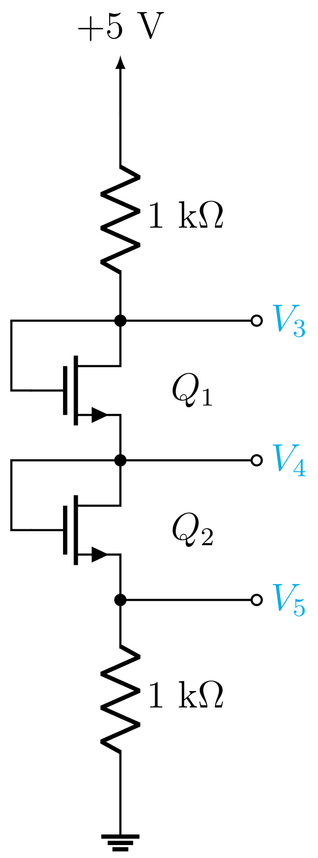
(15 marks) **[EEE 263 | L-2/T-1 | 2023–24]**

Q4. The NMOS and PMOS transistors in the circuit are matched with $k_n = k_p = 1\,\text{mA/V}^2$ and $V_{tn} = -V_{tp} = 1\,\text{V}$. Assuming $\lambda = 0$ for both MOSFETs, find the drain currents i_{DN} and i_{DP} and the voltage v_0 for $v_1 = 0\,\text{V}$ and $v_1 = +2.5\,\text{V}$.



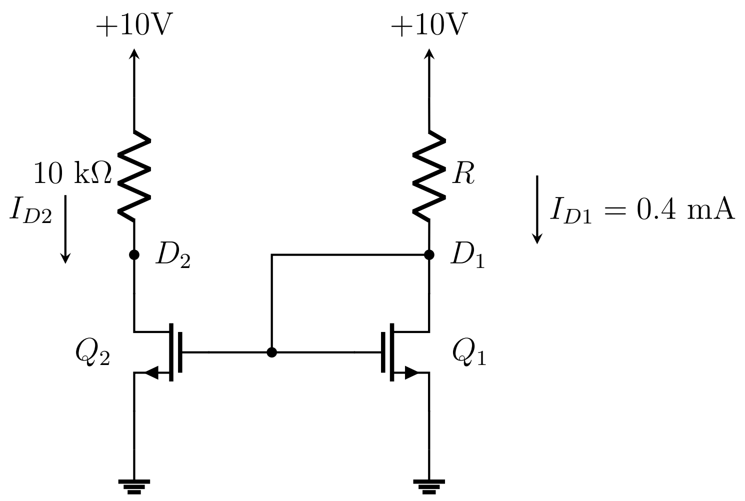
(20 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Q5. For the circuit shown, find the node voltages V_3, V_4, and V_5. The NMOS transistors have $V_t = 0.9\,\text{V}$ and $k_n'(W/L) = 1.5\,\text{mA/V}^2$.



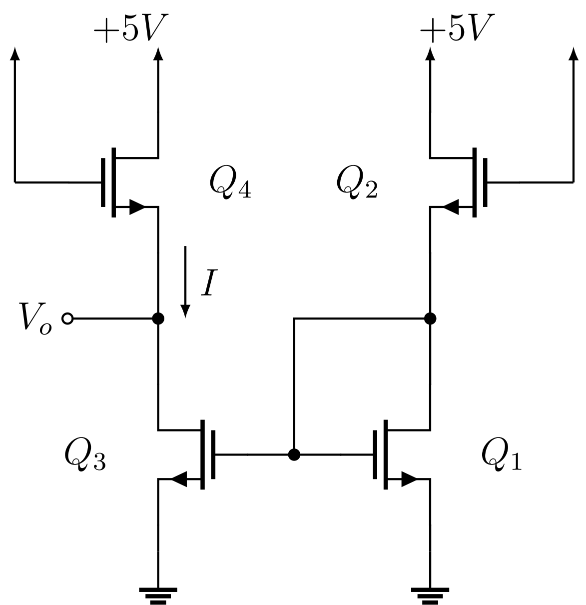
(28 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Q6. For the circuit shown, find R so that $I_{D1} = 0.4\,\text{mA}$. Also find V_{D1}, V_{D2}, and I_{D2}. Q_1 and Q_2 are identical. Given: $V_t = 2\,\text{V}$, $\mu_n C_{ox} = 20\,\mu\text{A/V}^2$, $L = 10\,\mu\text{m}$, $W = 100\,\mu\text{m}$.



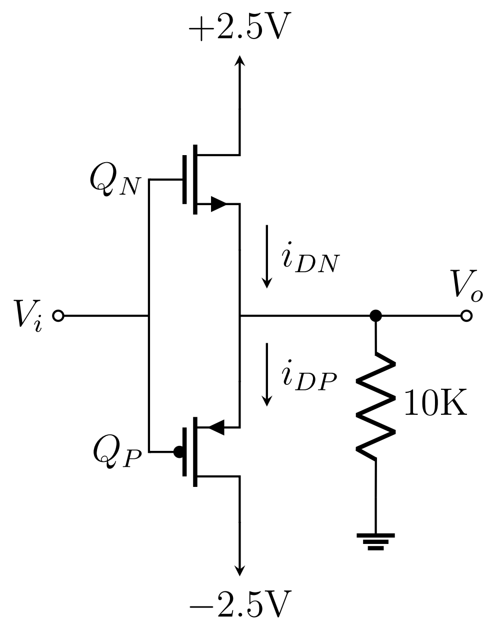
(20 marks) [EEE 263 | L-2/T-1 | 2018–19]

Q7. For the circuit shown, for all MOSFETs $|V_t| = 1\,\text{V}$, $\mu_n C_{ox} = 50\,\mu\text{A/V}^2$, $L = 1\,\mu\text{m}$, $W = 10\,\mu\text{m}$. Find V and I. If Q_3 and Q_4 are made to have $W = 100\,\mu\text{m}$, find V and I in that case.



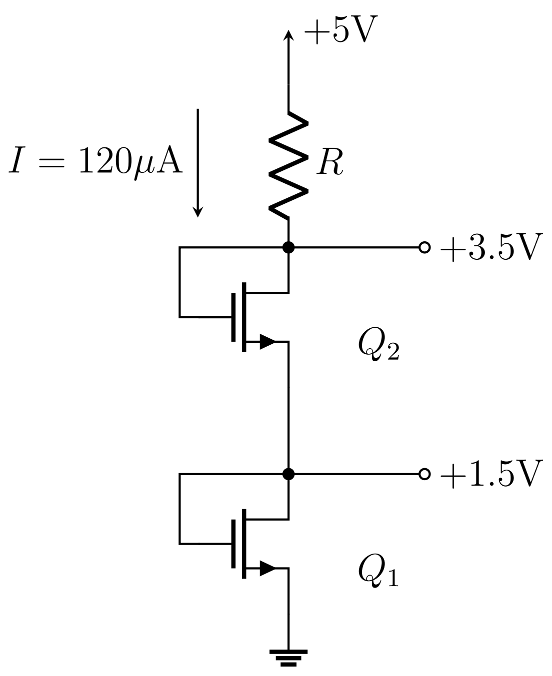
(26 marks) [EEE 263 | L-2/T-1 | 2016–17]

Q8. The NMOS and PMOS transistors in the circuit shown have $k_n(W/L) = k_p(W/L) = 1\,\text{mA/V}^2$ and $V_{tn} = -V_{tp} = 1\,\text{V}$. Assume $\lambda = 0$. Find i_{DN}, i_{DP}, and v_o for $V_i = +2.5\,\text{V}$ and $V_i = -2.5\,\text{V}$.



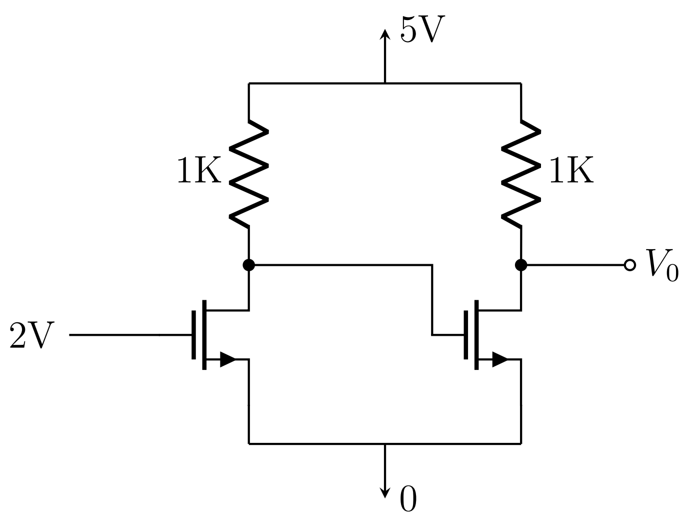
(20 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Q9. The NMOS transistors in the circuit have $V_t = 1\,\text{V}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, $L_1 = L_2 = 1\,\mu\text{m}$. Find W_1, W_2, and R, where W_1 and W_2 are the gate widths of Q1 and Q2 respectively.



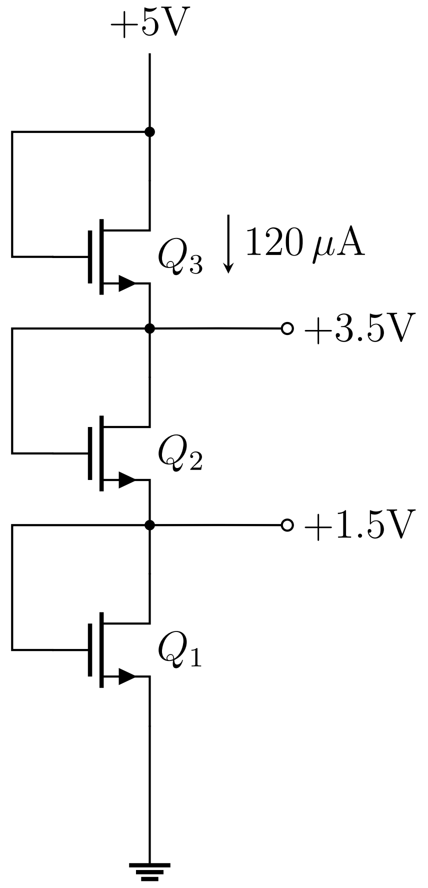
(16 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Q10. For the circuit shown, find the value of V_0. Both NMOS transistors have $V_t = 1\,\text{V}$, $\lambda = 0$, and $\mu_n C_{ox}(W/L) = 1\,\text{mA/V}^2$. Supply voltage is $5\,\text{V}$.



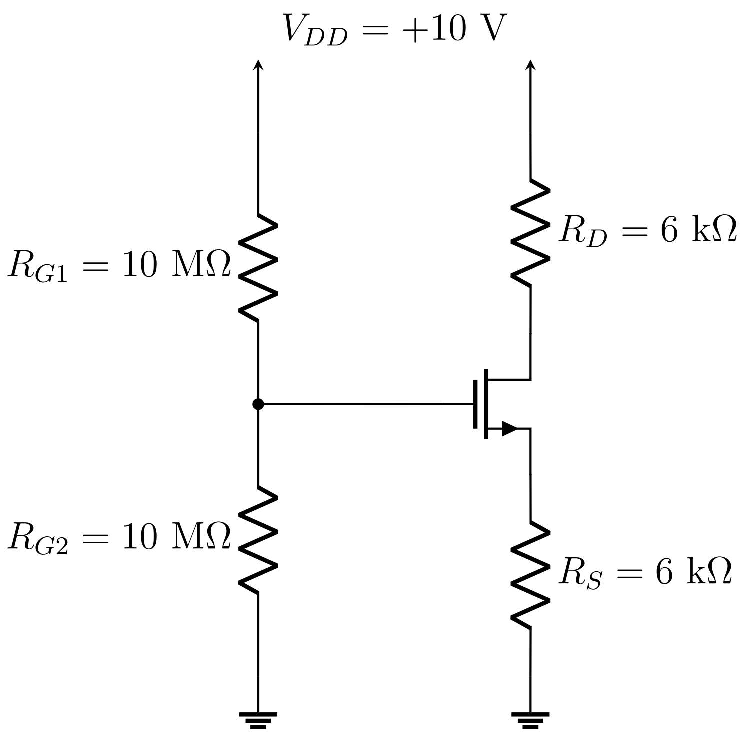
(16 marks) **[EEE 263 | L-2/T-1 | 2013–14]**

Q11. The NMOS transistors in the circuit have $V_t = 1\,\text{V}$, $\mu_n C_{ox} = 120\,\mu\text{A/V}^2$, and $L_1 = L_2 = L_3 = 1\,\mu\text{m}$. Find the required values of gate width for each of Q1, Q2, and Q3 to obtain the voltage and current values indicated in the figure.



(14 marks) **[EEE 263 | L-2/T-1 | 2012–13]**

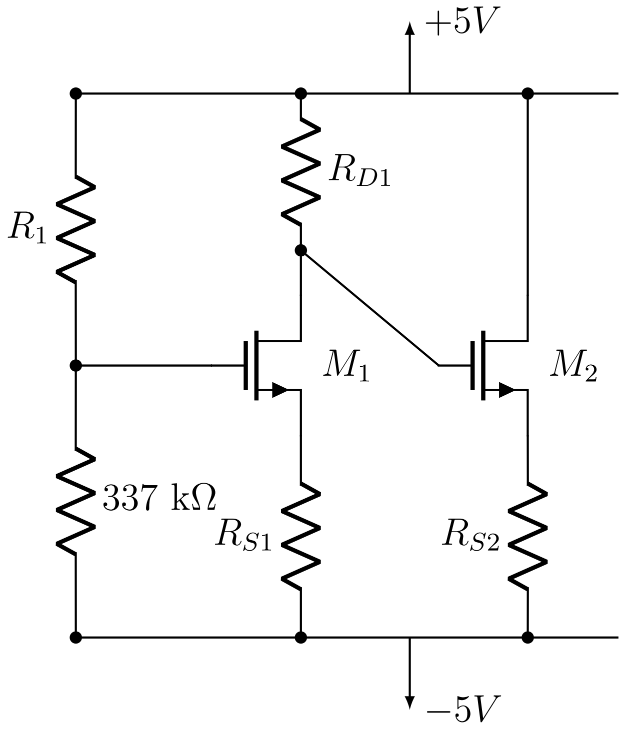
Q12. Determine all node voltages and drain current of the circuit. Assume $V_t = 1\,\text{V}$, $K_n'W/L = 1\,\text{mA/V}^2$, $\lambda = 0$, $V_{DD} = +10\,\text{V}$, $R_D = 6\,\text{k}\Omega$.



(20 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Multi-stage MOSFET Circuit Design

Q1. For the circuit shown with transistor parameters $K_{n1} = 500\,\mu\text{A/V}^2$, $K_{n2} = 200\,\mu\text{A/V}^2$, $V_{T1} = V_{T2} = 1.2\,\text{V}$, design the circuit such that $I_{D1} = 0.1\,\text{mA}$, $I_{D2} = 0.3\,\text{mA}$, $V_{DS1} = V_{DS2} = 5\,\text{V}$.



(20 marks) [EEE 263 | L-2/T-1 | 2016–17]

MOSFET Transconductance

Q1. How can the MOSFET transconductance g_m be varied with the design parameters W/L ratio and overdrive voltage V_{ov}? Explain. **(10 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Voltage Transfer Characteristics

Q1. Draw the voltage transfer characteristic of a MOSFET. In which part of this graph is a MOSFET used as an amplifier, and why? What are the benefits and drawbacks of increasing bias voltage? **(19 marks)** [EEE 263 | L-2/T-1 | 2012–13]

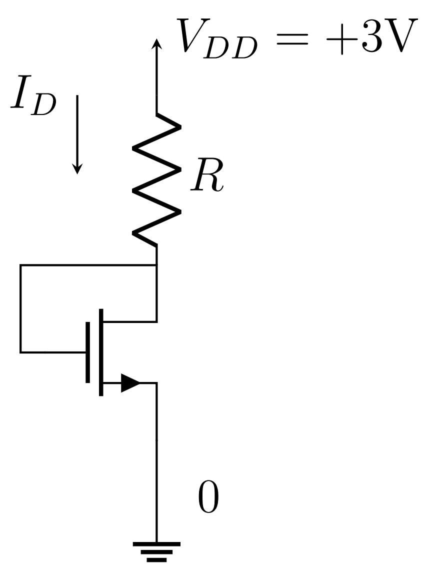
MOSFET as Small-Signal Amplifier

Q1. Briefly explain the operation of a MOSFET as a small-signal amplifier and deduce its VTC from the I_D–V_{DS} graph. **(15 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Biasing Stability

Q1. What is the reason for biasing a MOSFET? Write a short note on the biasing technique of a discrete MOSFET using a fixed voltage at the gate and a resistance R_s in the source. **(11 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Q2. Design the circuit shown to obtain a current $I_D = 80\,\mu$A. Given: $V_t = 0.6\,$V, $\mu_n C_{ox} = 200\,\mu$A/V^2, $L = 0.8\,\mu$m, $W = 4\,\mu$m, and $V_A = 50\,$V.

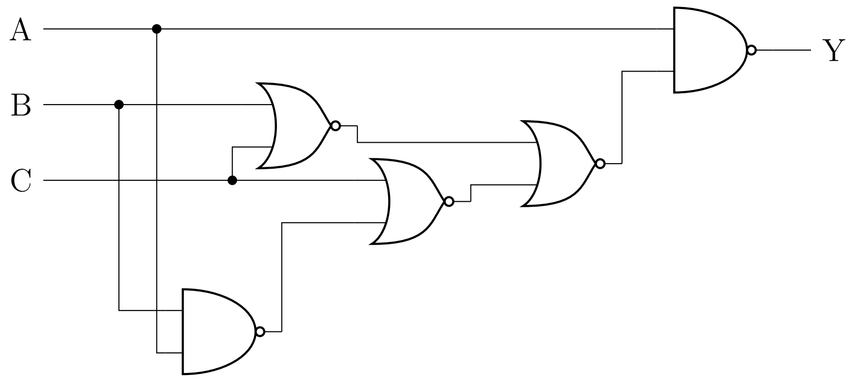


(16 marks) [EEE 263 | L-2/T-1 | 2013–14]

Q3. Explain graphically the benefit of using a resistance at the MOSFET source terminal. How does it counteract the problem of unpredictable variation in process parameters such as V_t, C_{ox}, μ_n? **(20 marks)** [EEE 263 | L-2/T-1 | 2012–13]

CMOS Logic Implementation

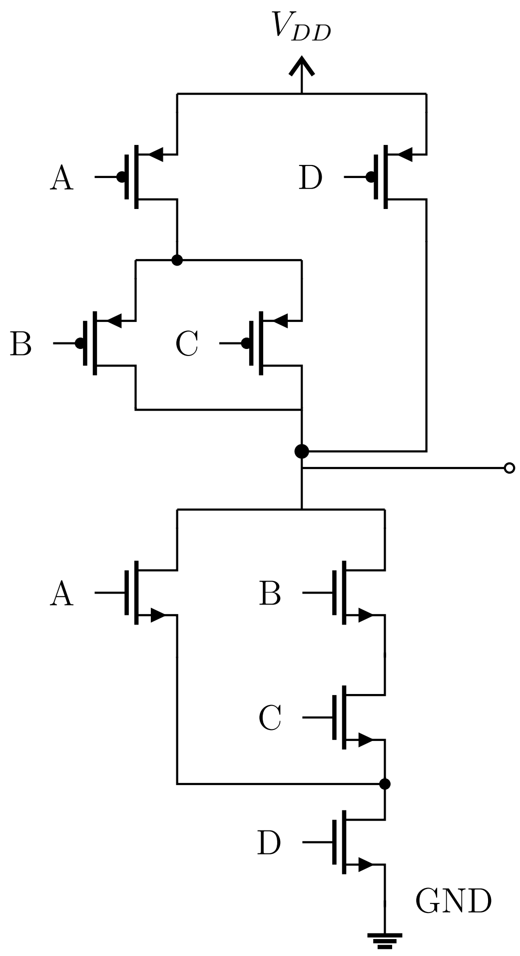
Q1. The diagram shows a Boolean function Y being implemented as a combinational circuit. Design a CMOS transistor-level implementation of the Boolean function using the minimum number of transistors.



(20 marks) [EEE 263 | L-2/T-1 | 2020–21]

CMOS Sizing and Delay

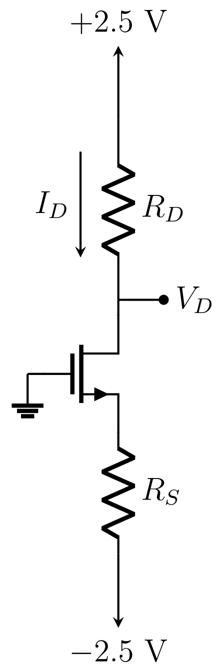
Q1. For the logic circuit shown, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume $\mu_n = 3\mu_p$. What is the minimum and maximum propagation delay in the circuit?



(26 marks) [EEE 263 | L-2/T-1 | 2020–21]

NMOS Circuit Design

Q1. Design the circuit shown so that the transistor operates at $I_D = 0.3\,\text{mA}$ and $V_D = 0.4\,\text{V}$. The NMOS transistor has a threshold voltage of 1 V, aspect ratio of 40, and a process transconductance parameter of $60\,\mu\text{A/V}^2$. Neglect channel-length modulation.



(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

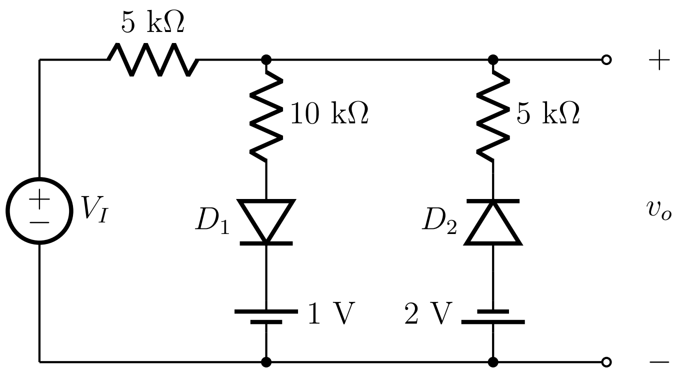
Part II — Wave Shaping Circuits

Clipping, clamping, comparator, and switching circuits using diodes and BJTs

S4 Clipping, Clamping & Switching Circuits

Clipping Circuits — Waveform & Transfer Characteristics

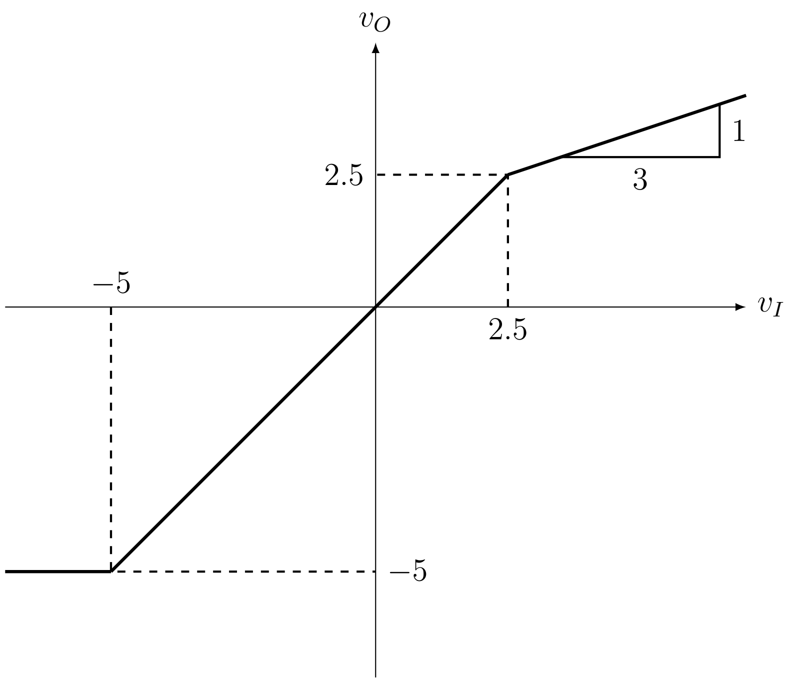
Q1. For the circuit given, plot v_0 vs time. The input v_1 is a sine wave with 5 V amplitude and 1 kHz frequency. Assume 0.7 V voltage drop across diodes in forward bias.



(14 marks) [EEE 263 | L-2/T-1 | 2022–23]

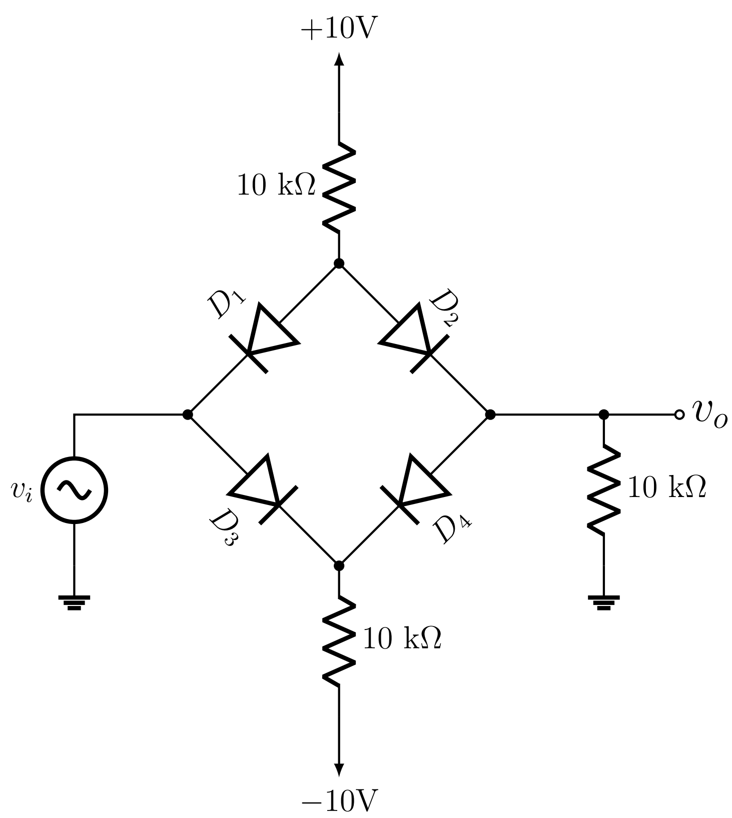
Q2. Design a limiter circuit using only diodes and 10 kΩ resistors to provide an output signal limited to the range ±2.1 V. Assume each diode has a 0.7 V drop when conducting. **(15 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q3. Design a parallel-based clipper that will yield the voltage transfer function shown in the figure. Assume 0.7 V as diode cut-in voltage.



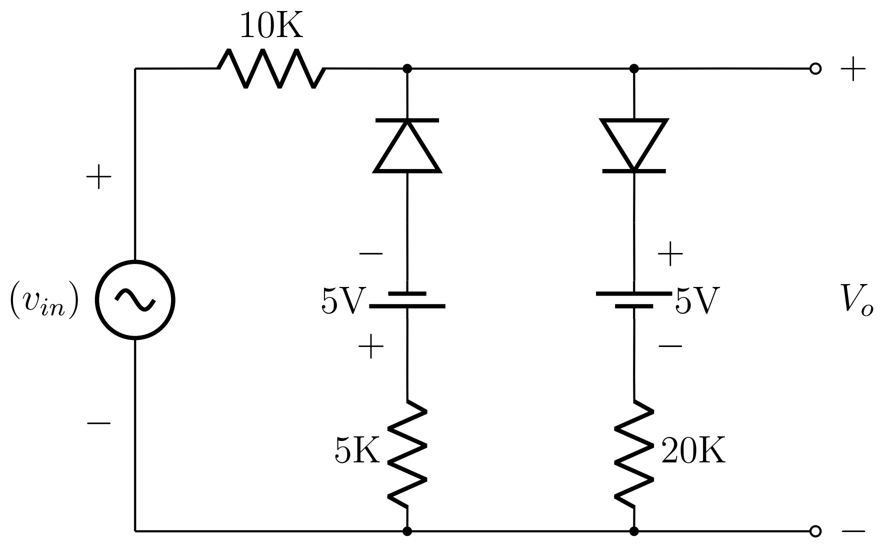
(18 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q4. For the circuit shown in Figure , find and sketch the voltage transfer characteristics with proper labelling. Utilize the constant-voltage drop model (0.7 V) for each conduction diode.



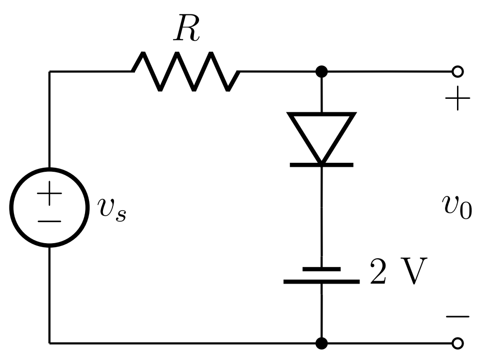
(13 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Q5. For the circuit shown, neatly draw the output (v_o) waveform corresponding to the input (v_{in}). Here, $v_{in} = 10\sin(200\pi t)$ and diodes are non-ideal.



(16 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

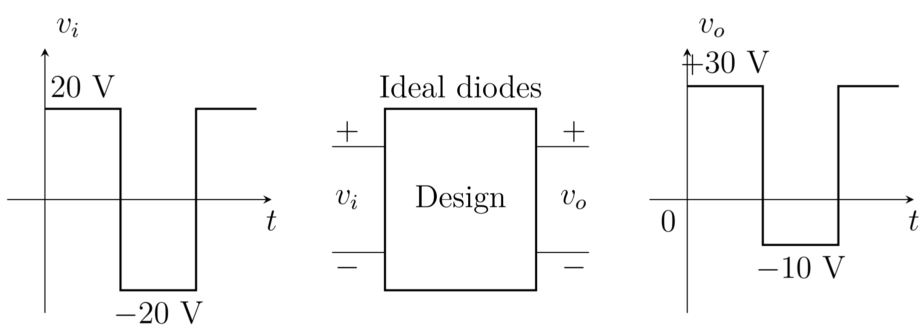
Q6. Sketch the output waveform and transfer characteristics for the circuit shown. Assume the diode is ideal and V_s is a sinusoid with 5 V peak amplitude and frequency 5 kHz.



(15 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

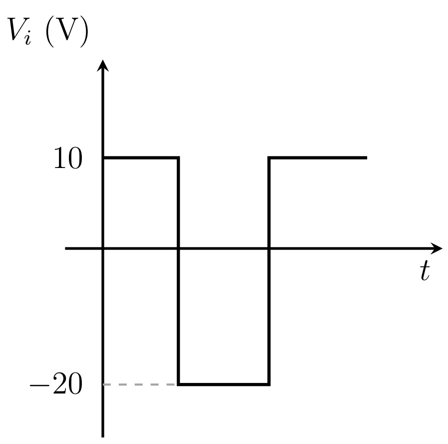
Clamping Circuits

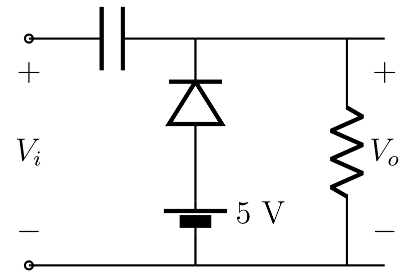
Q1. Design a clamper circuit to perform the function indicated in the figure (input v_i, output v_0). Assume ideal diodes.



(12 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

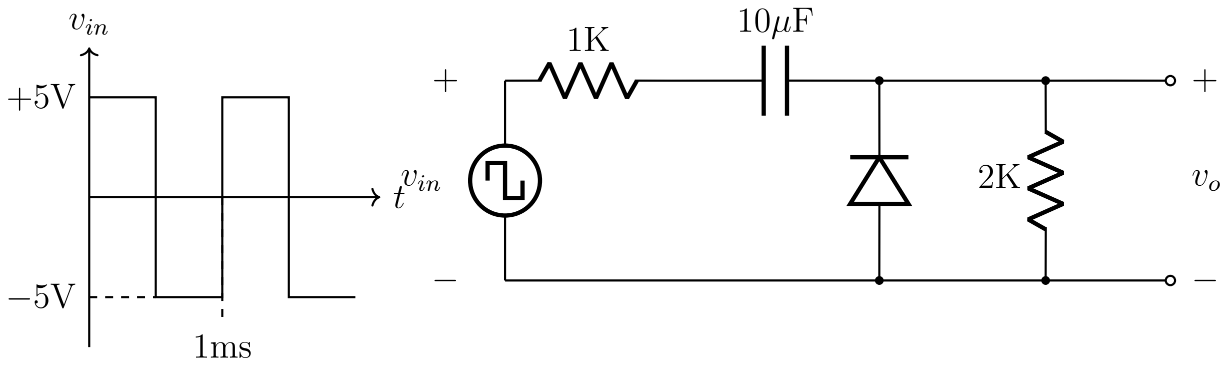
Q2. For the following circuit, draw the output waveshape for the given square input waveshape.





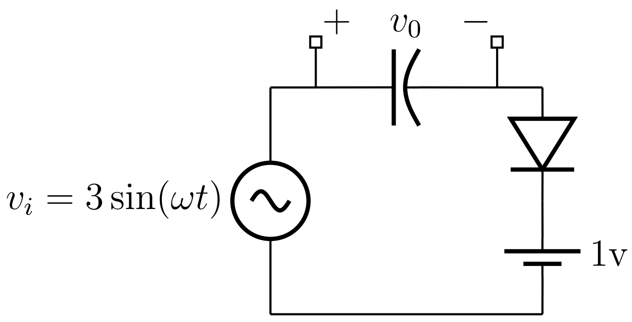
(16 marks) **[EEE 263 | L-2/T-1 | Jan 2021]**

Q3. For the circuit shown, neatly draw the output waveform (v_o) corresponding to the input waveform (v_{in}) in steady state. Assume the diode is non-ideal.



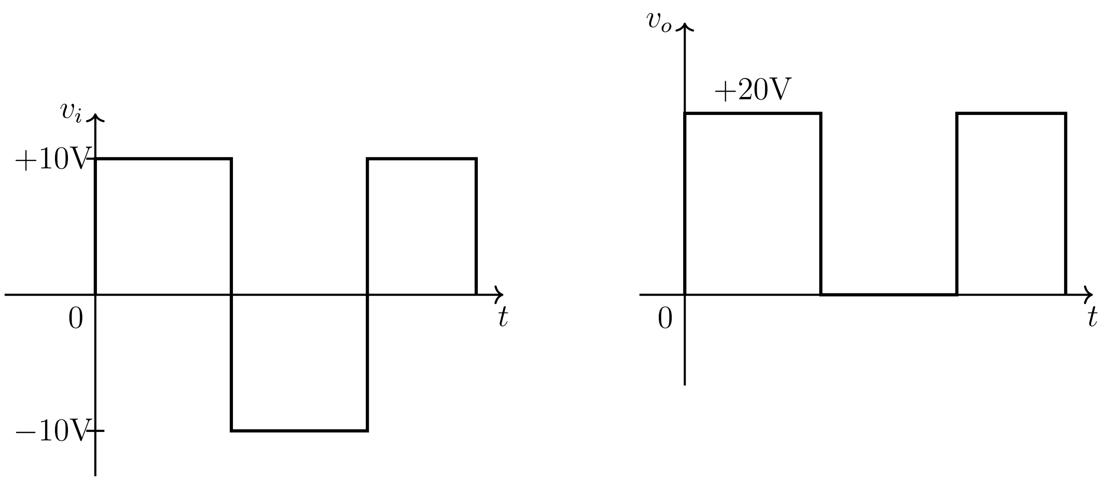
(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

Q4. Draw the output wave shape (v_0) and also the V_L–V_o graph for the circuit shown. Assume a constant 0.7 V forward voltage drop across the diode.



(20 marks) [EEE 263 | L-2/T-1 | 2013–14]

Q5. Design a clamper circuit to perform the function indicated in the figure using an ideal diode.



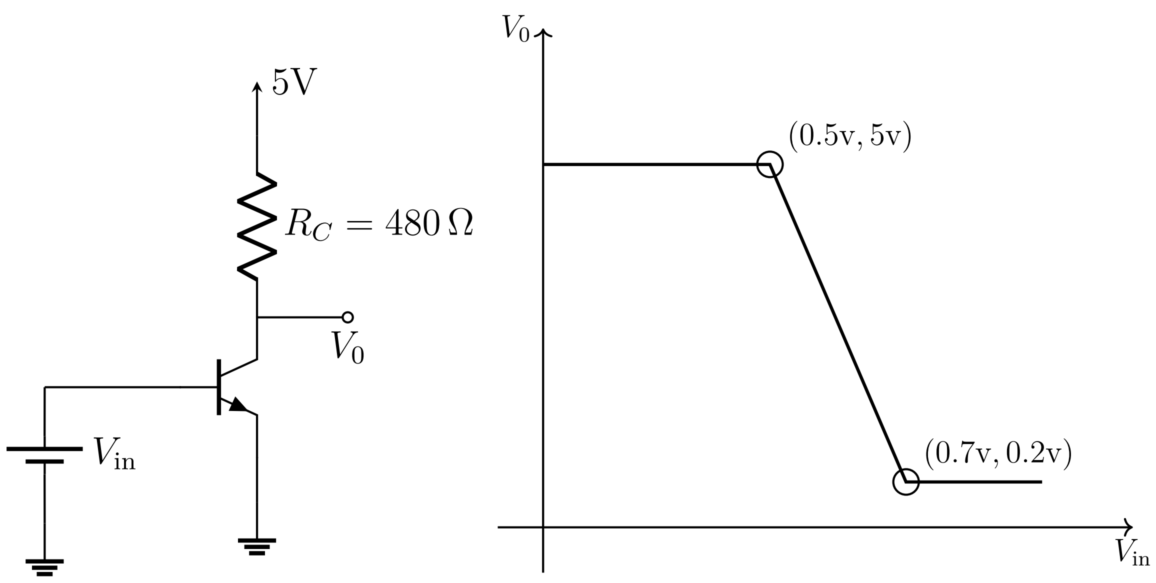
(12 marks) [EEE 263 | L-2/T-1 | 2011–12]

Switching Circuits — Logic Gates using BJTs

Q1. Make a NAND and a NOR gate using BJTs. **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

BJT Inverter Analysis

Q1. (a) Modify both the V_o–V_{in} and I_c–V_{in} graphs if supply voltage is decreased from 5 V to 2.5 V ($R_c \approx 480\,\Omega$). **(10 marks)** [EEE 263 | L-2/T-1 | 2013–14]



(b) The figure shows an inverter gate built by an npn BJT and its V_o–V_{in} characteristics. Copy the V_o–V_{in} graph to your answer script exactly to scale and then draw the collector current (I_c) vs. V_{in} over the same graph. Mention the co-ordinates of the bending points of the I_c–V_{in} graph as shown for V_o–V_{in}.
(16 marks) [EEE 263 | L-2/T-1 | 2012–13]

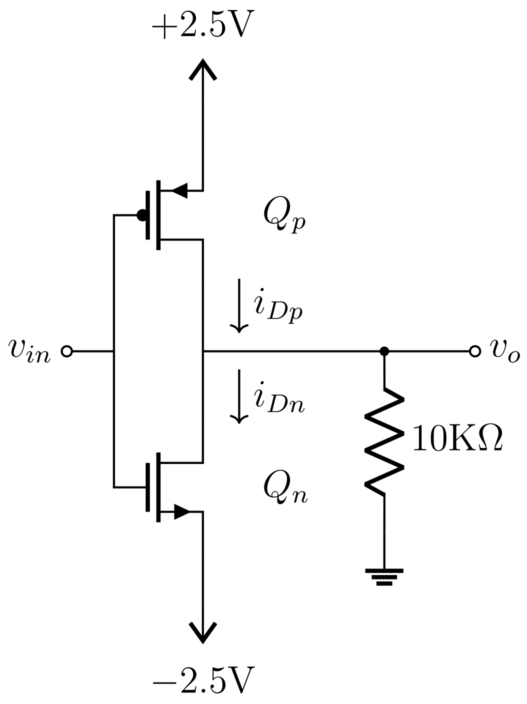
(c) Modify both the V_o–V_{in} and I_c–V_{in} graphs if R_c is infinity. **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

(d) Modify both the V_o–V_{in} and I_c–V_{in} graphs if supply voltage is increased from 5 V to 10 V ($R_c = 480\,\Omega$). **(10 marks)** [EEE 263 | L-2/T-1 | 2012–13]

(e) If the BJT in active mode shows a linear I_c–V_{BE} behaviour instead of exponential, does voltage gain change with change in bias voltage V_{in}? Does output voltage swing change with the change in V_{in}? Explain why. **(5 marks)** [EEE 263 | L-2/T-1 | 2012–13]

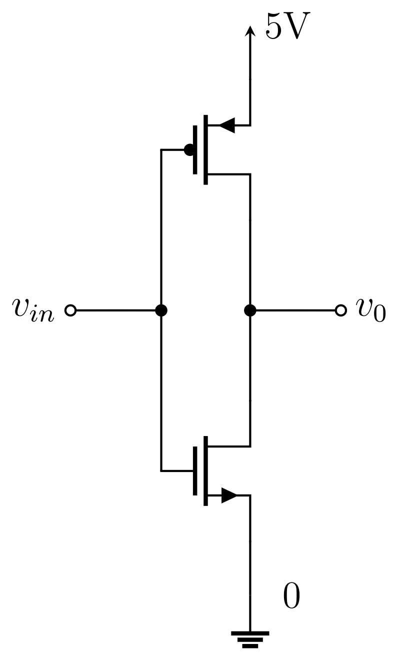
CMOS Inverter Analysis

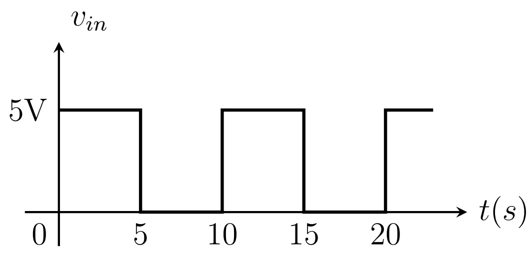
Q1. The inverter circuit with CMOS shown has $k_n'(W_n/L_n) = k_p'(W_p/L_p) = 1\,\text{mA/V}^2$ and $V_{tn}' = -V_{tp}' = 1\,\text{V}$. Find (i) i_{DN}, (ii) i_{DP}, (iii) V_o for $V_{in} = -2.5\,\text{V}$.



(15 marks) [EEE 263 | L-2/T-1 | 2015–16]

Q2. For the CMOS inverter, draw V_o for the given input V_{in} waveform.





(15 marks) [EEE 263 | L-2/T-1 | 2013–14]

Digital vs. Analog Systems

Q1. Why is digital processing of data advantageous over analog processing? **(6 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Q2. Why do we need to miniaturize a transistor as much as possible? If you continue to miniaturize the transistor (switch), what problems may surface? **(5 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Q3. Write a script in C language that takes a sinusoidal input and provides a full-wave rectified output. **(5 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

Q4. How does a computer program rectify the sinusoidal input mentioned above? Is there any diode inside the processor? If not, what makes that act of rectification possible? **(10 marks)** **[EEE 263 | L-2/T-1 | 2013–14]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

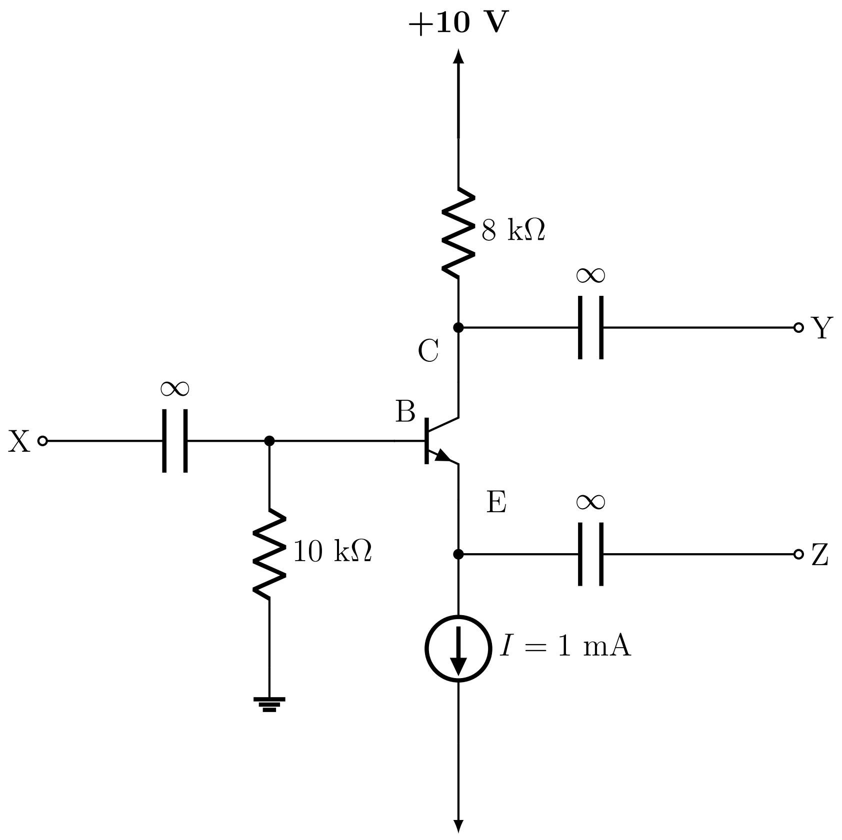
Part III — Amplifiers

BJT and MOSFET amplifier configurations, small-signal analysis, gain derivation

S5 BJT Amplifiers

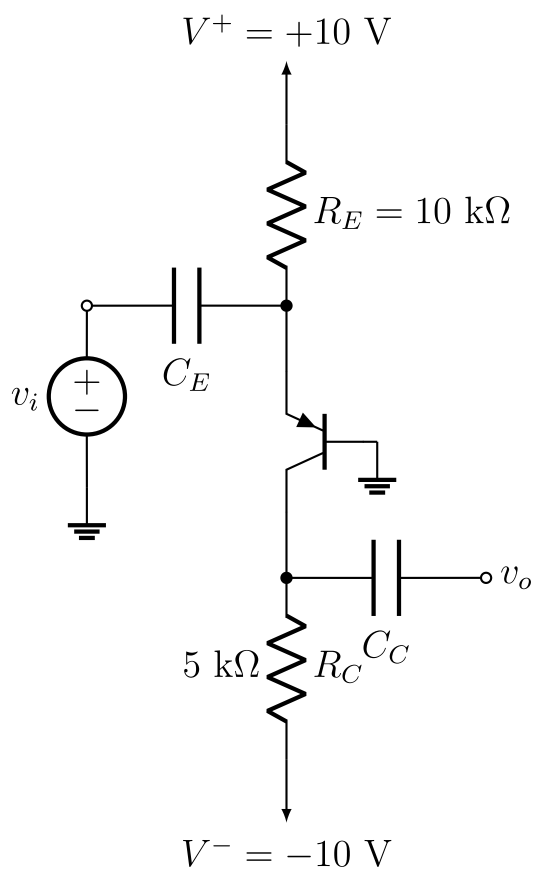
Common Emitter / Trans-conductance Amplifier

Q1. The transistor shown is biased with a constant current $I = 1\,\text{mA}$ and has $\beta = 100$ and $V_A = 100\,\text{V}$. (i) Find the DC voltages at the Base, Emitter, and Collector terminals. (ii) Find the small-signal equivalent circuit parameters r_o, r_π, g_m and draw the hybrid-π model. (iii) If the Z terminal is connected to ground, X is connected to v_{sig} through $R_{sig} = 2\,\text{k}\Omega$, and Y is connected to $R_L = 8\,\text{k}\Omega$, find the overall voltage gain v_y/v_{sig}.



(20 marks) [EEE 263 | L-2/T-1 | 2024–25]

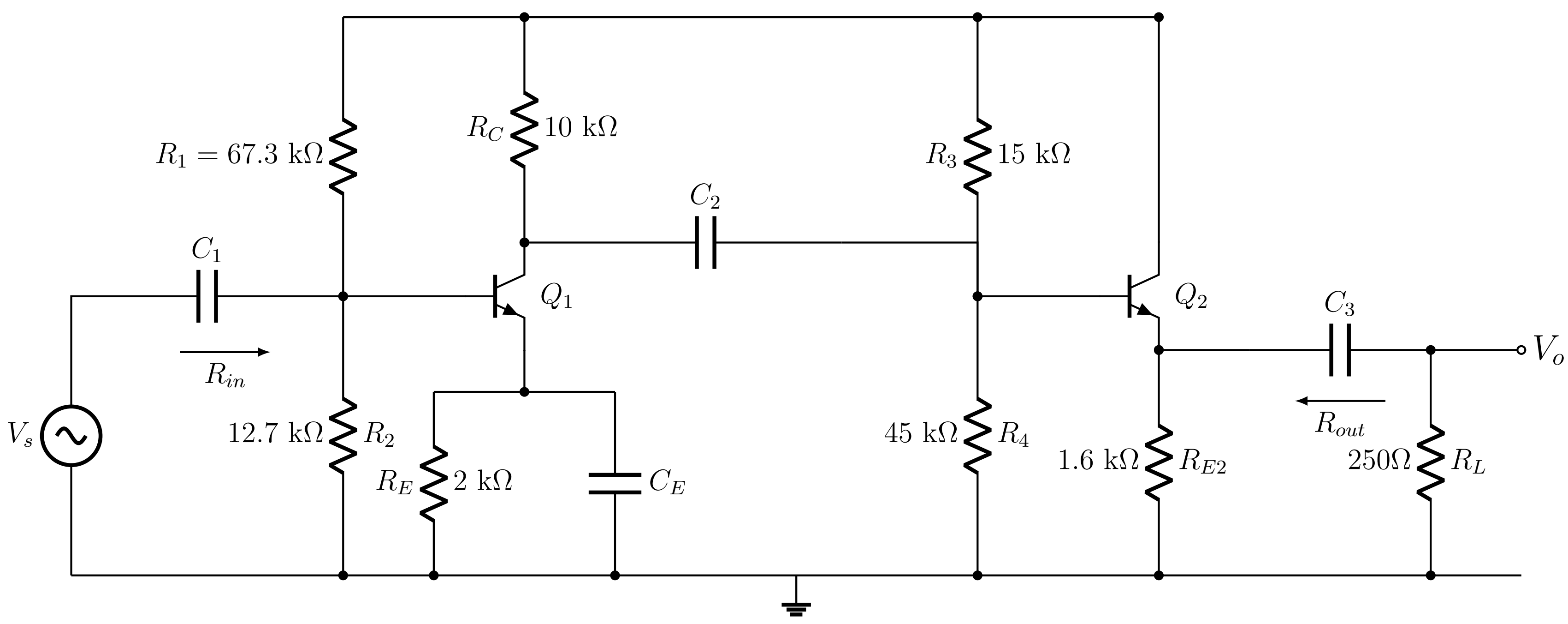
Q2. Determine the voltage gain v_0/v_i of the transistor amplifier shown in the figure. Assume $\beta = 100$.



(26 marks) [EEE 263 | L-2/T-1 | 2017–18]

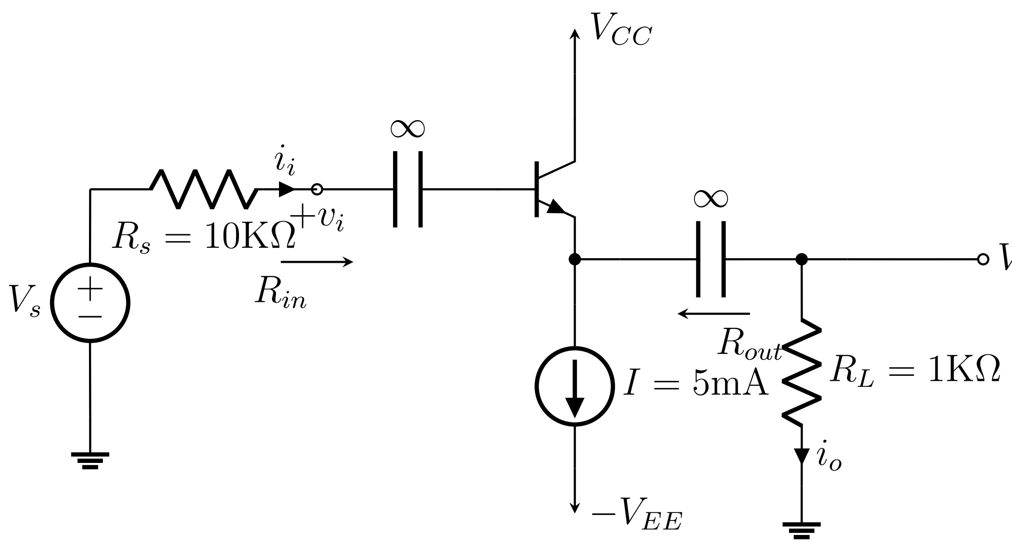
Q3. A CE amplifier utilizes a BJT with $\beta = 100$ and $V_A = 100\,\text{V}$, is biased at $I_C = 1\,\text{mA}$ and has a collector resistance $R_C = 5\,\text{k}\Omega$. Find R_{in}, R_o, and A_{vo}. If the amplifier is fed with a signal source having a resistance of $5\,\text{k}\Omega$ and a load resistance $R_L = 5\,\text{k}\Omega$ is connected to the output terminal, find the resulting A_v and G_v. If v_π is to be limited to $5\,\text{mV}$, what are the corresponding v_{sig} and v_o with the load connected? **(20 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Q4. For the circuit shown, determine small-signal parameters g_m, r_π, and r_e for both transistors, the overall small-signal voltage gain $A_v = v_0/v_s$, input resistance R_{in}, and output resistance R_{out}. Here $\beta = 120$.



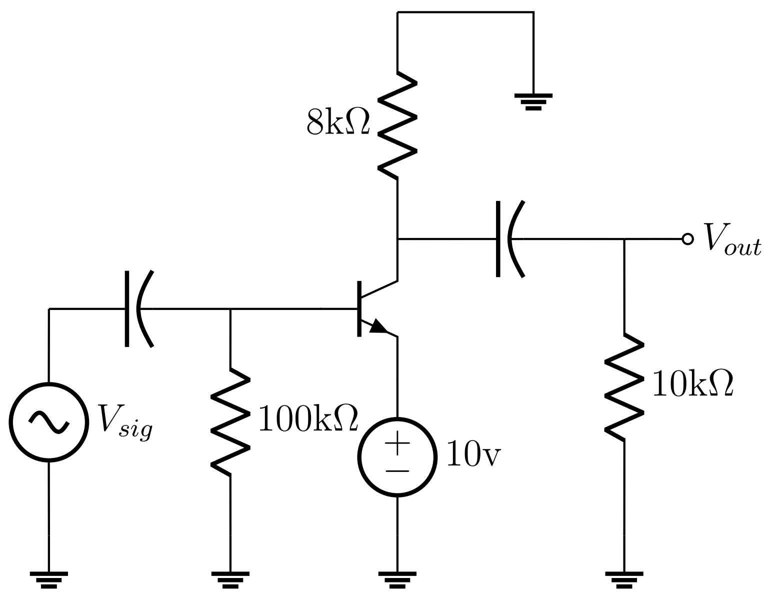
(36 marks) [EEE 263 | L-2/T-1 | 2016–17]

Q5. For the circuit shown, determine R_{in}, R_{out}, V_o/V_i, and i_o/i_i. The BJT is biased with $I = 5\,\text{mA}$ and has $\beta = 100$ and $V_A = 100\,\text{V}$.



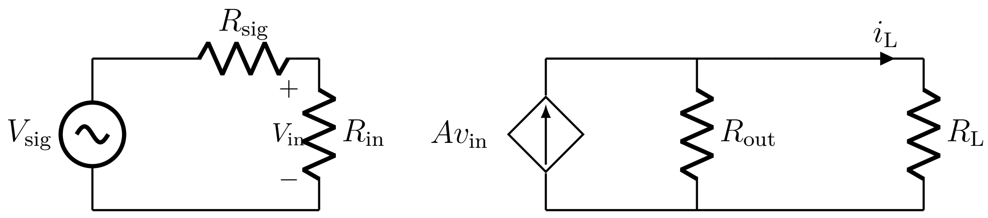
(20 marks) [EEE 263 | L-2/T-1 | 2014–15]

Q6. Draw the small signal circuit diagram of the CE circuit shown. Why is the circuit called a "Small Signal" circuit? Determine the values of input resistance R_{in}, output resistance R_{out}, and voltage gain A_v. If $v_{sig} \approx 2\,\text{mV}$, find the value of V_{out}.



(36 marks) [EEE 263 | L-2/T-1 | 2013–14]

Q7. The figure shows the basic structure of a trans-conductance amplifier to be implemented by a BJT. Given $AV_{in} = I_c R_{out}$ along with $R_{sig} \approx 0$ and the load (R_L) can be much greater than R_{out}; which BJT configuration (common emitter, common base, or common collector) will you choose for best performance? Explain why. Comment on its gain and loading effect.



(20 marks) **[EEE 263 | L-2/T-1 | 2012–13]**

Q8. Consider the emitter degenerated CE circuit of Fig. for Q. No. 8(b).

(i) Find the DC current I_C and DC voltage V_C, V_B and V_E. Assume $\beta = 100$, $V_{BE} = 0.7$ V.

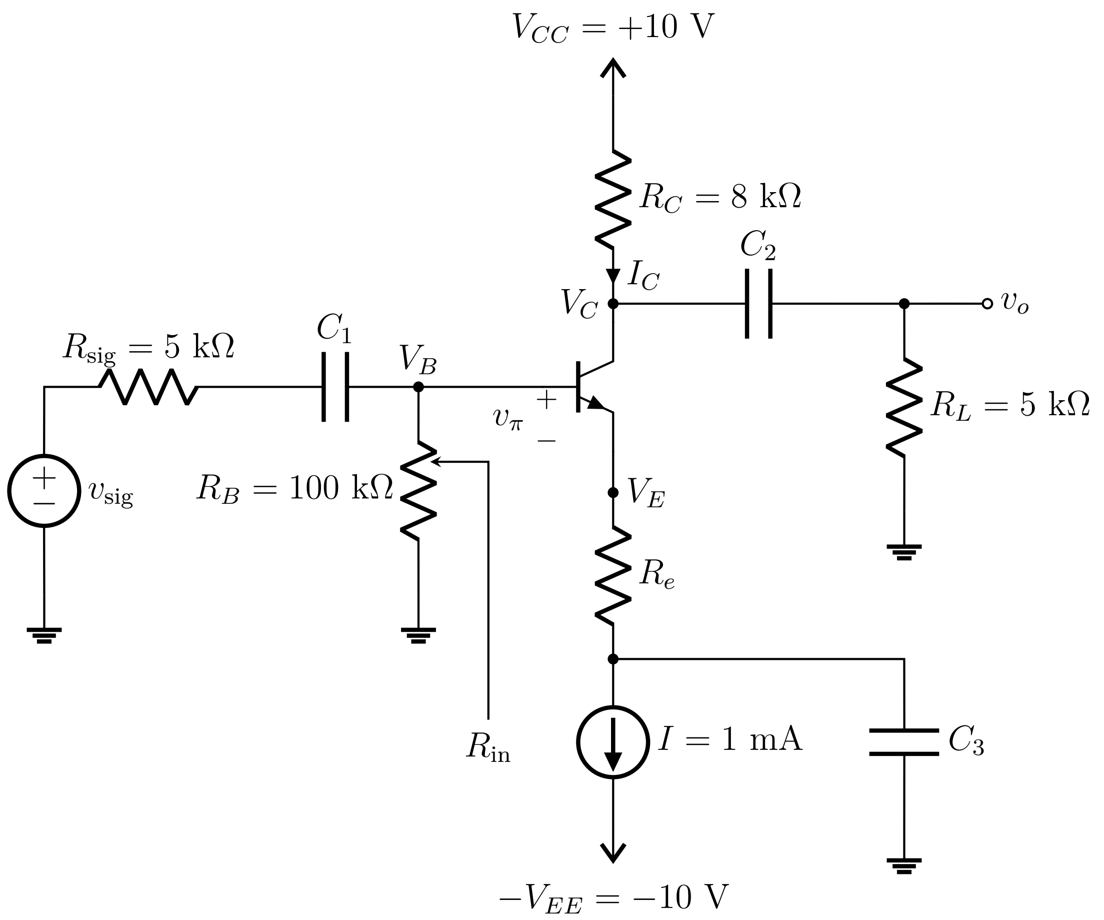
(ii) What are the allowable signal swings at the collector in both directions?

(iii) Determine the values of the BJT small signal model parameter: r_π, r_e, g_m at this bias point.

(iv) Draw the complete small signal equivalent circuit of the amplifier using suitable model of BJT. Neglect r_0.

(v) Find the value of R_e that results in R_{in} equal to four times the source resistance R_{sig}.

(vi) For this value of R_e, find A_{vo}, R_{out}, A_v, G_v and A_{is} of the amplifier.

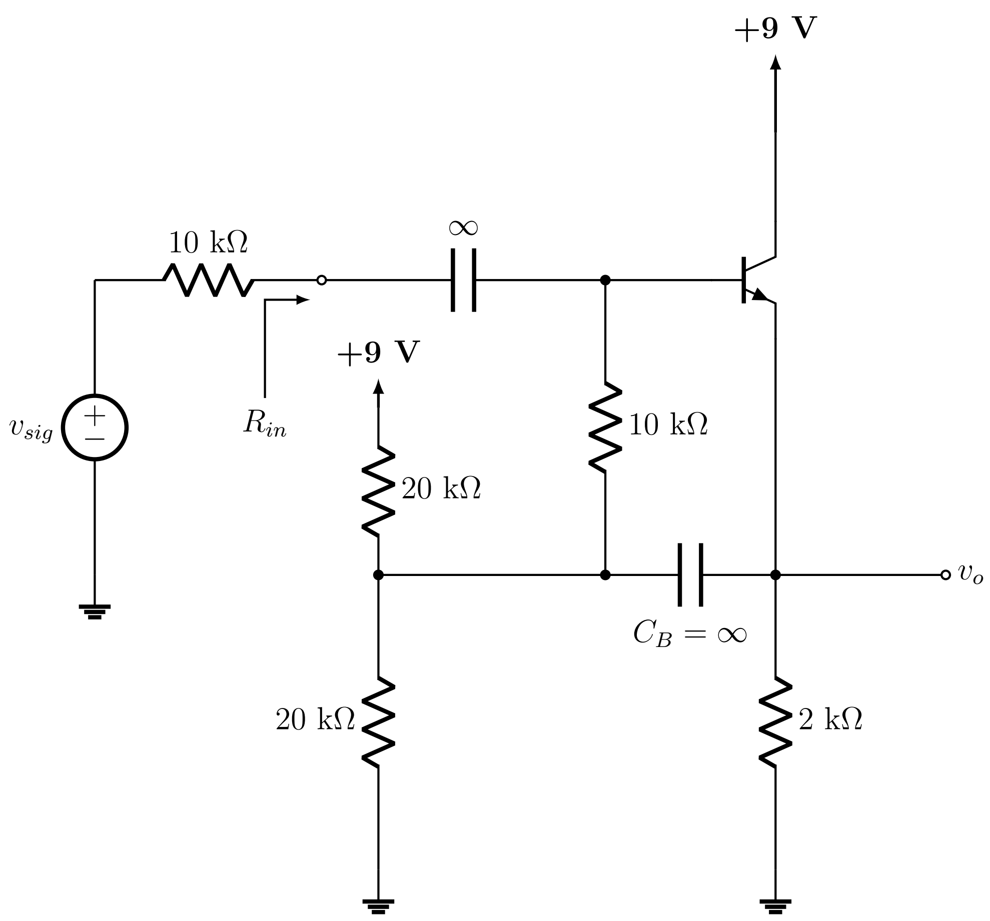


(36 marks) **[EEE 263 | L-2/T-1 | 2011–12]**

Common Base Amplifier

Q1. For a Common Base Amplifier, using the appropriate small-signal equivalent model, derive the expressions of input resistance, output resistance, open-circuit voltage gain, and overall voltage gain. **(14 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

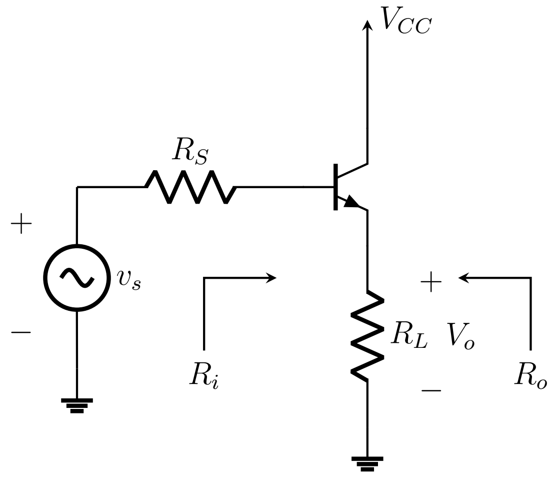
Q1. For the bootstrapped follower circuit shown: (i) Find the DC emitter current and g_m, r_e, r_π. Assume $\beta = 100$. (ii) Replace the BJT with its T model (neglecting r_o) and determine R_{in} and the voltage gain v_o/v_{sig}. (iii) Repeat (ii) for the case when capacitor C_B is open-circuited. Compare the results with (ii) and comment on the advantage of bootstrapping.



(15 marks) [EEE 263 | L-2/T-1 | 2024–25]

Q1. For a common collector amplifier, derive the expressions for R_i, R_o, A_v, and A_i using the T model with early effect for the transistor. Where is the common collector amplifier used? **(26 marks)** [EEE 263 | L-2/T-1 | 2020–21]

Q2. For the common collector circuit shown, derive the expressions for A_v, A_i, R_i, and R_o using the T model for the transistor.



(23.67 marks) [EEE 263 | L-2/T-1 | 2018–19]

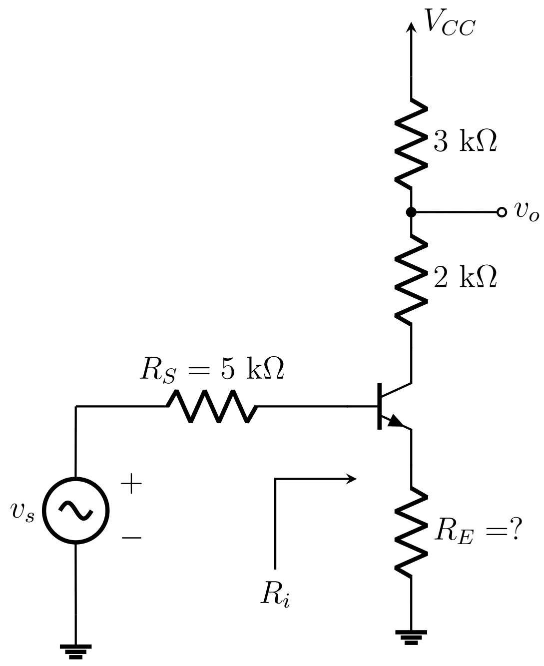
Q3. The small-signal voltage gain A_v of a BJT CE amplifier is $-I_C R_C/V_T$ without the Early effect. Show that if the Early effect is considered using $i_C = I_S e^{v_{BE}/V_T}(1 + v_{CE}/V_A)$, the gain becomes $A_v = -\frac{I_C R_C/V_T}{1+I_C R_C/(V_A+V_{CE})} = -\frac{(V_{CC}-V_{CE})/V_T}{1+(V_{CC}-V_{CE})/(V_A+V_{CE})}$. For $V_{CC} = 5\,\text{V}$, $V_{CE} = 2.5\,\text{V}$, $V_A = 100\,\text{V}$, calculate the gain with and without the Early effect. **(20 marks)** [EEE 263 | L-2/T-1 | 2024–25]

Q4. Derive the expression of R_i, R_o, A_v, and G_v for a common-emitter amplifier with emitter resistance R_e using the T model. **(26 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q5. For a BJT common-emitter amplifier with resistance in the emitter, derive the expressions for input impedance, output impedance, voltage gain, and current gain. What are the advantages and disadvantages of adding resistance in the emitter? **(30 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Biasing and Input Impedance

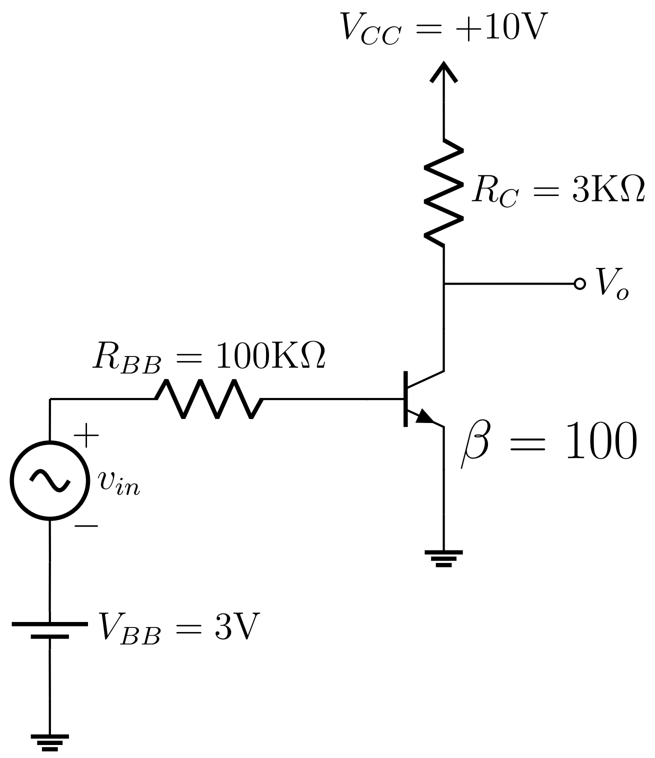
Q1. For the circuit shown, find the value of R_E so that R_i is five times the value of R_s. Then using these values find R_i, R_o, A_v, and A_i for the circuit. Neglect early effect. Given: $\beta = 100$ and r_e (T-model) $= 40\,\Omega$.



(26.67 marks) [EEE 263 | L-2/T-1 | 2018–19]

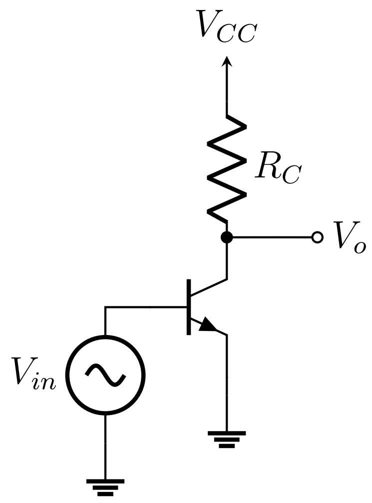
BJT Input-Output Characteristics

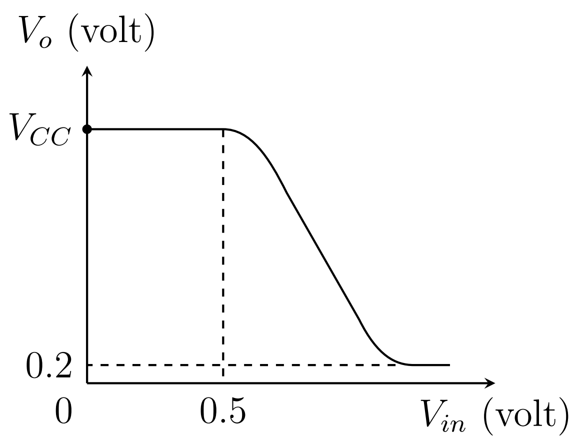
Q1. For the circuit shown, calculate (i) DC bias points, (ii) small-signal voltage gain, (iii) maximum allowable input AC voltage.



(26 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Q2. The input-output characteristics of a BJT circuit are given. What will happen (i) if V_{cc} is increased, (ii) if R_c is increased? Explain with the characteristic curve.





(10 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

S6 — MOSFET Amplifiers

Common-source configuration, small-signal model, gain derivation

S6 MOSFET Amplifiers

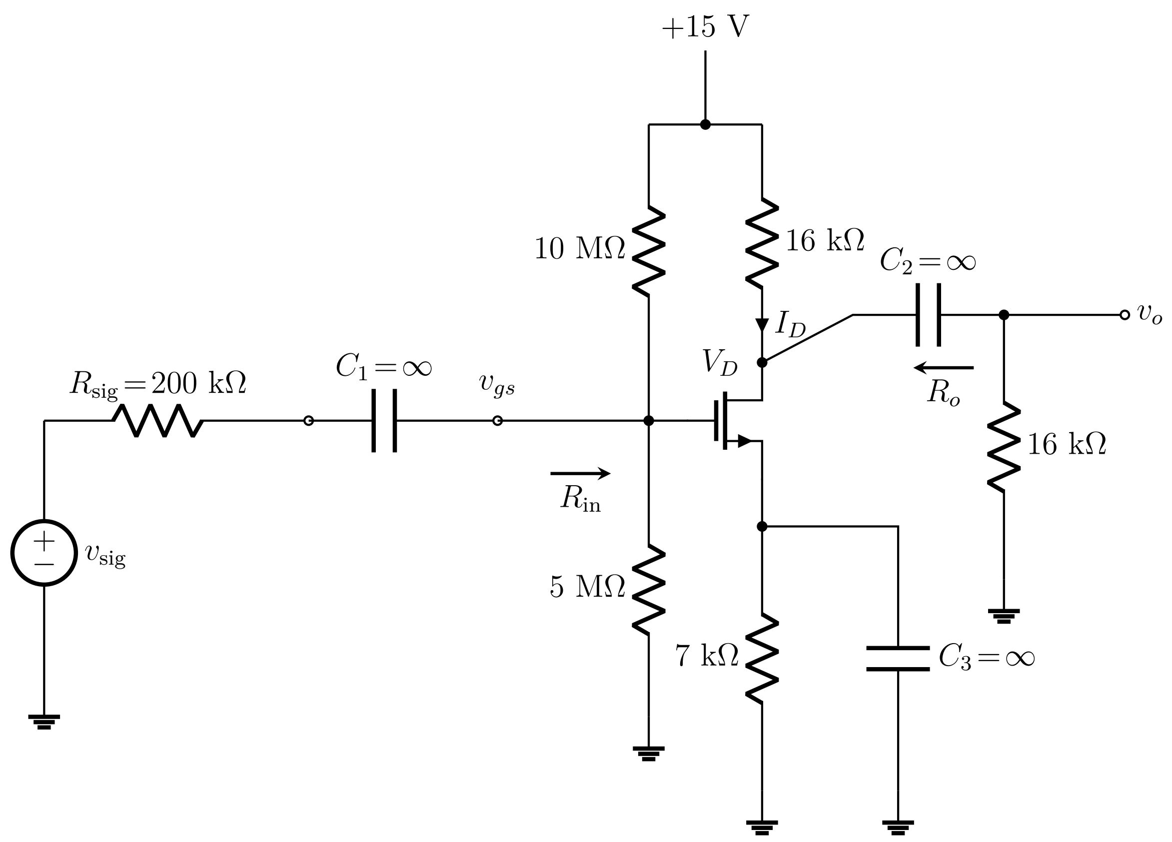
Common Gate Amplifier

Q1. Derive the expression for input resistance, output resistance, and overall voltage gain of a common-gate (CG) amplifier. **(26 marks)**
[EEE 263 | L-2/T-1 | 2013–14]

Common Source Amplifier

Q1. For the Common Source amplifier circuit shown, the transistor has $V_t = 1\,\text{V}$, $\mu_n C_{ox}(W/L) = 4\,\text{mA/V}^2$, and $V_A = 100\,\text{V}$. All capacitors behave as open circuits under DC and short circuits under AC.

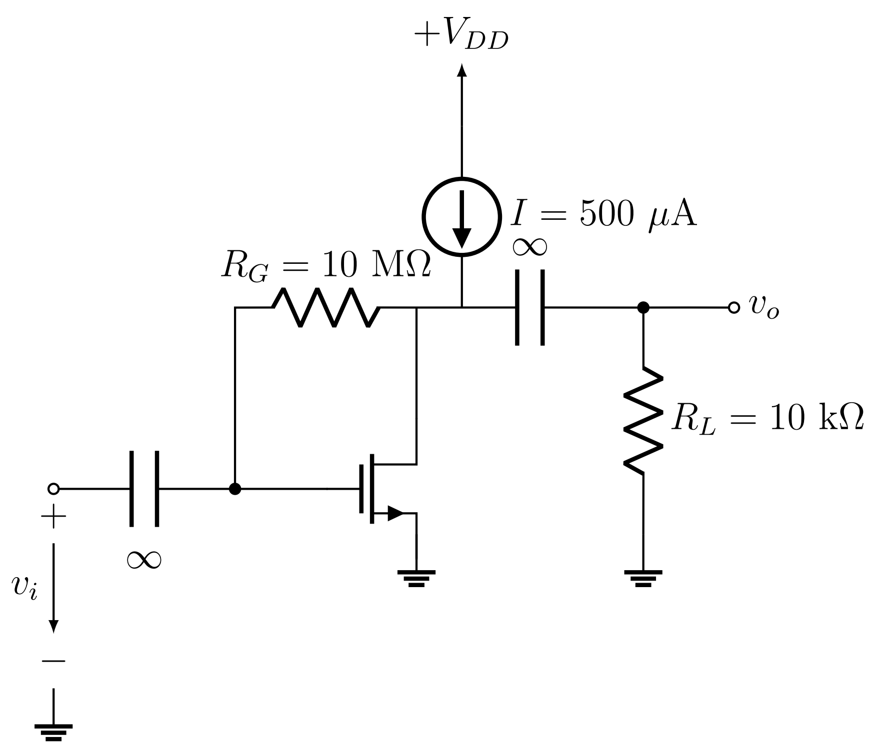
(i) Find the DC bias voltage V_D and drain current I_D (ignore Early effect for bias).
(ii) Find g_m, r_o and draw the small-signal equivalent circuit.
(iii) Determine R_{in}, R_o, and the overall voltage gain v_o/v_{sig}.
(iv) Calculate the maximum peak value of v_{sig} such that $v_{gs} \le 0.1\,V_{OV}$ (small-signal approximation valid).
(v) Suggest a circuit modification that will double the maximum v_{sig} found in (iv) without changing the DC bias point.



(35 marks) [EEE 263 | L-2/T-1 | 2024–25]

Q2. Explain why adding a source resistor R_S to a Common-Source MOSFET amplifier improves its bias stability compared to directly grounding the source terminal. Detail how R_S actively compensates for manufacturing variations in the threshold voltage V_t and transconductance parameter k_n. **(12 marks)** [EEE 263 | L-2/T-1 | 2024–25]

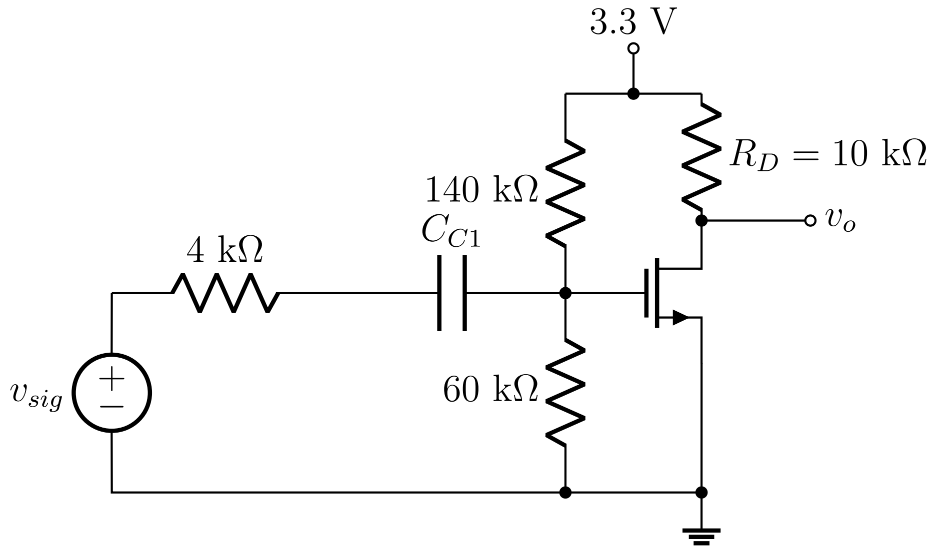
Q3. In the circuit shown, the NMOS transistor has $|V_t| = 0.9\,\text{V}$ and $V_A = 50\,\text{V}$ and is operated with $V_D = 2\,\text{V}$. What is the voltage gain v_0/v_i? What do V_D and the gain become for $I = 1\,\text{mA}$?



(17 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

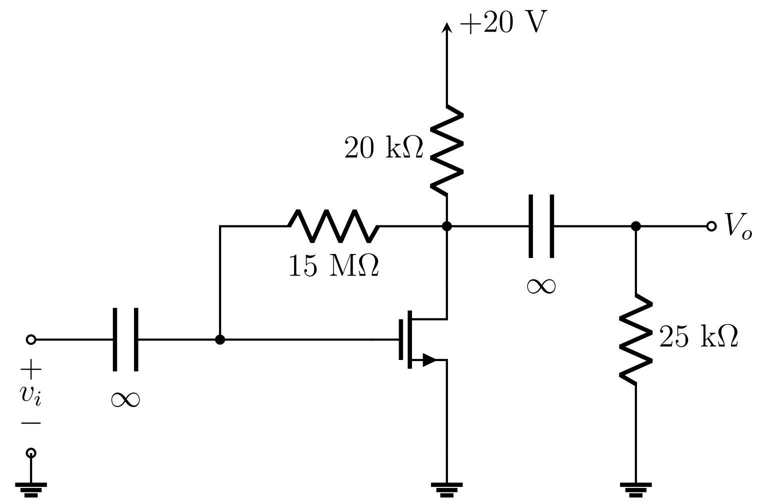
Q4. A CS amplifier utilizes a MOSFET biased at $I_D = 0.25\,\text{mA}$ with $V_{ov} = 0.25\,\text{V}$ and $R_D = 20\,\text{k}\Omega$. The amplifier is fed with a signal source having $R_{sig} = 100\,\text{k}\Omega$, and a 20-k$\Omega$ load is connected to the output. Find R_{in}, A_{vo}, R_o, A_v, and G_v. **(20 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Q5. Determine the overall small-signal voltage gain of the common source amplifier shown. The transistor parameters are: $V_t = 0.4\,\text{V}$, $k_n = 0.5\,\text{mA/V}^2$, $\lambda = 0.02\,\text{V}^{-1}$.



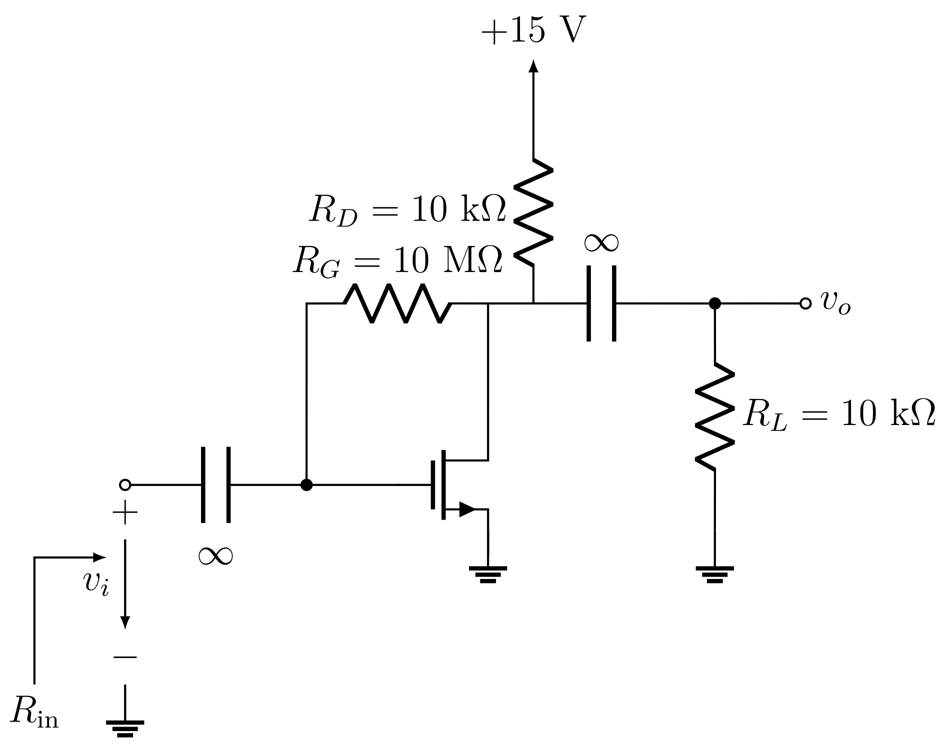
(26 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Q6. For the circuit shown, find the small-signal input impedance and voltage gain. Given: $V_t = 1.2\,\text{V}$, $k_n'(W/L) = 0.5\,\text{mA/V}^2$, and $V_A = 55\,\text{V}$.



(30 marks) **[EEE 263 | L-2/T-1 | Jan–2021]**

Q7. Determine the voltage gain v_0/v_i of the common-source MOSFET amplifier shown in the figure. The transistor has $V_t = 1.5\,\text{V}$, $k_n'(W/L) = 0.25\,\text{mA/V}^2$, and $V_A = 50\,\text{V}$.



(20 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

Q8. For the circuit shown:

(a) Calculate DC bias point (I_D and V_D).

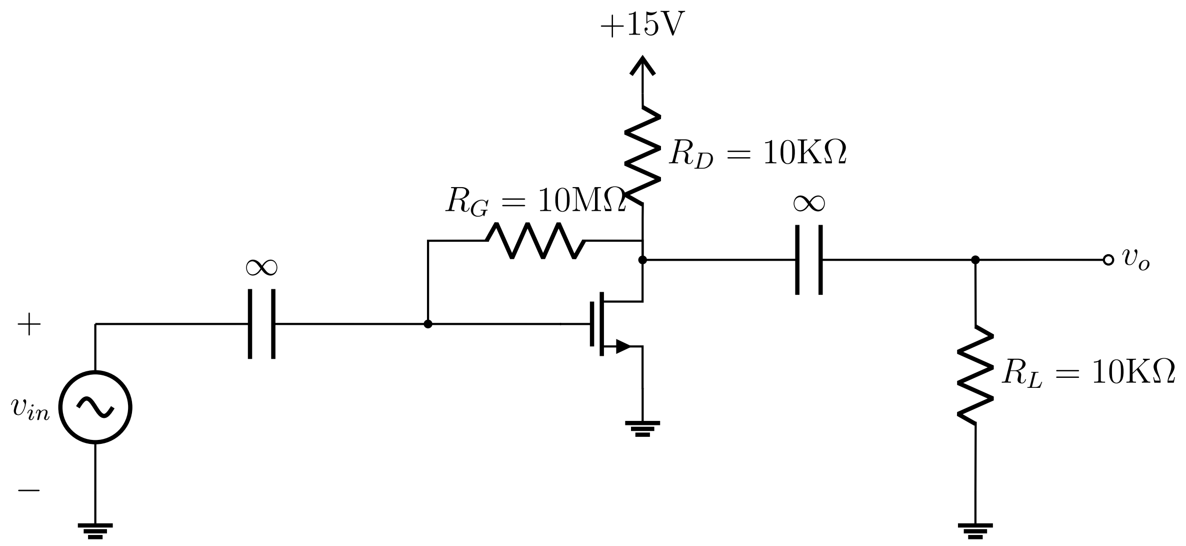
(b) Draw the small-signal equivalent circuit.

(c) Calculate the small-signal voltage gain.

(d) Find the input resistance.

(e) Determine the largest allowable input signal voltage.

The transistor has $V_t = 1.5\,\text{V}$, $k_n'(W/L) = 0.25\,\text{mA/V}^2$, and $V_A = 50\,\text{V}$.



(31 marks) [EEE 263 | L-2/T-1 | 2015–16]

Q9. For the CS amplifier, derive the expressions for input resistance (R_{in}), output resistance (R_{out}), open-circuit voltage gain (A_{vo}), and overall voltage gain (G_v). **(26 marks)** [EEE 263 | L-2/T-1 | 2012–13]

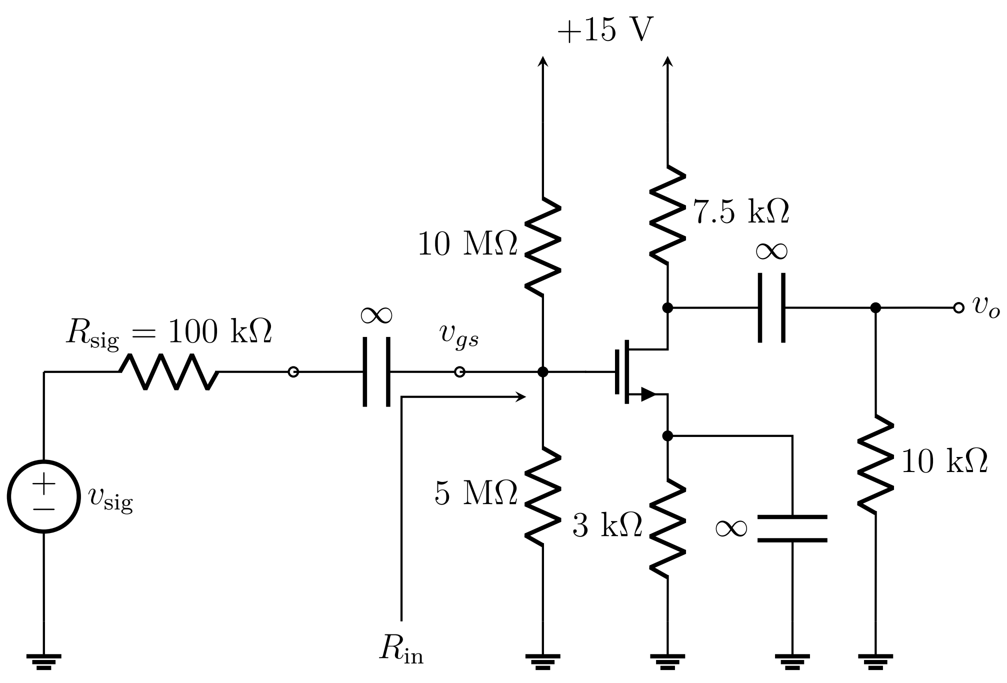
Q10. A common source amplifier circuit is shown in the figure.

(a) Prove that this circuit is operating in the saturation region.

(b) Draw the small-signal equivalent circuit.

(c) Find g_m, R_{in}, R_{out}, A_v, and G_v.

Assume $V_t = 1\,\text{V}$, $K_n' W/L = 2\,\text{mA/V}^2$, $V_A = 100\,\text{V}$.



(46 marks) [EEE 263 | L-2/T-1 | 2011–12]

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

Part IV — Linear Integrated Circuits

Op-amp ideal characteristics, mathematical operations, active filters, ADC

S7 Operational Amplifiers (Op-Amps)

Ideal Op-Amp Characteristics

Q1. What are the characteristics of an ideal op-amp? **(5 marks)** [EEE 263 | L-2/T-1 | 2013–14]

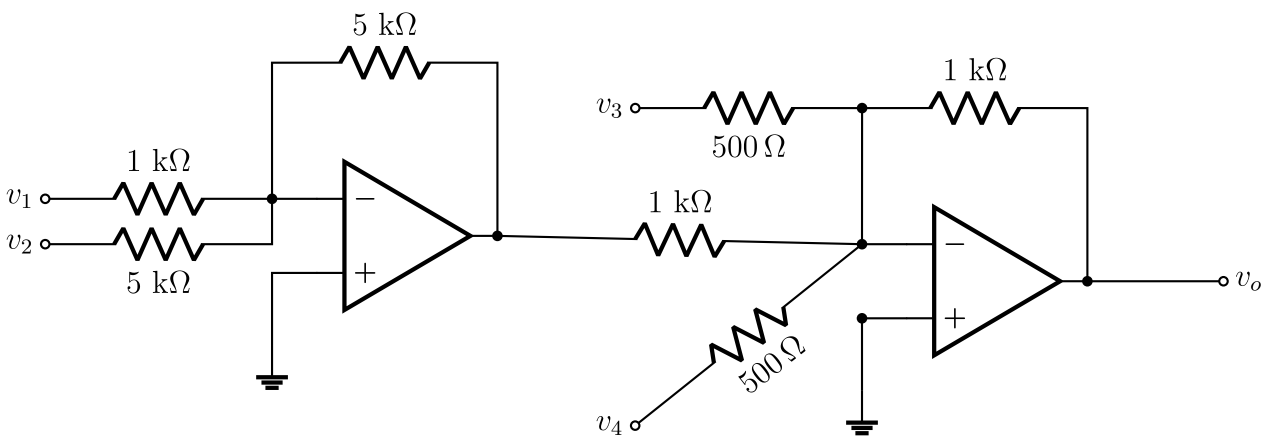
Q2. Explain graphically why in a closed-loop feedback circuit of an Op-Amp the differential voltage E_d is taken as zero. **(12 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Q3. Derive the closed-loop gain of an inverting Op-Amp amplifier as a function of its open-loop gain, provided that the open-loop gain is not infinity. **(13 marks)** [EEE 263 | L-2/T-1 | 2012–13]

Q4. Write down the properties of an ideal Op-Amp. **(10 marks)** [EEE 263 | L-2/T-1 | 2011–12]

Mathematical Operations

Q1. Find the expression of output voltage v_0 in terms of input voltages and resistances in the circuit shown. Also calculate the value of output voltage at $t = 1\,\text{sec}$ given that $v_1 = 2\,\text{V}$, $v_2 = 5\,\text{V}$, $v_3 = 1\,\text{V}$, and $v_4 = 5\sin(100\pi t)$.



(20 marks) [EEE 263 | L-2/T-1 | 2018–19]

Q2. For the circuit shown, find V_0. The MOSFET is in saturation with $k_n'(W/L) = 1\,\text{mA/V}^2$ and $V_t = 0.7\,\text{V}$. The op-amp is ideal.

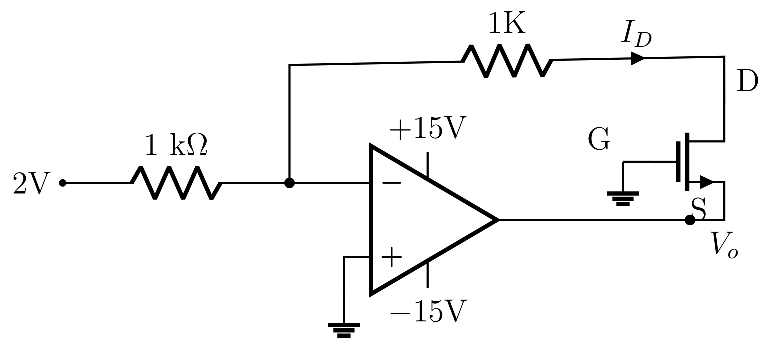
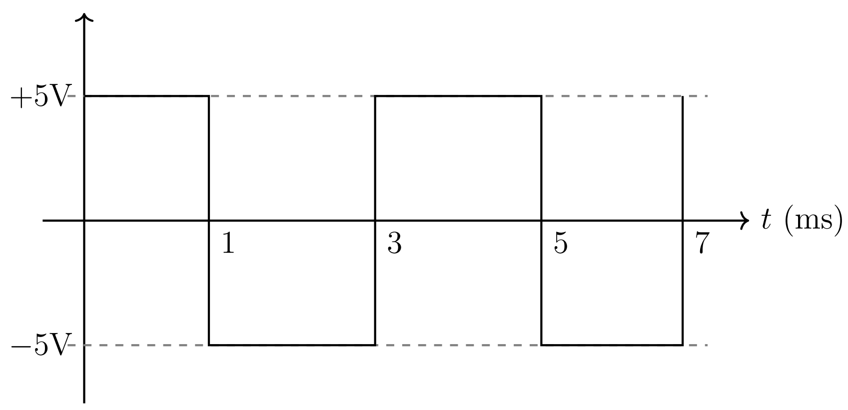
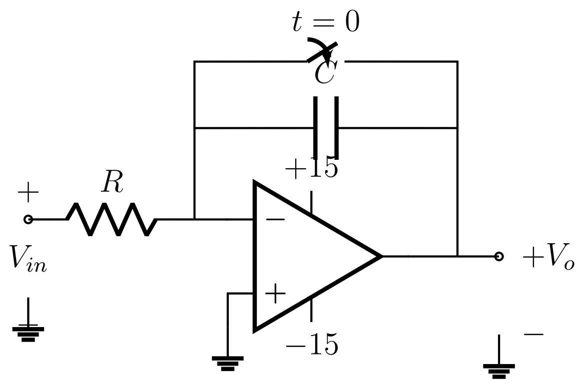


Fig for Q. 5(a)

(10 marks) [EEE 263 | L-2/T-1 | 2017–18]

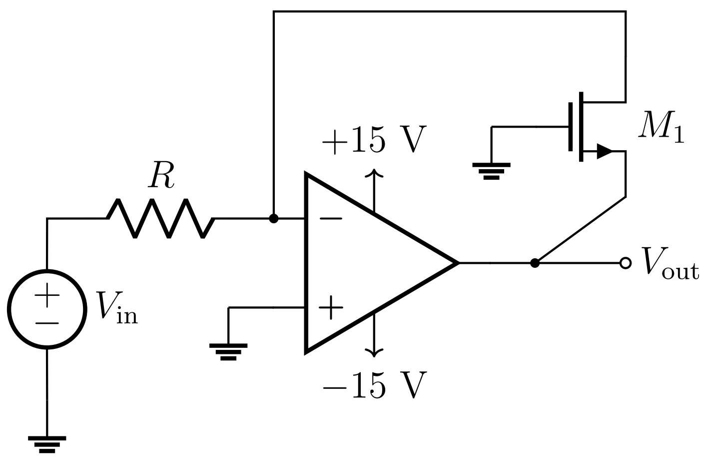
Q3. Consider the circuit and input voltage waveform shown in the figure. (i) If $R = 10\,\text{k}\Omega$, $C = 0.1\,\mu\text{F}$, and the Op-Amp is ideal, sketch the output voltage waveform to scale. (ii) If $R = 10\,\text{k}\Omega$, what value of C is required for the peak-to-peak output amplitude to be 2 V?





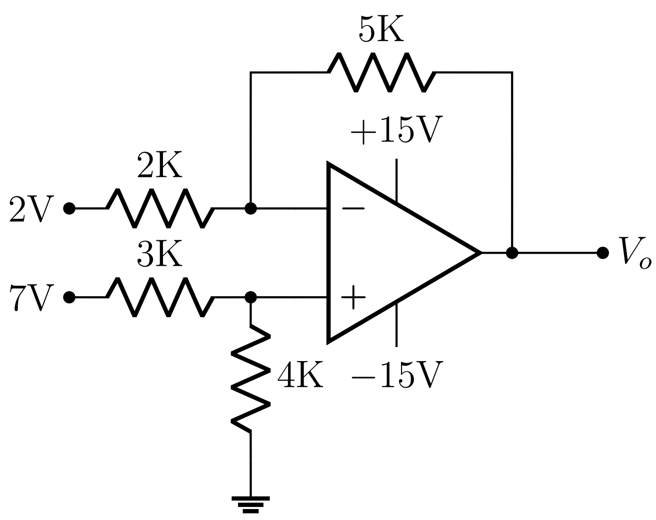
(16 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q4. For the circuit shown, (i) determine output voltage V_{out} in terms of input voltage V_{in}; (ii) using this circuit perform the following operation: $V_{out} = 3V_{in}^2$. Use additional Op-Amps and a DC voltage source if necessary. Specify values of all the parameters used. (Threshold voltage $V_{TH} = 0.5\,\text{V}$ for the MOSFET M1.)



(20 marks) [EEE 263 | L-2/T-1 | 2015–16]

Q5. What is the value of the output voltage V_o for the circuit shown?



(16.67 marks) [EEE 263 | L-2/T-1 | 2014–15]

Q6. Design an Op-Amp circuit that performs the following mathematical operation:

$$v_{out} = 5v_1 - 2\frac{dv_2}{dt} + 6\int v_3\,dt$$

where v_1, v_2, and v_3 are three input signals and v_{out} is the overall output signal. **(16 marks)** [EEE 263 | L-2/T-1 | 2012–13]

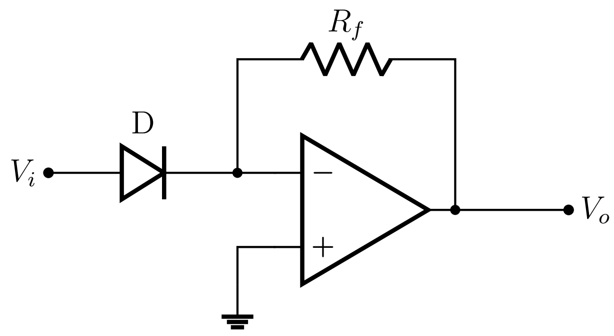
Q7. Propose a circuit using Op-Amp which will produce the following as output:

$$V_{out} = -4V_1 + 2V_2 + 7\frac{dV_1}{dt} - 8\int V_2\,dt$$

where V_1 and V_2 are two different input sources. **(20 marks)** [EEE 263 | L-2/T-1 | 2011–12]

Logarithmic / Exponential Amplifier

Q1. For the circuit shown , deduce the equation for output voltage V_0 and explain the function of this circuit.



(17 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Q2. Show that a proper connection of a diode, a resistor, and an op-amp can perform the exponential function. Mention the assumptions, if any. **(Unspecified marks)** [EEE 263 | L-2/T-1 | 2013–14]

Q3. Draw the circuit diagram of an inverting logarithmic amplifier and derive the expression of its output voltage. **(26 marks)** [EEE 263 | L-2/T-1 | 2011–12]

Difference Amplifier

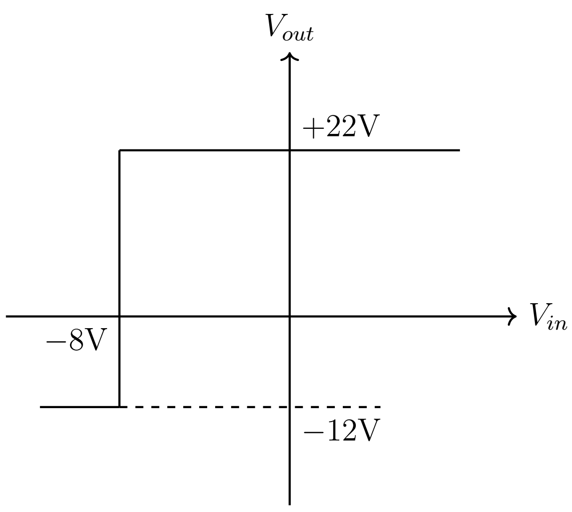
Q1. Design a subtractor circuit using op-amps so that the output voltage is $v_0 = v_1 - v_2$. **(15 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Q2. Design a circuit using a single Op-Amp that performs the following operation: $V_o = 3(V_1 - V_2)$ where V_1 and V_2 are input signals (AC or DC). Specify values of all resistors used in the design. **(14 marks)** [EEE 263 | L-2/T-1 | 2015–16]

Q3. Implement $z = 5V_1 - 3V_2$ using only one op-amp, where V_1 and V_2 are input signals. **(15 marks)** [EEE 263 | L-2/T-1 | 2013–14]

Comparator and Schmitt Trigger

Q1. Determine the op-amp circuit that generates the transfer characteristics shown in the figure.



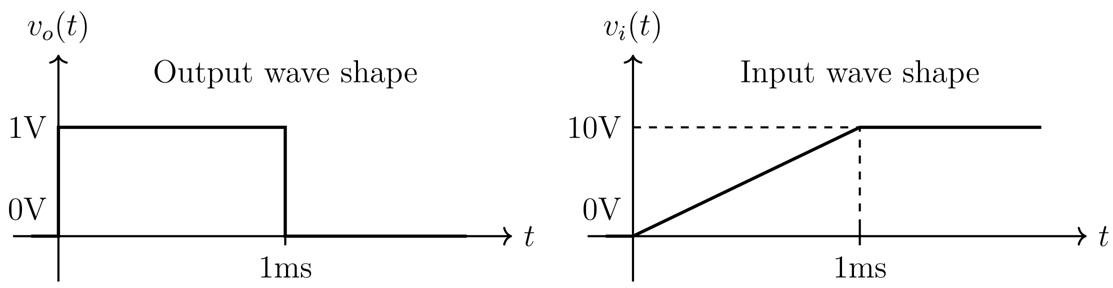
(11 marks) [EEE 263 | L-2/T-1 | 2016–17]

Q2. Design a Schmitt trigger that triggers from +16 V to −16 V at an input voltage of 12 V and triggers from −16 V to +16 V at an input voltage of 8 V. Specify all the values of resistors and reference DC voltage. **(15 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Q3. Draw a zero-crossing detector circuit using Op-Amp and briefly describe its operation. **(15 marks)** [EEE 263 | L-2/T-1 | 2012–13]

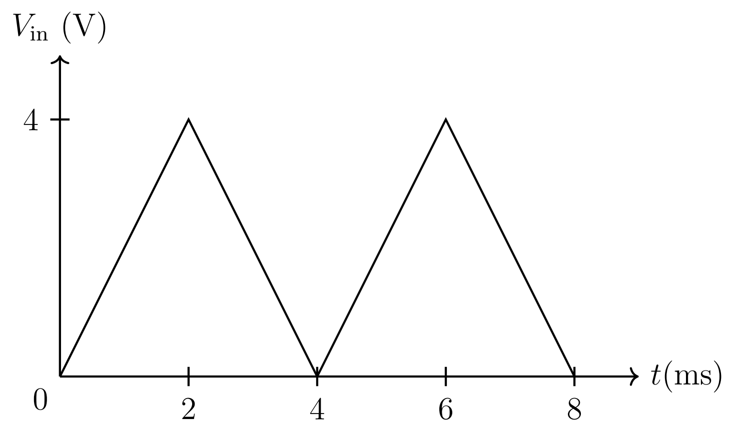
Waveform Generation

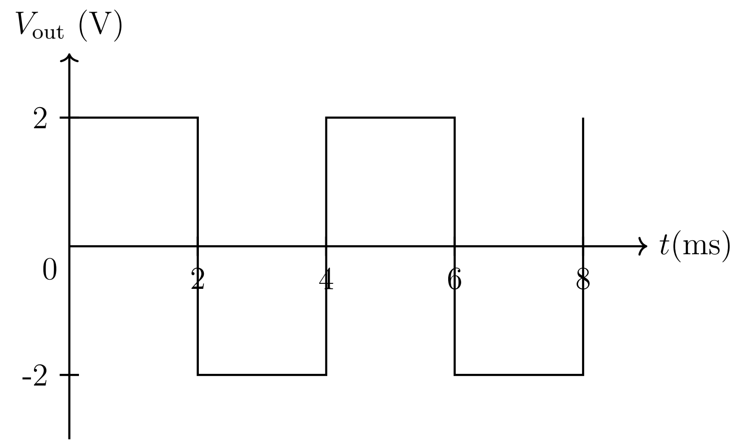
Q1. Design a circuit using op-amp(s) for given input $v_i(t)$ and output $v_o(t)$ waveforms shown in the figure.



(16 marks) [EEE 263 | L-2/T-1 | 2018–19]

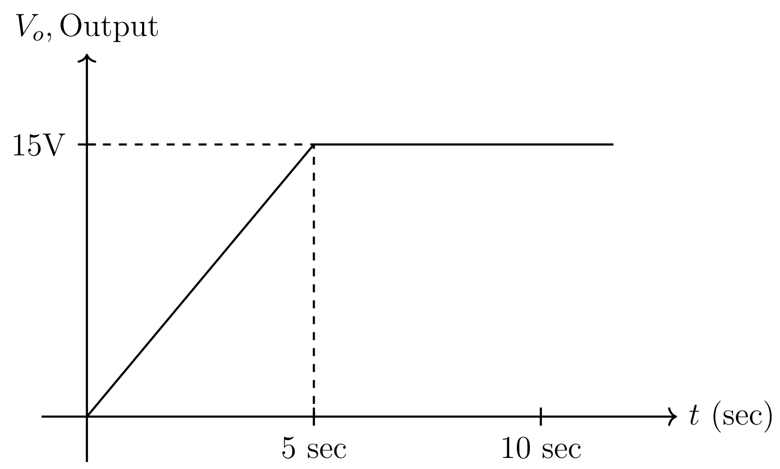
Q2. Input signal V_{in} and output signal V_{out} of a circuit are given in the figure. Design the circuit using Op-Amps. Show necessary calculations.





(12 marks) **[EEE 263 | L-2/T-1 | 2015–16]**

Q3. Design a circuit using Op-Amps that gives a specified output waveform from a 1 V DC input. Use +15 V and −15 V for DC biasing. Specify all the values of resistors and capacitors used.



(15 marks) **[EEE 263 | L-2/T-1 | 2014–15]**

Analog Computing (Differential Equations)

Q1. Design a circuit using op-amps and other passive elements to solve:

$$2\frac{d^2v}{dt^2} - 3\frac{dv}{dt} + 4v + 5 = 0$$

Draw the overall connection diagram and specify element values. Assume op-amps biased with ±15 V; all initial conditions are zero.
(17 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Q2. Using op-amps, design a circuit to solve the following differential equation where x is input and y is output:

$$\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + 2y = 8x$$

(20 marks) **[EEE 263 | L-2/T-1 | 2022–23]**

Q3. Design an electronic analog computer using op-amp to solve the following differential equation. Give a brief explanation of the operation of the designed computer.

$$\frac{d^2v}{dt^2} + 4\frac{dv}{dt} + 5v - 5 = 0$$

(26 marks) **[EEE 263 | L-2/T-1 | 2018–19]**

Q4. Design an analog computer circuit to solve the following differential equation for $t > 0$:

$$\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v + 4\cos(10t) = 0$$

with initial conditions $v(0) = 2$, $v'(0) = 0$. An oscillator is available to provide $\sin(10t)$ signal. **(20 marks)** **[EEE 263 | L-2/T-1 | 2015–16]**

Q5. Design a circuit using Op-Amps that can solve the following differential equation:

$$3\frac{d^2x(t)}{dt^2} + 4x(t) = 0$$

Point out the node that represents the solution $x(t)$. Specify all the values of resistors and capacitors used. **(20 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Q6. Design a circuit using op-amp that can provide solution to the following equation:

$$3\frac{d^2v}{dt^2} + K\frac{dv}{dt} - 3v + 8 = 0$$

where K = last digit of your student ID. Briefly discuss the operation of your designed circuit. **(29 marks)** **[EEE 263 | L-2/T-1 | Jan–2021]**

Multiplier Circuit Design

Q1. Suppose you have two input voltages X_1 and X_2. Design a circuit using Op-Amps that gives output $V_o = 10X_1X_2$. The diodes available maintain the characteristic equation $I_D = I_s e^{V_D/nV_T}$ where $I_s = 10^{-6}$ A. Specify all the values of resistors used. **(26 marks)** **[EEE 263 | L-2/T-1 | 2014–15]**

Frequency Multiplier

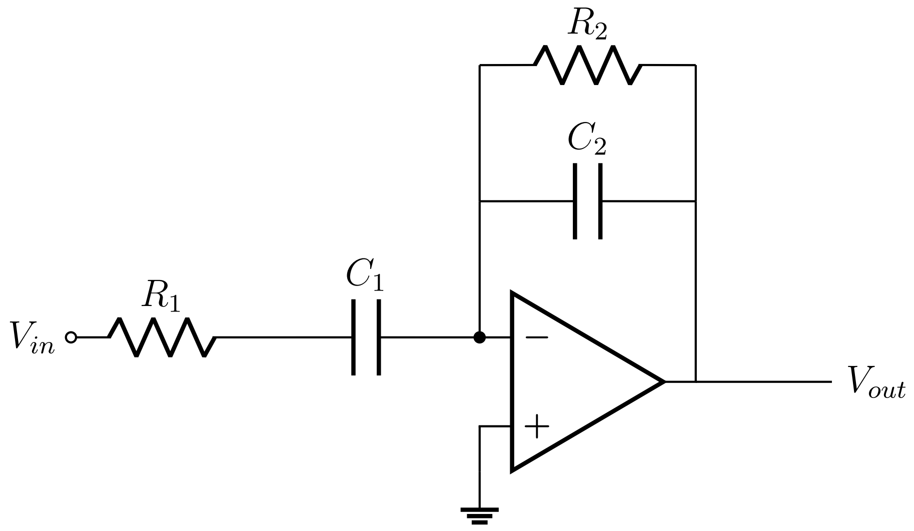
Q1. You have two supply voltage sources, $v_1 = \sin(100t)$ V and $v_2 = 1$ V. Design a circuit using op-amps that will produce an output voltage $v_o = (\sin(50t))^2$ V. Also show that this circuit produces this output. **(20 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

Q2. Suppose you have a sinusoidal input voltage $V_{in}(t) = \sin(4000\pi t)$ volt. Design a frequency multiplier circuit using Op-Amps that multiplies the frequency by a factor of 2 to produce $V_{out}(t) = \sin(8000\pi t)$ volt. Use resistors, capacitors, inductors, and diodes (having reverse saturation current 10^{-6} A) if necessary. Show necessary calculations and specify all design parameter values. **(26 marks)** **[EEE 263 | L-2/T-1 | 2015–16]**

Active Filters

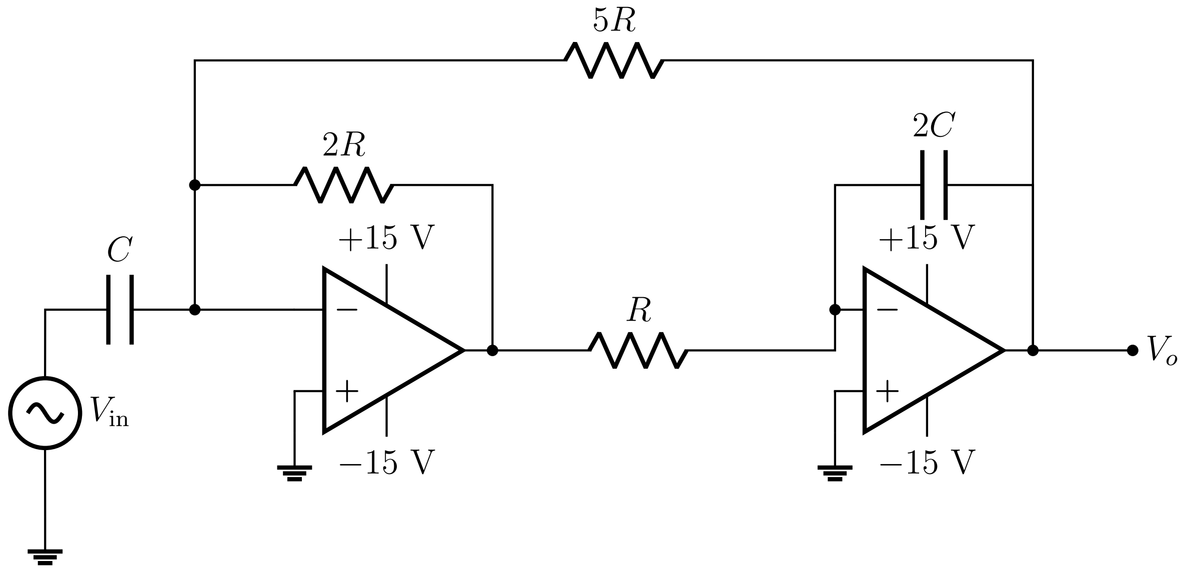
Q1. Design a 2nd order bandpass filter with lower cutoff frequency $f_{CL} = 20$ kHz and higher cutoff frequency $f_{CH} = 200$ kHz. **(10 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Q2. Determine the type of filter shown in the circuit by computing the transfer function in the frequency domain.



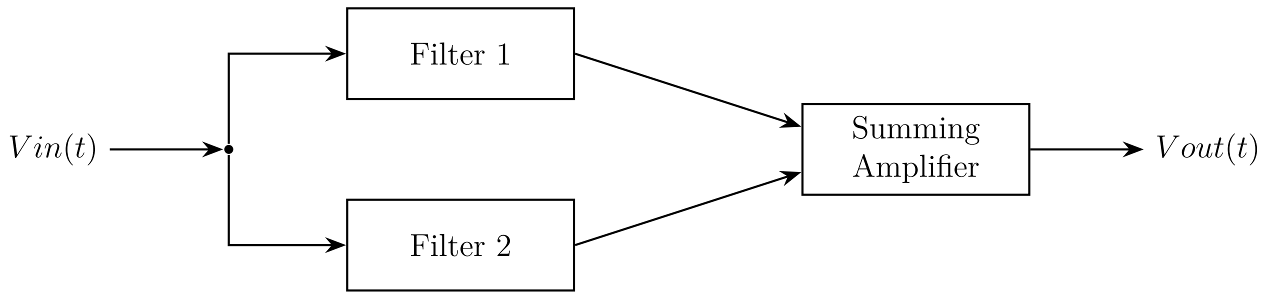
(20 marks) **[EEE 263 | L-2/T-1 | 2016–17, 2017–18]**

Q3. For the filter circuit given, determine (i) its type and comment on its order; derive the transfer function and find the cutoff frequency. (ii) If $R = 10\,\text{k}\Omega$ and $C = 0.1\,\mu$F, would it be possible to pass a 100 Hz signal through this filter without distortion? Justify your answer. Note that the first Op-Amp block performs summation of input and output signals.



(23 marks) [EEE 263 | L-2/T-1 | 2015–16]

Q4. In the circuit shown, input voltage is $v_{in}(t) = 5 + 3\cos(300\pi t - 30) + 8\sin(1000\pi t) + 7\sin(1600\pi t + 45)$ volt. Design the active filter blocks and summing amplifier to obtain output $V_{out}(t) = 7 + 12\sin(1000\pi t)$ volt. Sketch the frequency response and specify the cutoff frequency for each filter. Specify all values of resistors and capacitors/inductors.



(23 marks) [EEE 263 | L-2/T-1 | 2015–16]

Q5. Write the equation of magnitude of gain of a −20 dB/decade low pass filter. Why is it called a −20 dB/decade filter? Can it also be called a −6 dB/octave filter? **(20 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Q6. Design a notch filter that eliminates any signal between 8 kHz and 12 kHz using op-amps. Specify all the values of resistors and capacitors used. **(20 marks)** [EEE 263 | L-2/T-1 | 2014–15]

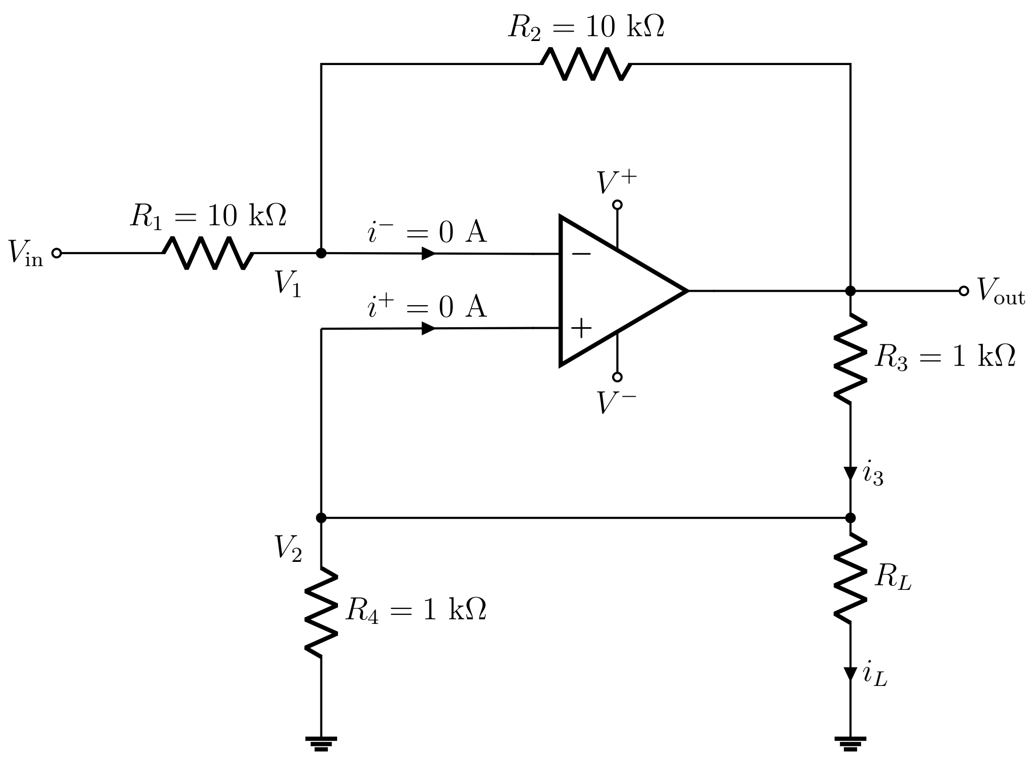
Q7. A system blocks 20 Hz and 20 kHz but passes 10 kHz. (i) Draw the frequency response of the system. (ii) Draw a block diagram of the system. **(15 marks)** [EEE 263 | L-2/T-1 | 2013–14]

Q8. Design a 40 dB low pass filter (LPF) with a cutoff frequency of 1 kHz. You have only one op-amp. **(15 marks)** [EEE 263 | L-2/T-1 | 2013–14]

Q9. Draw the circuit diagram of a −20 dB/dec low pass filter. Also write down the design procedure. Design a −20 dB/dec low pass filter with a cut-off frequency of 100 kHz. **(16 marks)** [EEE 263 | L-2/T-1 | 2011–12]

Voltage-to-Current Converter

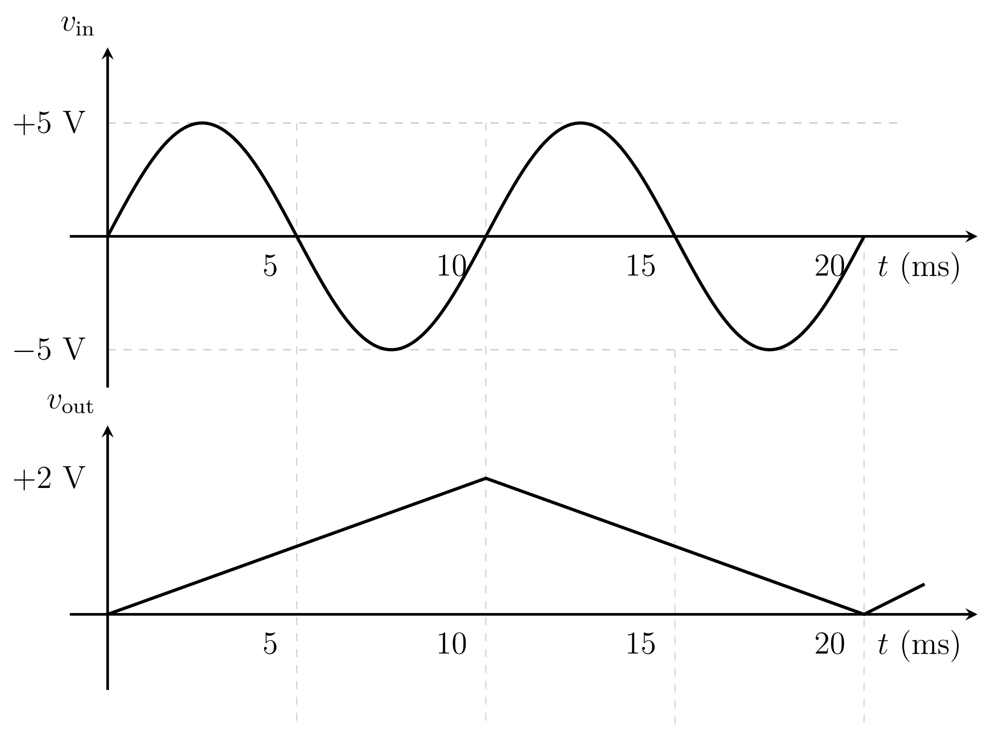
Q1. The circuit shown in Figure functions as a voltage-to-current converter. The op-amp used in this circuit is biased using ±20 V sources. The overall circuit operates in the linear region as long as V_{out} stays within the bias voltage range, and the concept of virtual short (i.e. $V_1 = V_2$) can be applied under such circumstances. If V_{out} attempts to exceed these limits, the amplifier enters saturation, and the virtual short assumption becomes invalid. Given the circuit parameters $R_1 = R_2 = 10$ kΩ, $R_3 = R_4 = 1$ kΩ and $R_L = 100$ Ω, calculate the load current i_L and output voltage V_{out} for and input voltage of $V_{\text{in}} = -5$ V.



(18 marks) [EEE 263 | L-2/T-1 | 2024–25]

System Design (Analog to Digital Interface)

Q1. Design a circuit using Op-Amp and Digital Circuits for given input wave shape v_{in} and output wave shape v_{out} in Figure . Assume a 5 V level to be digital 1 and −5 V level to be digital 0.



(26.67 marks) [EEE 263 | L-2/T-1 | 2017–18]

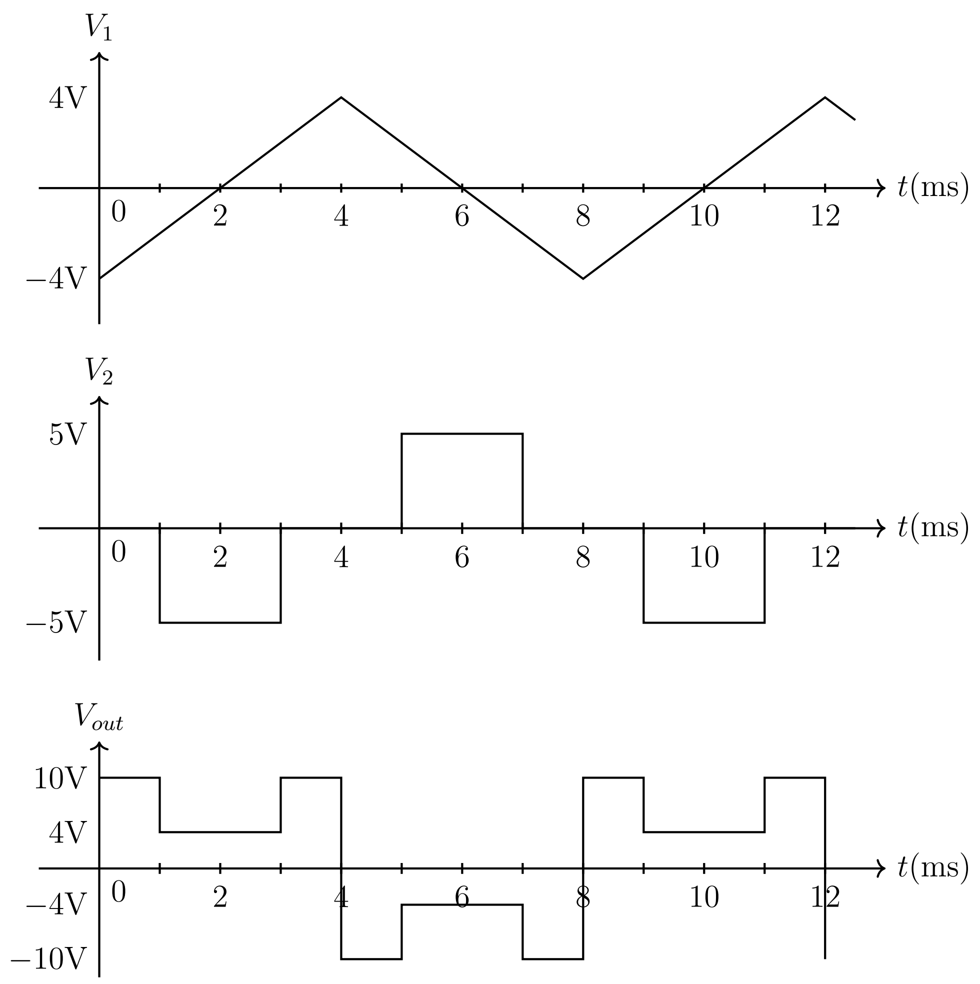
Q2. A circuit is to be designed where depending on the range of input voltage a digital sequence is passed to a computer. The ranges and corresponding digital sequences are:

Range	Digital Sequence
0 – 1.2 V	010
1.2 – 3.2 V	110
3.2 – 3.5 V	001
3.5 – 5 V	111

Design a circuit using op-amps and digital circuits (gates, multiplexers) to perform the task. Assume 5 V = digital 1 and 0 V = digital 0. **(25 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Waveform Realization

Q1. Determine an op-amp circuit realization for given input wave shapes V_1 and V_2 and output wave shape V_{out}.



(15 marks) [EEE 263 | L-2/T-1 | 2016–17]

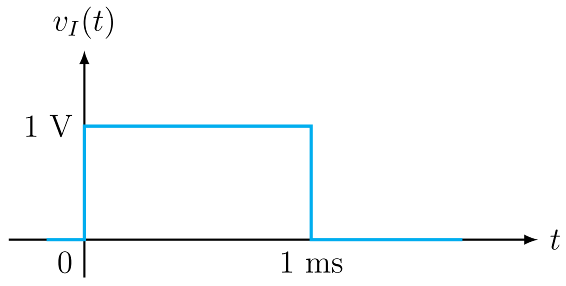
Integrator

Q1. Design a circuit using ideal operational amplifiers that implements the following equation:

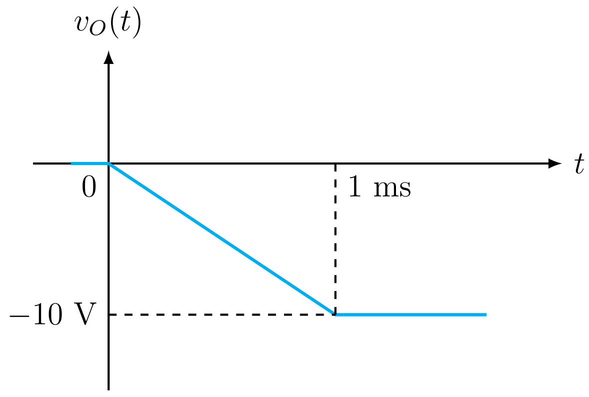
$$v_0 \propto \int (v_1 + v_2)\, dt$$

Write the final output expression in terms of the circuit components used in your design. **(20 marks)** **[EEE 263 | L-2/T-1 | 2020–21]**

Q2. The input voltage v_I and output voltage v_O of an op-amp based circuit are given. Design the circuit.



(a)

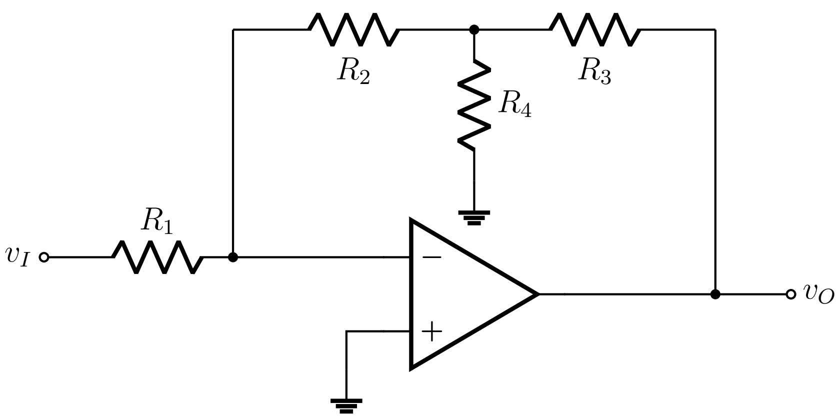


(b)

(17 marks) **[EEE 263 | L-2/T-1 | Jan–2021]**

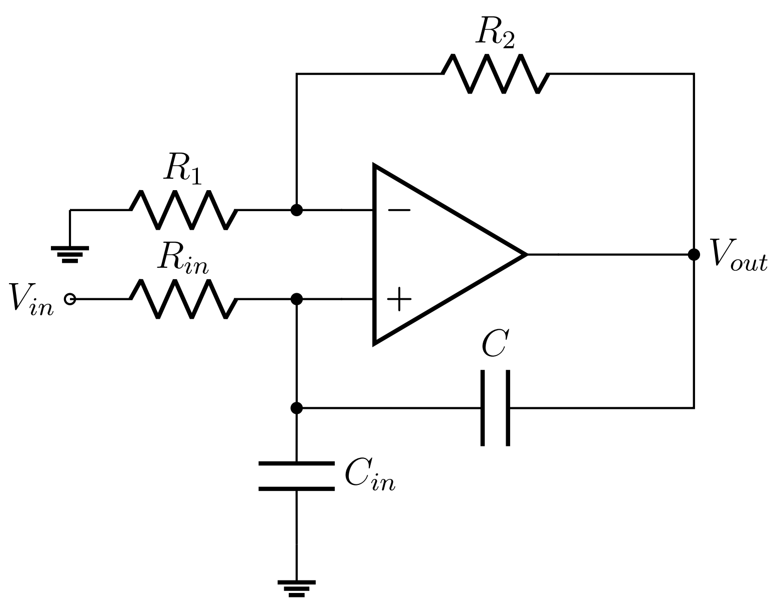
Op-Amp Network Analysis

Q1. Determine the closed-loop voltage gain and current gain of the ideal op-amp network shown in the figure.



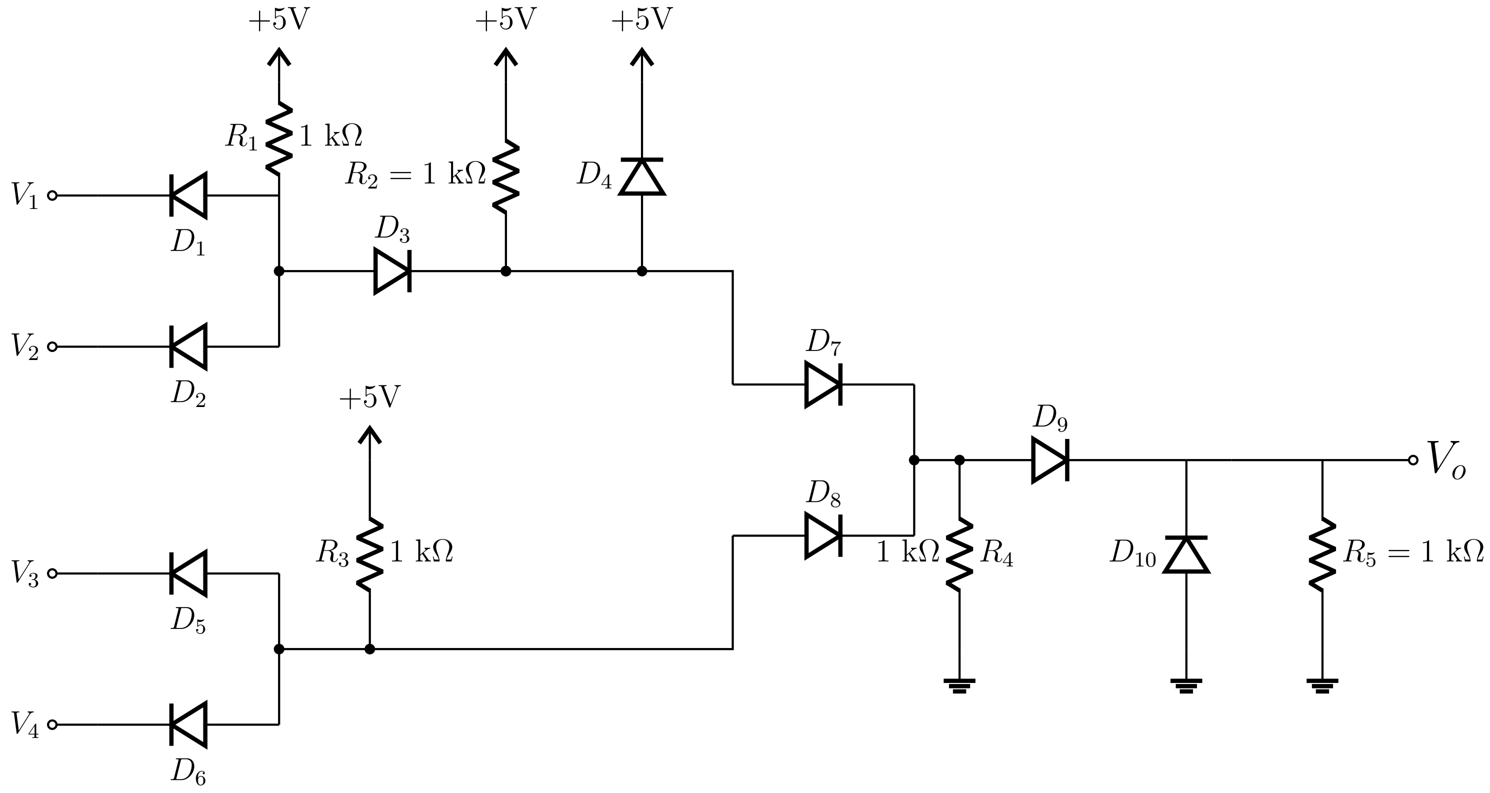
(23 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Q2. Determine the differential equation solved by the circuit shown: (i) when the capacitor C is absent; (ii) when the capacitor C is present. At what value of C does the circuit become a non-inverting amplifier?



(21 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Q3. Determine the Boolean expression for V_0 in terms of the four input voltages for the circuit shown (diode logic).



(10 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Op-Amp Practical Imperfections

Q1. What are the AC imperfections of a practical op-amp? How can we find out the value of input offset voltage and input offset current?
(15 marks) **[EEE 263 | L-2/T-1 | 2013–14]**

Dual Slope A/D Converter

Q1. By drawing appropriate circuits, explain the operation of a dual slope A/D converter. **(20 marks)** **[EEE 263 | L-2/T-1 | 2020–21]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

Part V — Oscillators & Timers

Colpitts oscillator, 555 Timer astable design, triangular wave generators

S8 Oscillators, Timers & Function Generators

Barkhausen Stability Criterion

Q1. Write down the Barkhausen criterion. **(10 marks)** [EEE 263 | L-2/T-1 | 2018–19]

Q2. Derive expression of Desensitivity. Explain positive and negative feedback from desensitivity. **(10+5 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Q3. Write down the Barkhausen stability criterion for an oscillation in a linear electronic circuit to sustain. **(6 marks)** [EEE 263 | L-2/T-1 | 2014–15]

Wien Bridge Oscillator

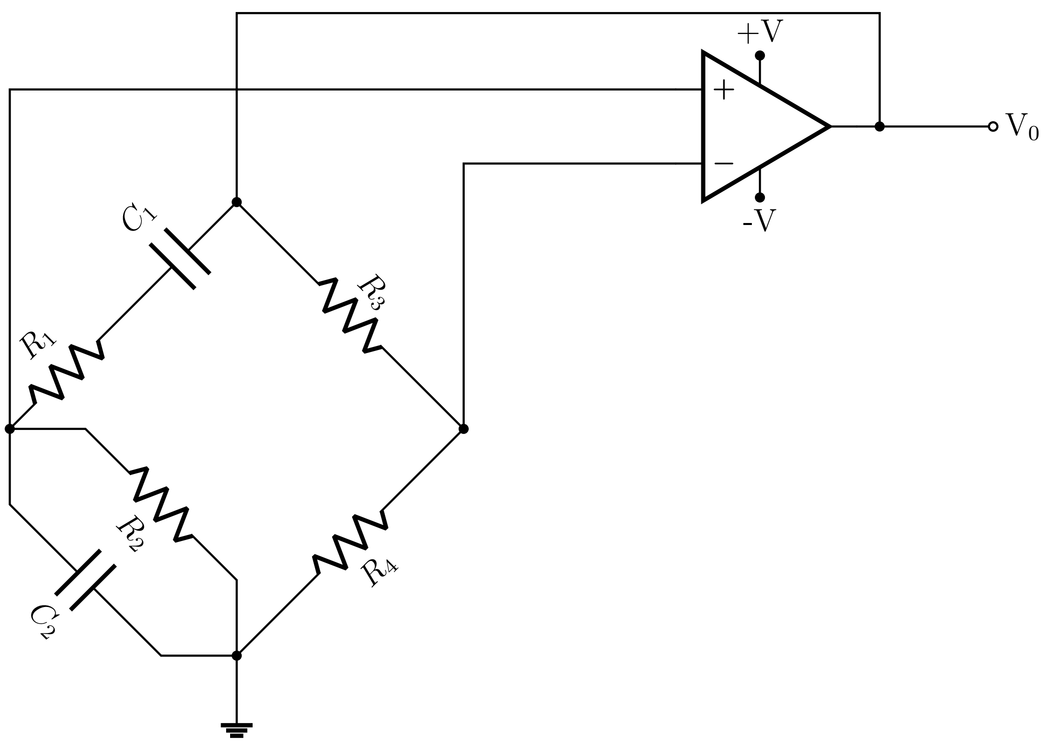
Q1. For the Wien bridge circuit connected to an op-amp shown:

(a) state conditions for sinusoidal oscillation;

(b) derive the equation for oscillation frequency;

(c) show that the voltage gain A_V must be $\geq \left(1 + \frac{R_1}{R_2} + \frac{C_1}{C_2}\right)$ for oscillation to sustain;

(d) if $R_1 = R_2 = 10\,\text{k}\Omega$ and $C_1 = C_2 = 0.1\,\mu\text{F}$, design the values of R_3 and R_4 so that oscillation sustains.



(28 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Q2. Design a Wien Bridge Oscillator with an oscillation frequency of 1 kHz. What is the gain of the feedback network of your design? Specify all the values of resistors and capacitors used. **(20 marks)** [EEE 263 | L-2/T-1 | 2014–15]

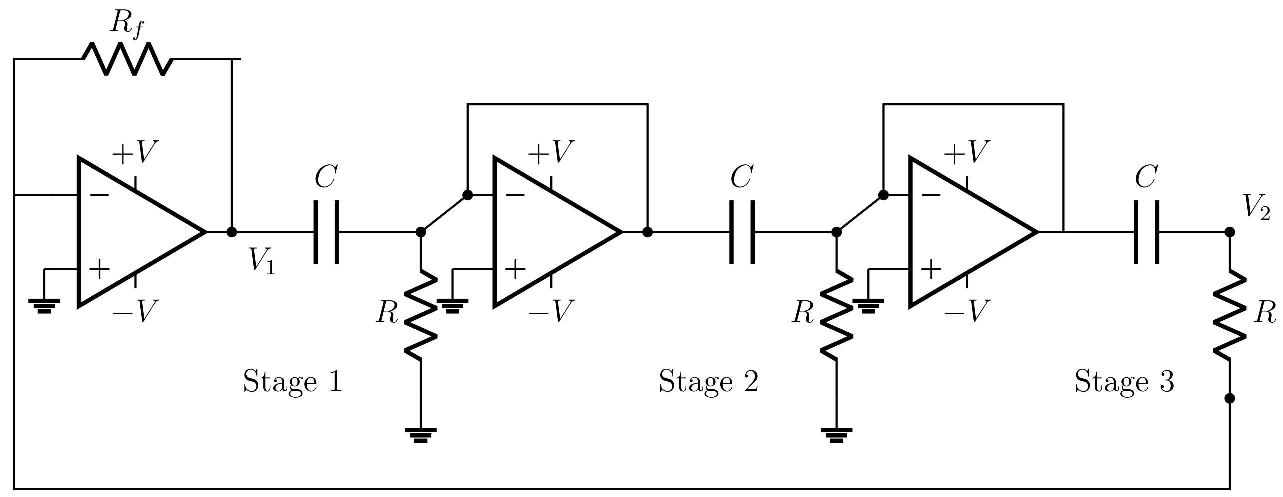
Q3. Design a Wien Bridge Oscillator with an oscillation frequency of 1000 rad/sec. Available capacitors are $0.1\,\mu$F, $1\,\mu$F, and 100 pF. **(Unspecified marks)** [EEE 263 | L-2/T-1 | 2013–14]

Q4. Draw a Wien Bridge oscillator circuit and describe its operating principle. **(17 marks)** [EEE 263 | L-2/T-1 | 2012–13]

RC Phase Shift Oscillator

Q1. For an RC phase shift oscillator, derive the expression for oscillation frequency. Also derive the conditions for the Barkhausen criterion to be satisfied. Assume all resistors in the feedback network are of the same value and all capacitors are of the same value. **(26 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q2. For the oscillator circuit shown below, find oscillation frequency (w), β-network gain (β), and amplifier gain (A). Also design this oscillator for an oscillation frequency of 1 kHz.Note that, each state is separated by a source follower circuit. Hence each stage will give a phase shift of 60° and 3 stages will give in total 180 phase shift. Here, $\beta = \frac{v_2}{v_1}$.

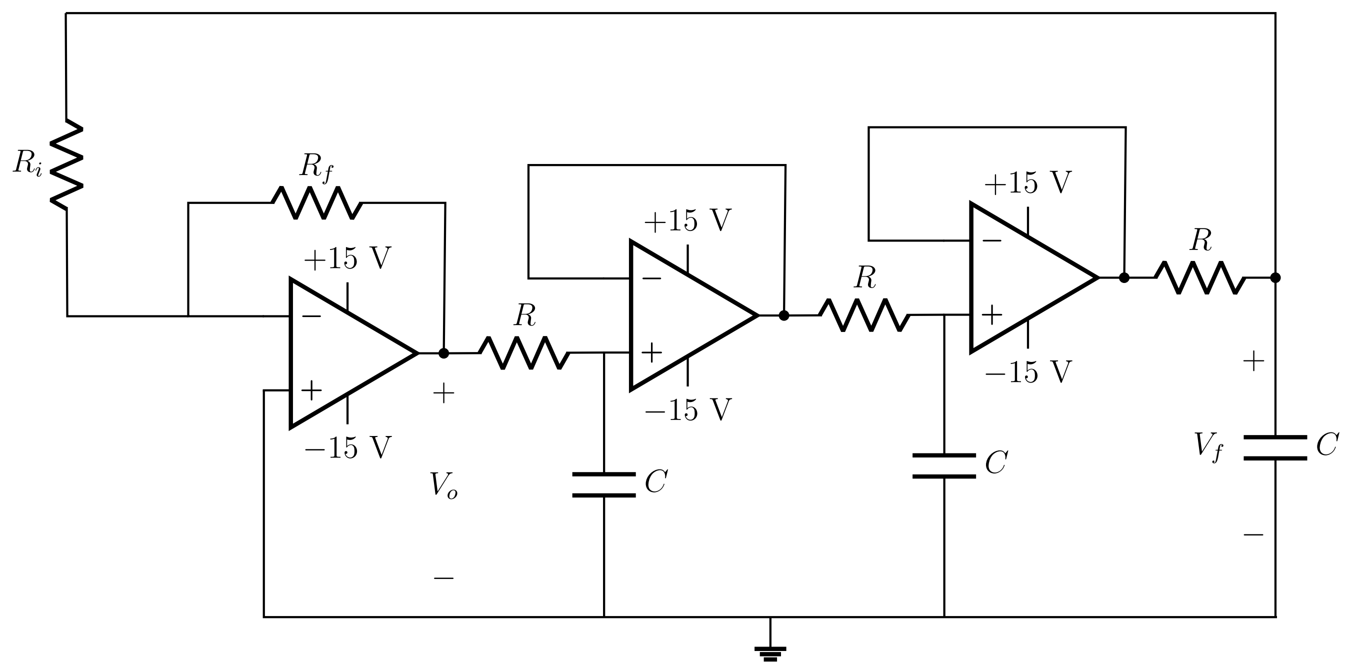


(18 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q3. For the RC phase shift oscillator shown:

(a) Determine the oscillation frequency f_0 and obtain the relationship between R_i and R_f for sustainable oscillation.

(b) How can you generate a 100 kHz signal from this oscillator?



(20 marks) [EEE 263 | L-2/T-1 | 2015–16]

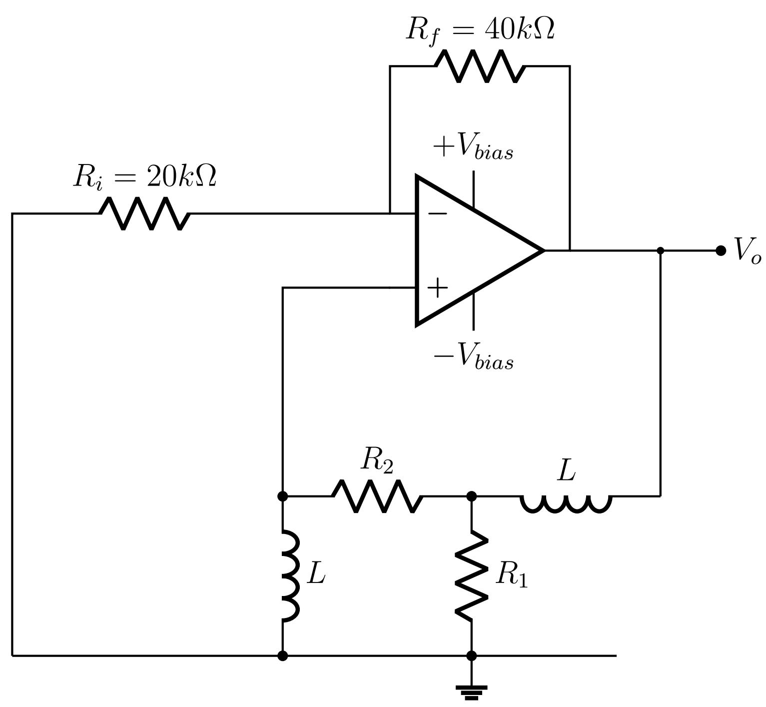
LC Oscillator

Q1. For the oscillator circuit given:

(a) Derive the expression for oscillation frequency f_0.

(b) Obtain the relationship between R_1 and R_2 for sustainable oscillation.

(c) Specify the bias conditions and values of L, R_1, R_2 to obtain a 20 V peak-to-peak, 500 kHz sinusoidal signal from this oscillator.



(26 marks) [EEE 263 | L-2/T-1 | 2015–16]

Colpitts Oscillator

Q1. With an appropriate circuit diagram, describe the operation of an op-amp based Colpitts oscillator. Derive the frequency of oscillation and explain how this circuit satisfies the conditions for sustained oscillation. **(18 marks)** **[EEE 263 | L-2/T-1 | 2024–25]**

Q2. Derive the expression of oscillation frequency for a Colpitts oscillator. Then derive the conditions for the Barkhausen criterion to be satisfied. **(20 marks)** **[EEE 263 | L-2/T-1 | 2022–23]**

Q3. Draw the circuit of the Colpitts oscillator. Then, using the small-signal model of the MOSFET, show that the frequency of oscillation is:

$$\omega_0 = \frac{1}{\sqrt{L \cdot \dfrac{C_1 C_2}{C_1 + C_2}}}$$

and that $\dfrac{C_1}{C_2} = g_m R$. **(26 marks)** **[EEE 263 | L-2/T-1 | 2018–19, 2020–21]**

Crystal Oscillator

Q1. Draw the equivalent circuit of a piezoelectric crystal. Qualitatively draw the crystal reactance versus frequency and identify the inductive and capacitive reactance regions. **(10 marks)** **[EEE 263 | L-2/T-1 | 2018–19]**

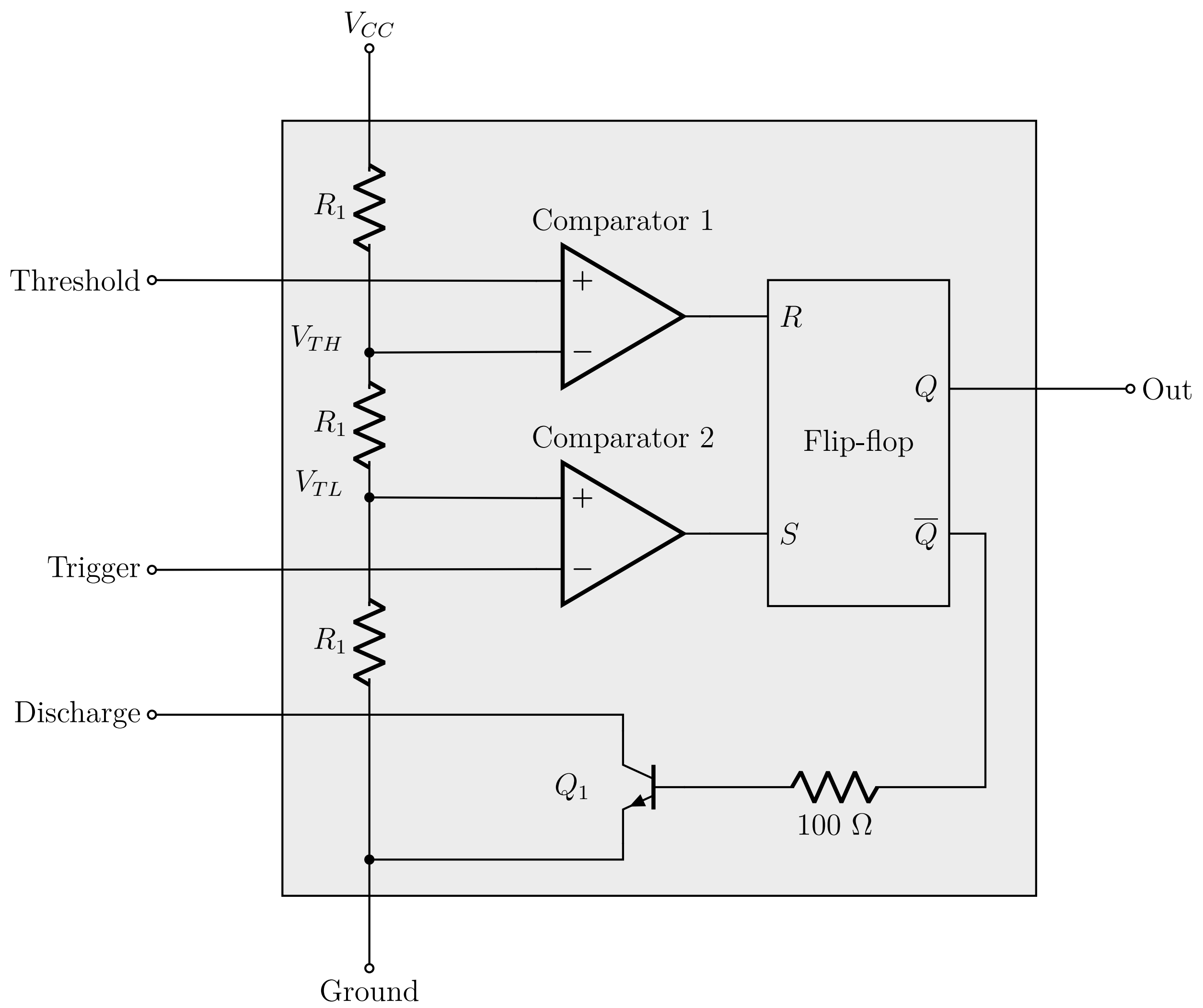
One-Shot (Monostable) Multivibrator

Q1. Draw a one-shot multivibrator circuit using an op-amp. Explain the operation of the circuit. **(26 marks)** **[EEE 263 | L-2/T-1 | 2021–22]**

555 Timer—Oscillator Design

Q1. Design an oscillator circuit using a 555 Timer IC which has a square wave output with oscillation frequency of 20 kHz and duty cycle of 25%. Provide a necessary illustration of the designed circuit and draw the wave shape.
(20 marks) **[EEE 263 | L-2/T-1 | 2020–21]**

Q2. Design a circuit using a 555 timer to generate a 100 kHz square wave with a duty cycle of 40%. Draw the final connection diagram with proper component values.



(15 marks) **[EEE 263 | L-2/T-1 | 2024–25]**

Q3. Design a square wave generator with 10 ms period and 75% duty cycle using a 555 timer. **(15 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Q4. To get an output pulse of 10 ms width when a trigger is applied, an external resistor and a 1 μF capacitor are connected with a 555 timer. What should be the value of the resistor? **(10 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q5. Design an astable multivibrator using a 555 timer with a period of 10 ms and 75% duty cycle. **(10 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q6. Design a circuit using a 555 IC which results in a square wave with oscillation frequency of 100 kHz and 75% duty cycle. Draw your designed circuit and output wave shape. **(20 marks)** [EEE 263 | L-2/T-1 | 2018–19]

Q7. For the 555 IC based astable multivibrator circuit shown with $R_1 = 7\,\text{k}\Omega$, $R_2 = 3\,\text{k}\Omega$, $C = 0.5\,\mu\text{F}$:

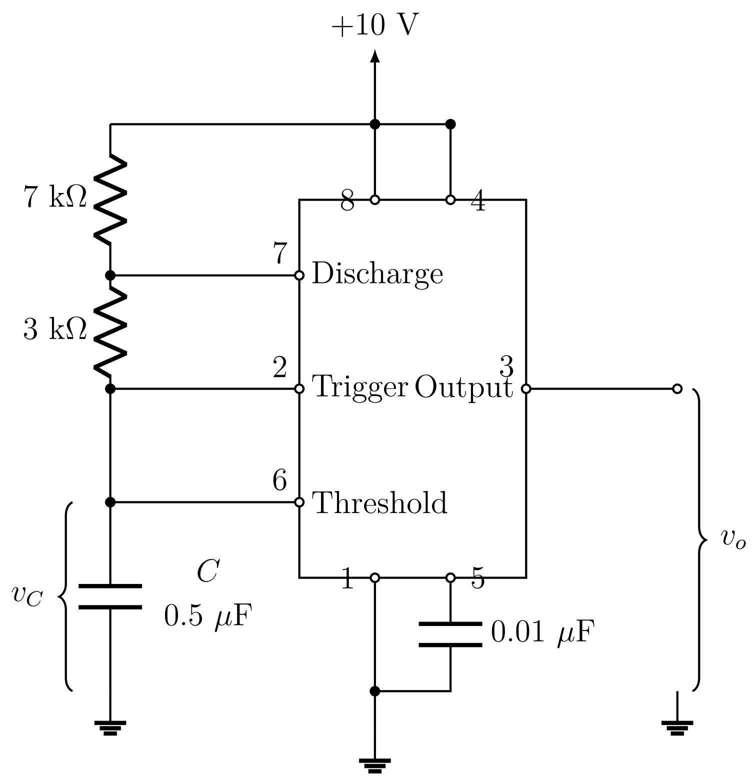
(a) draw waveshapes of V_C and V_O;

(b) determine t_{low} and t_{high};

(c) find duty cycle and free-running frequency;

(d) discuss how frequency depends on C with a graph;

(e) design resistors and capacitors so that duty cycle becomes 0.40.



(29 marks) [EEE 263 | L-2/T-1 | Jan–2021]

Triangular Wave Generator

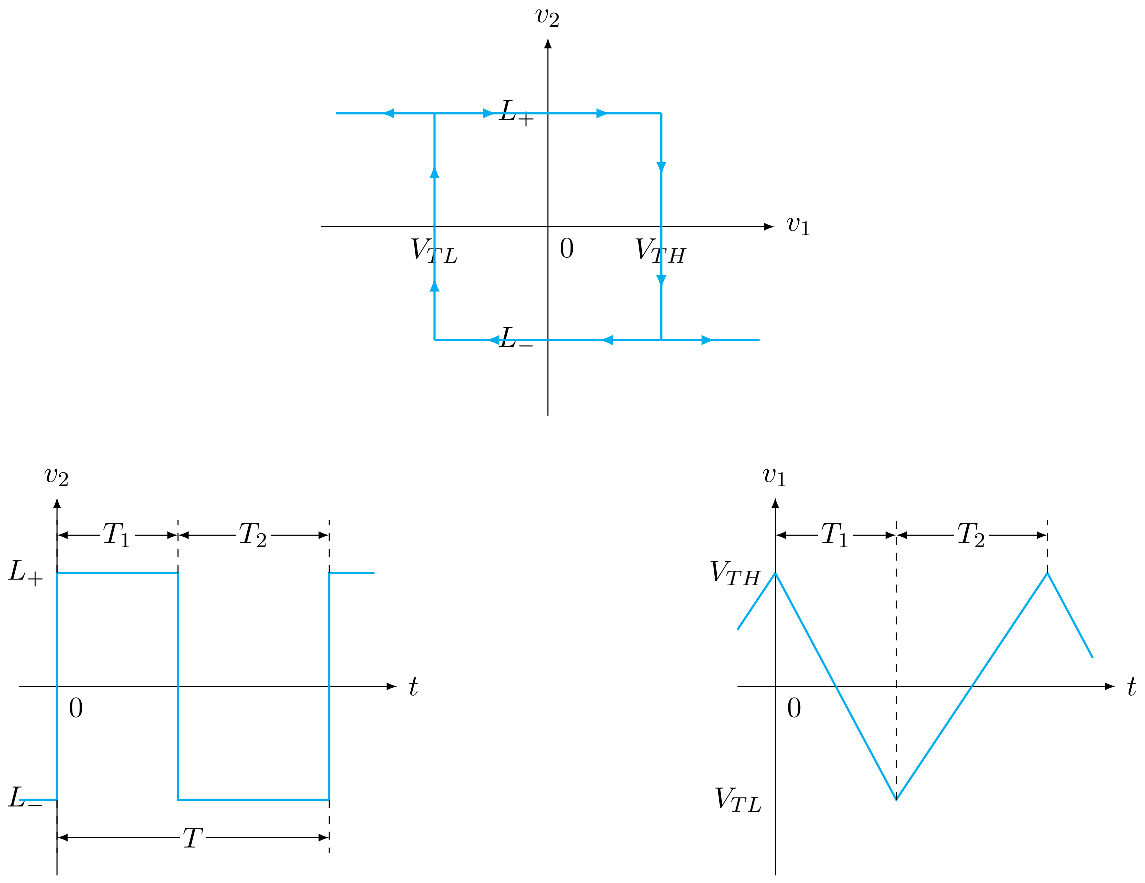
Q1. Draw and explain the operation of a triangular wave generator circuit using op-amps and a Schmitt trigger. Derive the frequency of oscillation for the circuit. **(20 marks)** [EEE 263 | L-2/T-1 | 2022–23]

Q2. Draw a sawtooth wave generator circuit using an op-amp. Explain the operation of the circuit. **(23 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q3. Drawing appropriate circuits, show how you can generate triangular waves using a bistable multivibrator and appropriate Op-amp circuits. **(26 marks)** [EEE 263 | L-2/T-1 | 2020–21]

Q4. Design a circuit using a bistable multivibrator which generates a triangular waveform with oscillation frequency of 2 kHz and 10 V peak-to-peak amplitude. **(16 marks)** [EEE 263 | L-2/T-1 | 2018–19]

Q5. A bistable multivibrator with a given hysteresis transfer characteristic is provided. Design a signal generator based on the bistable multivibrator that generates square and triangular waveforms. Choose values of your designed circuit components for which the frequency of oscillation is 1 MHz.



(18 marks) **[EEE 263 | L-2/T-1 | Jan–2021]**

BUET · EEE 263 · Electronic Circuits

Electronic Circuits

Part VI — Digital Logic and Sequential Circuits

CMOS logic design, flip-flops, counters, and digital circuit implementation

S9 Digital Logic & Sequential Circuits

CMOS Logic Design and Implementation

Q1. Discuss basic operational characteristics and parameters of CMOS and TTL logic families. **(23 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q2. Implement the function $f(A, B, C, D) = \sum m(0, 4, 8, 9, 10, 11, 12)$ using CMOS. **(13 marks)** [EEE 263 | L-2/T-1 | 2018–19]

Q3. Draw a CMOS complex gate for the logic function $f(a, b, c, d) = \sum m(0, 1, 2, 4, 6, 8, 10, 12, 14)$. Use as few transistors as possible. **(15 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Q4. Derive complex gates for the following logic functions using as few transistors as possible:

(a) $f(X_1, X_2, X_3, X_4) = \sum m(0, 4, 6, 8, 9, 15) + D(3, 7, 11, 13)$

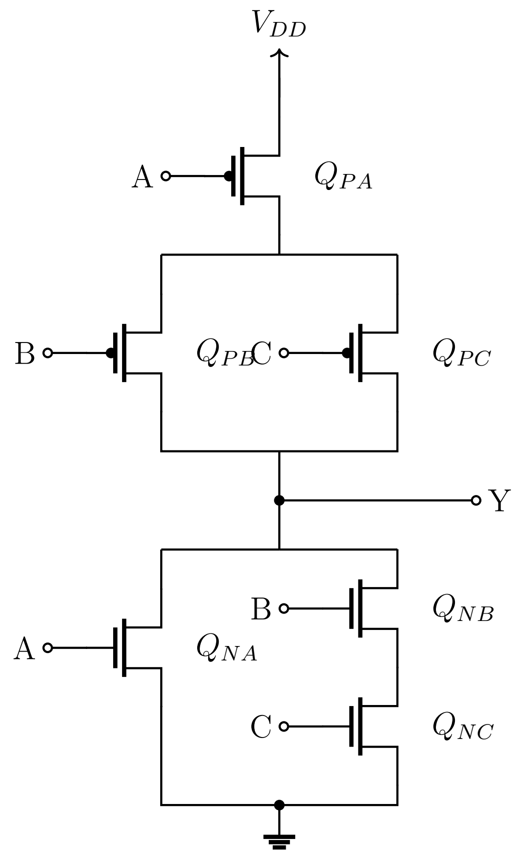
(b) $\bar{f} = (A + B)(C + D)(\bar{A} + B + D)$

(21 marks) [EEE 263 | L-2/T-1 | 2016–17]

Q5. Implement the function $f(A, B, C, D) = \sum m(1, 2, 4, 7, 14, 15)$ using (i) CMOS circuit and (ii) pseudo NMOS circuit. Which circuit would you prefer between your two designed circuits and why? **(25 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

CMOS Transistor Sizing

Q1. Calculate transistor W/L ratios for the logic circuit shown. Assume that for the basic inverter $n = 1.5$ and $P = 5$ and the channel length is $0.25\,\mu$m.

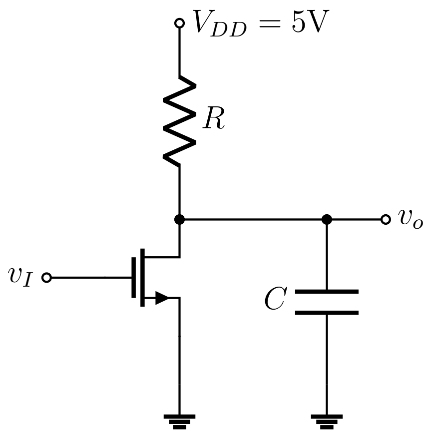


(16 marks) [EEE 263 | L-2/T-1 | 2018–19]

Q2. Design transistor W/L ratios for the logic $Y = \overline{(AB + D)C}$ implemented using CMOS circuit. Assume that for the basic inverter $(W/L)_n = n = 2$ and $(W/L)_p = p = 7$ and the channel length is $0.15\,\mu$m. **(21 marks)** [EEE 263 | L-2/T-1 | Jan–2021]

Inverter Propagation Delay

Q1. For the inverter shown with a capacitor C connected between the output node and ground: if at $t = 0$ the input goes low to high, find the equation of time for v_0 to reach $\frac{1}{2}(V_{OL} + V_{OH})$. Also calculate the high-to-low propagation time for $R = 25\,\text{k}\Omega$ and $C = 20\,\text{pF}$.



(10.67 marks) [EEE 263 | L-2/T-1 | 2018–19]

Transmission Gate Implementation

Q1. Implement the following function Y using CMOS transmission gate configuration:

$$Y = AB + \bar{A}C$$

(7 marks) [EEE 263 | L-2/T-1 | 2018–19]

CMOS Flip-Flop Implementation

Q1. Show a CMOS implementation of edge-triggered D flip-flops with asynchronous reset. **(25 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Flip-Flop Construction

Q1. Show how D flip-flop and T flip-flop can be constructed from J-K flip-flops. **(15 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Counter Design

Q1. Design a MOD-3 counter using JK flip-flops and verify the design showing the truth table. **(15 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q2. Design a modulo-12 up counter with synchronous reset. **(15 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Q3. Design a 4-bit up/down counter using T flip-flops. It should include a control pin $\overline{UP}/DOWN$. When $\overline{UP}/DOWN = 0$ the circuit behaves as an up counter; when $\overline{UP}/DOWN = 1$ it behaves as a down counter. The circuit should also have a synchronous active-low $\overline{RESET}$ pin. **(20 marks)** [EEE 263 | L-2/T-1 | 2016–17]

Combinational Logic Design

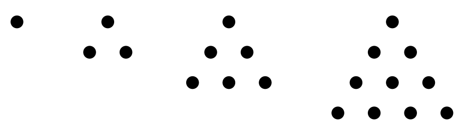
Q1. Draw a PLA programmed to implement the following functions:

$$Y_1 = A\bar{B}C + \bar{A}C + B$$
$$Y_2 = \bar{B}AC + AB\bar{C} + \bar{B}$$
$$Y_3 = \bar{A}C + B\bar{C}A$$

(15 marks) [EEE 263 | L-2/T-1 | 2017–18]

Q2. Implement the function $f(x_1, x_2, x_3) = \sum m(2, 3, 4, 6, 7)$ using two-input LUTs (Look Up Tables). You can use as many LUTs as needed. **(16 marks)** [EEE 263 | L-2/T-1 | 2017–18]

Q3. Triangular numbers are sequences formed by continuous summation of natural numbers (1, 3, 6, 10, ...). Design the simplest possible logic circuit that outputs 1 whenever it detects a 4-bit triangular number.



The triangular numbers
1, 3, 6, 10, ...

(11 marks) **[EEE 263 | L-2/T-1 | 2016–17]**

Priority Encoder

Q1. Design a four-line to two-line priority encoder circuit with active high inputs and outputs, with priority assigned to the higher-order data input line ($I_4 > I_3 > I_1$). Use NOR gates only for the circuit implementation. **(16 marks)** **[EEE 263 | L-2/T-1 | 2017–18]**

Shift Register Design

Q1. Design a 4-bit shift register using D flip-flops and a four-to-one multiplexer with mode selection inputs S_1 and S_0. The register operates according to the following function table:

S_1	S_0	**Register Operation**
0	0	No change
0	1	Complement the four outputs
1	0	Clear register to 0 (synchronous with clock)
1	1	Shift Right

(15 marks) **[EEE 263 | L-2/T-1 | 2017–18]**

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Electronic Circuits

Part VII — Data Converters, PLL & Memory

DAC, ADC, Phase-Locked Loop, DRAM, Flash Memory

S10 Data Converters, PLL & Memory

Digital-to-Analog Converter (DAC)

Q1. What are DAC circuits? How does an R-2R ladder DAC circuit operate? Explain using a circuit diagram and mathematical analysis. **(17 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Analog-to-Digital Converter (ADC)

Q1. A 6-bit A/D converter has reference voltage $V_{REF} = 8\,\text{V}$ and reference count $n_{REF} = 2^6$. Values of some consecutive analog sampled voltages $|V_A|$ are: 0 V, 3 V, 5.5 V, 7.125 V. If these voltages are given as inputs to the converter, find the corresponding digital outputs. **(12 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Phase-Locked Loop (PLL)

Q1. Draw a simplified block diagram of a PLL and briefly explain its operation. **(17 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Memory: DRAM and Flash

Q1. Explain the read and write operation in a DRAM cell. **(16 marks)** [EEE 263 | L-2/T-1 | 2021–22]

Q2. Explain briefly the programming and erase operation in a Flash Memory. **(15 marks)** [EEE 263 | L-2/T-1 | 2021–22]